

BT101

Introductory Biology

(Lecture 1)

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Course syllabus (BT 101, Introductory Biology)

- ❑ **Nutrients, bioenergetics and cell metabolism:** Essential nutrients to sustain life; biological energy and laws of thermodynamics, basics of aerobic and anaerobic glycolysis and citric acid cycle. (Dr. Priyadarshi Satpati)
- ❑ **Genes and chromosomes:** DNA, DNA replication; Central dogma of molecular biology: Transcription and translation; Mendelian Genetics; Genetic engineering/Cloning and its applications. (Dr. Kusum K Singh)
- ❑ **Biological systems:** Body systems required to sustain human physiology, special sense organs including hearing, taste, smell and visual receptors. (Dr. Souptick Chanda)
- ❑ **Evolution of life:** Origin of Life; Darwin's concepts of evolution; Biodiversity.
- ❑ **Cell, the structural and functional unit of life:** Three domains of life; cell types, cell organelles and structure; Basic biomolecules of cell. (Dr. Rajkumar P Thummer)

Text Books

1. J. L. Tymoczko, J. M. Berg and L. Stryer, Biochemistry, 8th Ed, W. H. Freeman & Co, 2015
2. D. L. Nelson and M. M. Cox, Lehninger Principles of Biochemistry, 7th Ed, Macmillan Worth, 2017.

Reference Books

1. N. Hopkins, J. W. Roberts, J. A. Steitz, J. Watson and A. M. Weiner, Molecular Biology of the Gene, 7th Ed, Benjamin Cummings, 1987.
2. C. R. Cantor and P. R. Schimmel, Biophysical Chemistry (Parts I, II and III), W.H. Freeman & Co., 1980.
3. C. C. Chatterjee, Human Physiology, Vol 1 & 2, 11th Ed, Medical Allied Agency, 1987.
4. Hall, B.K., Evolution: Principles and Processes, 1st Ed, Jones & Bartlett, 2011.

Course syllabus

Evolution of life:

- ☐ Origin of Life
- ☐ Darwin's concepts of evolution
- ☐ Biodiversity

Cell, the structural and functional unit of life:

- ☐ Three domains of life; cell types, cell organelles and structure
- ☐ Basic biomolecules of cell

Origin of life

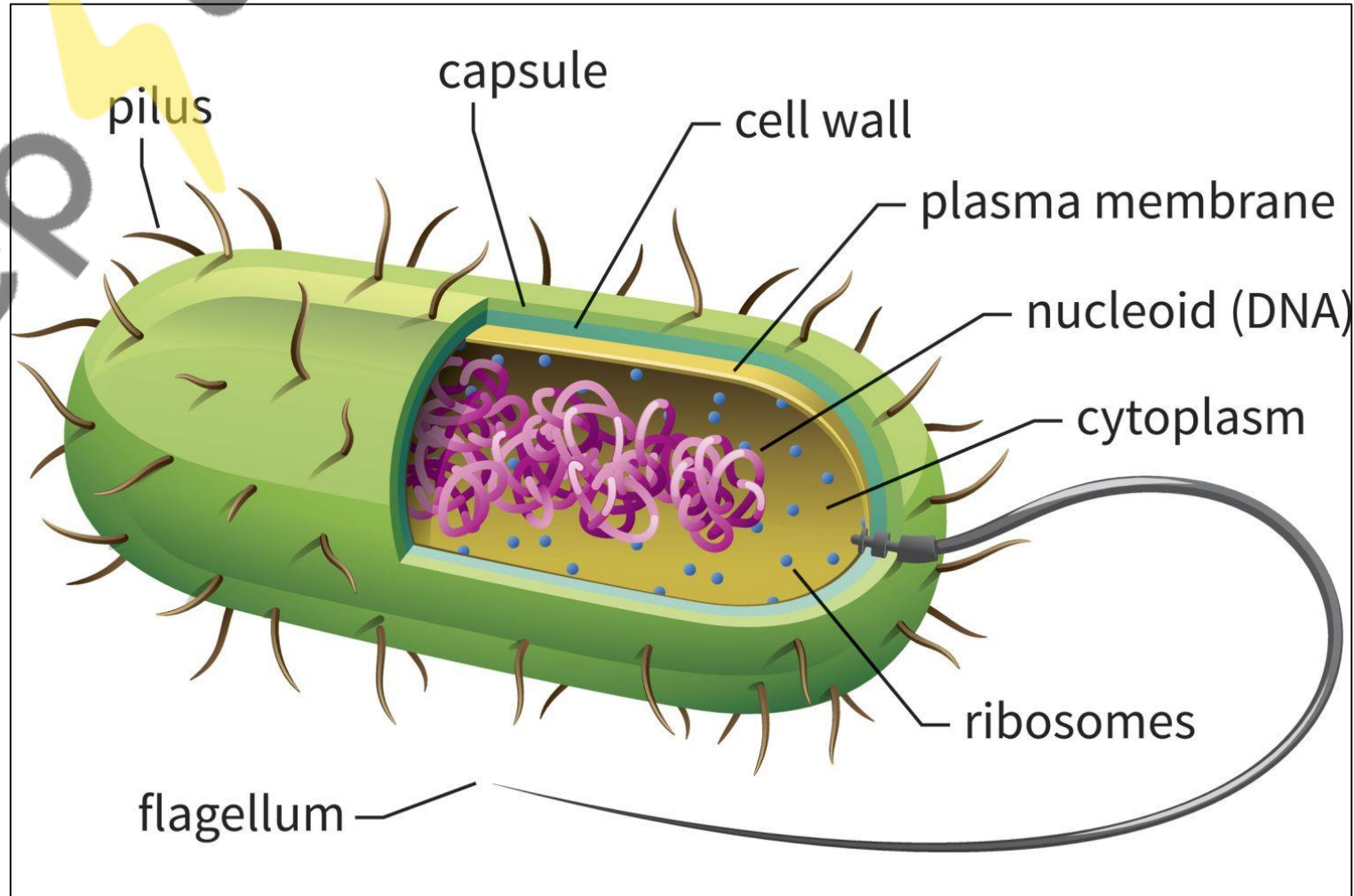
- ❑ It is believed that life first emerged at least **3.8 billion (3.8×10^9) years ago**, approximately 750 million years after Earth was formed.
- ❑ How life originated **and** how the first cell came into being are matters of speculation, since these events cannot be reproduced in the laboratory.
- ❑ Nonetheless, several types of experiments provide important evidence bearing on some steps of the process.

Types of organisms

- ☐ Prokaryotes
- ☐ Eukaryotes

Prokaryotes

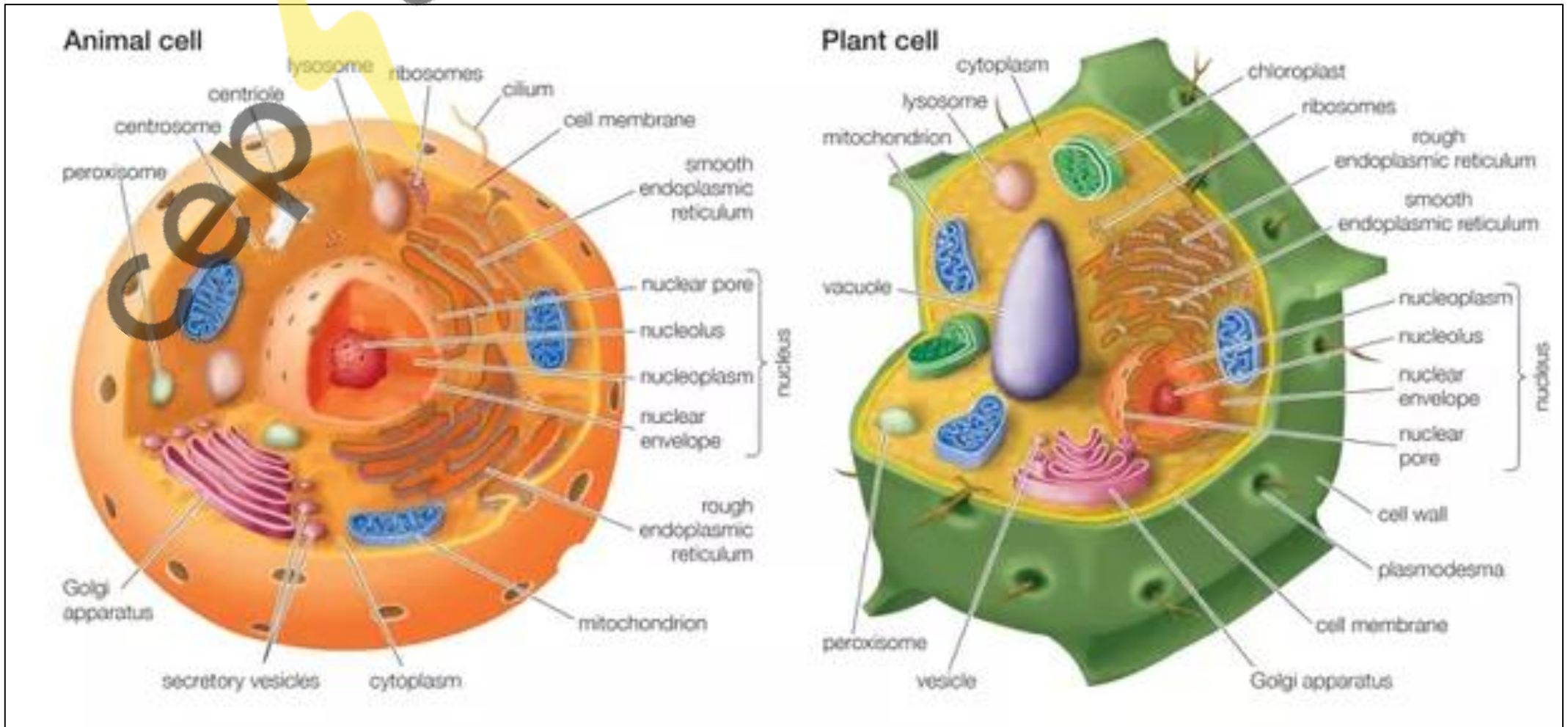
- ❑ It is a single-celled organism that lacks a membrane-bound nucleus (karyon), mitochondria or any other membrane-bound organelle.



(Structure of *Escherichia coli* cell, a Prokaryotic cell)

Eukaryotes

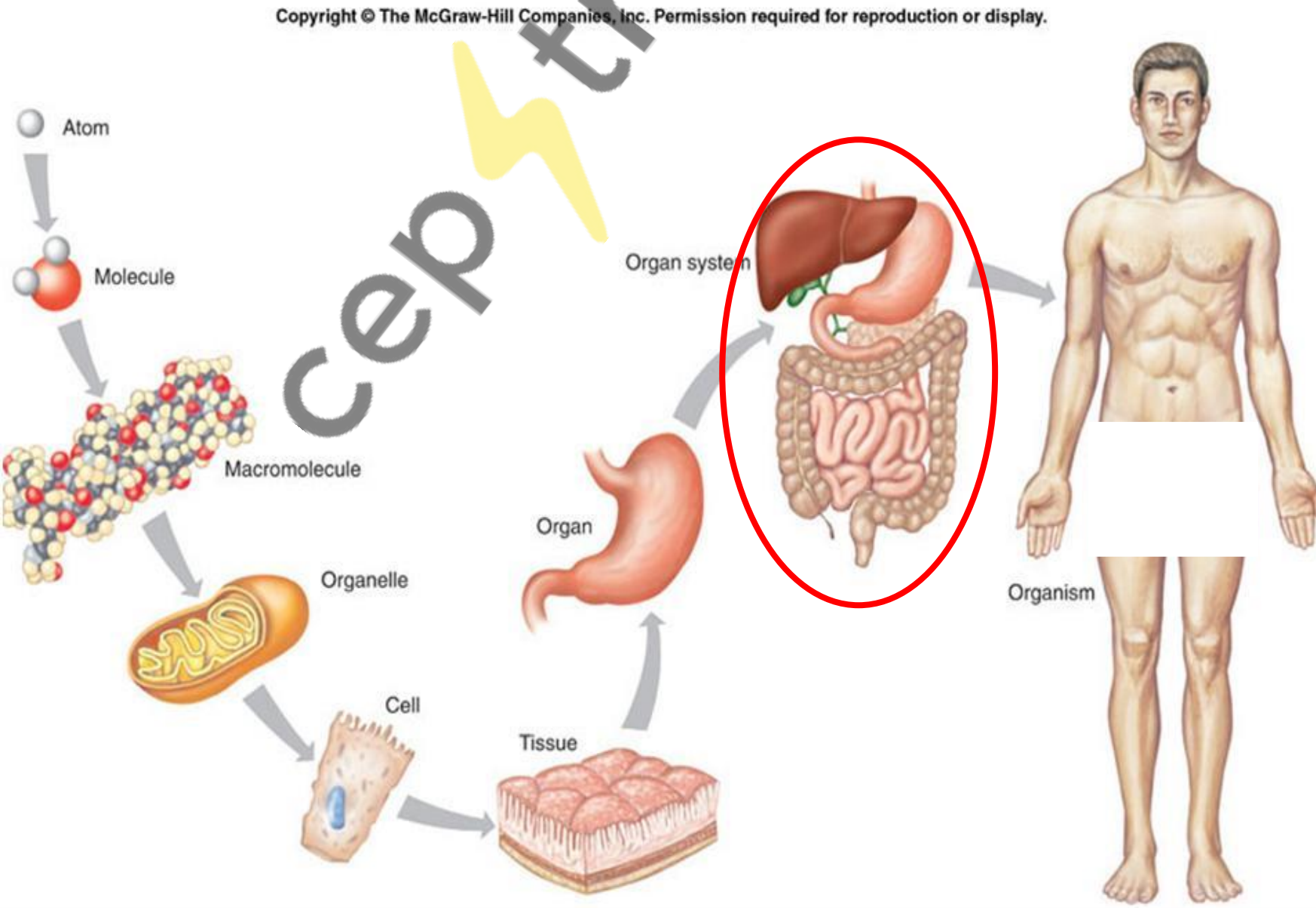
- ❑ It is any organism whose cells contain a nucleus and other organelles enclosed within membranes.



(Structure of a typical animal and plant cell; a Eukaryotic cell)

Eukaryote – Homo sapiens (Human)

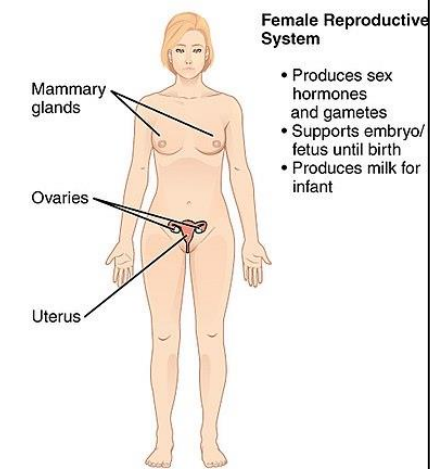
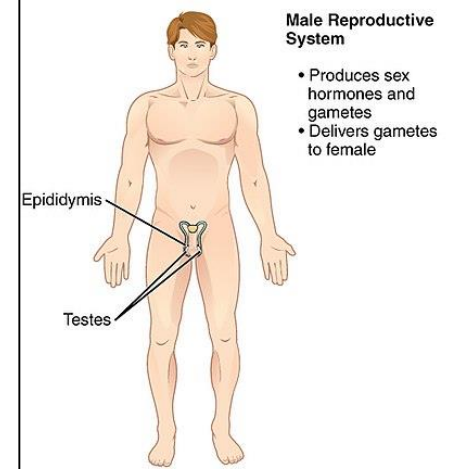
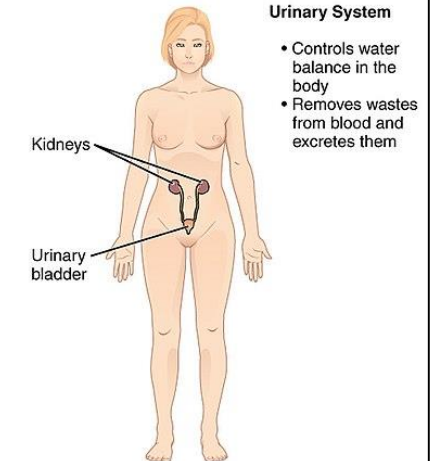
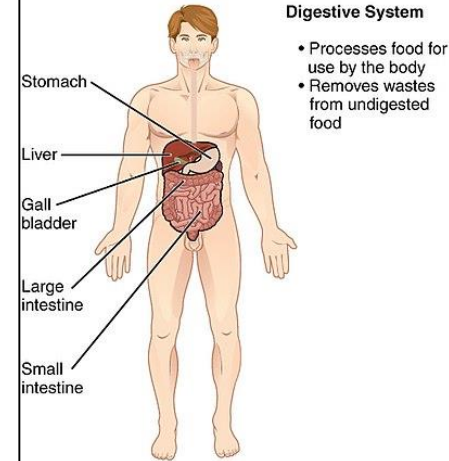
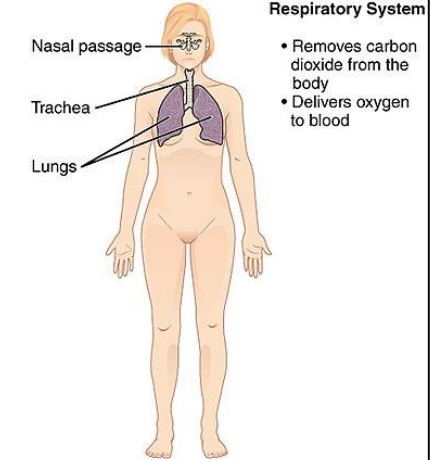
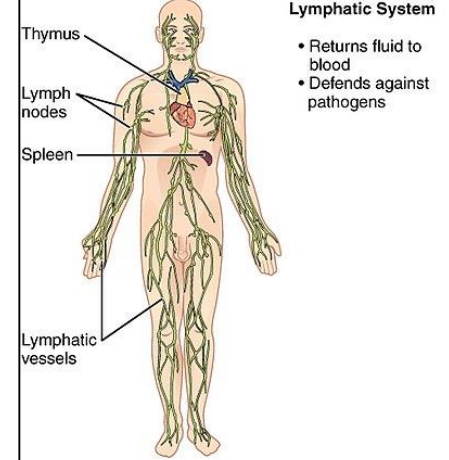
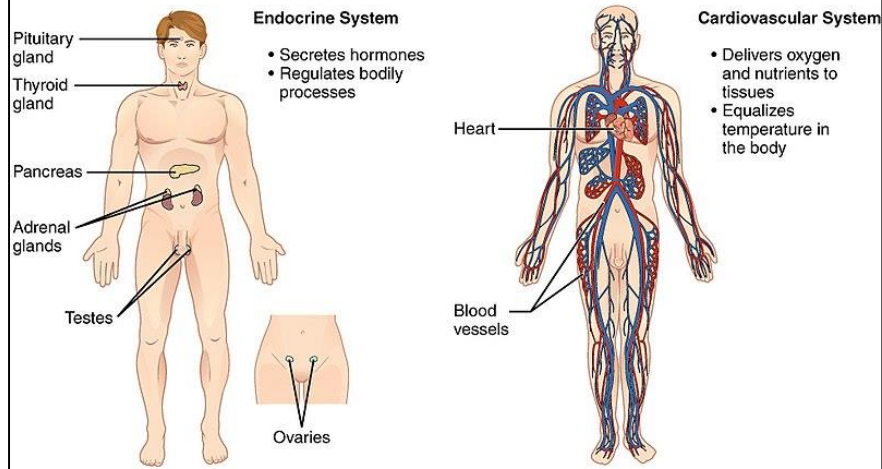
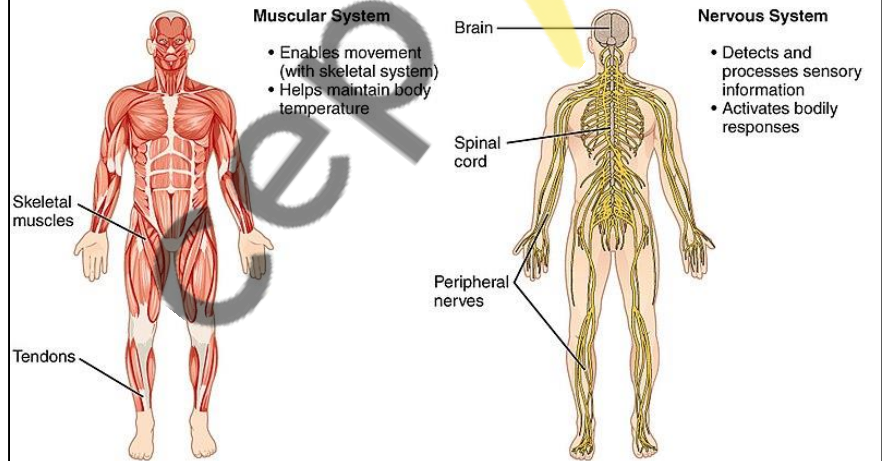
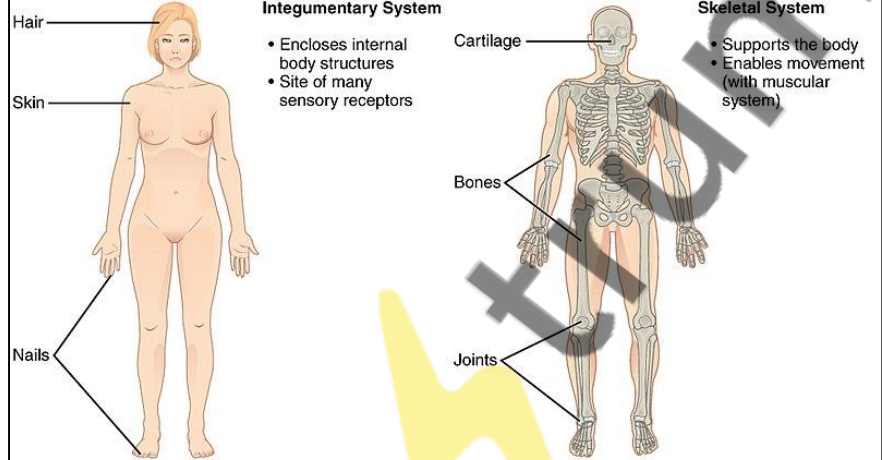
Fig. 1.03



- ☐ The human body is the structure of a human being.
- ☐ It is composed of many different types organ systems.
- ☐ They ensure homeostasis and the viability of the human body.

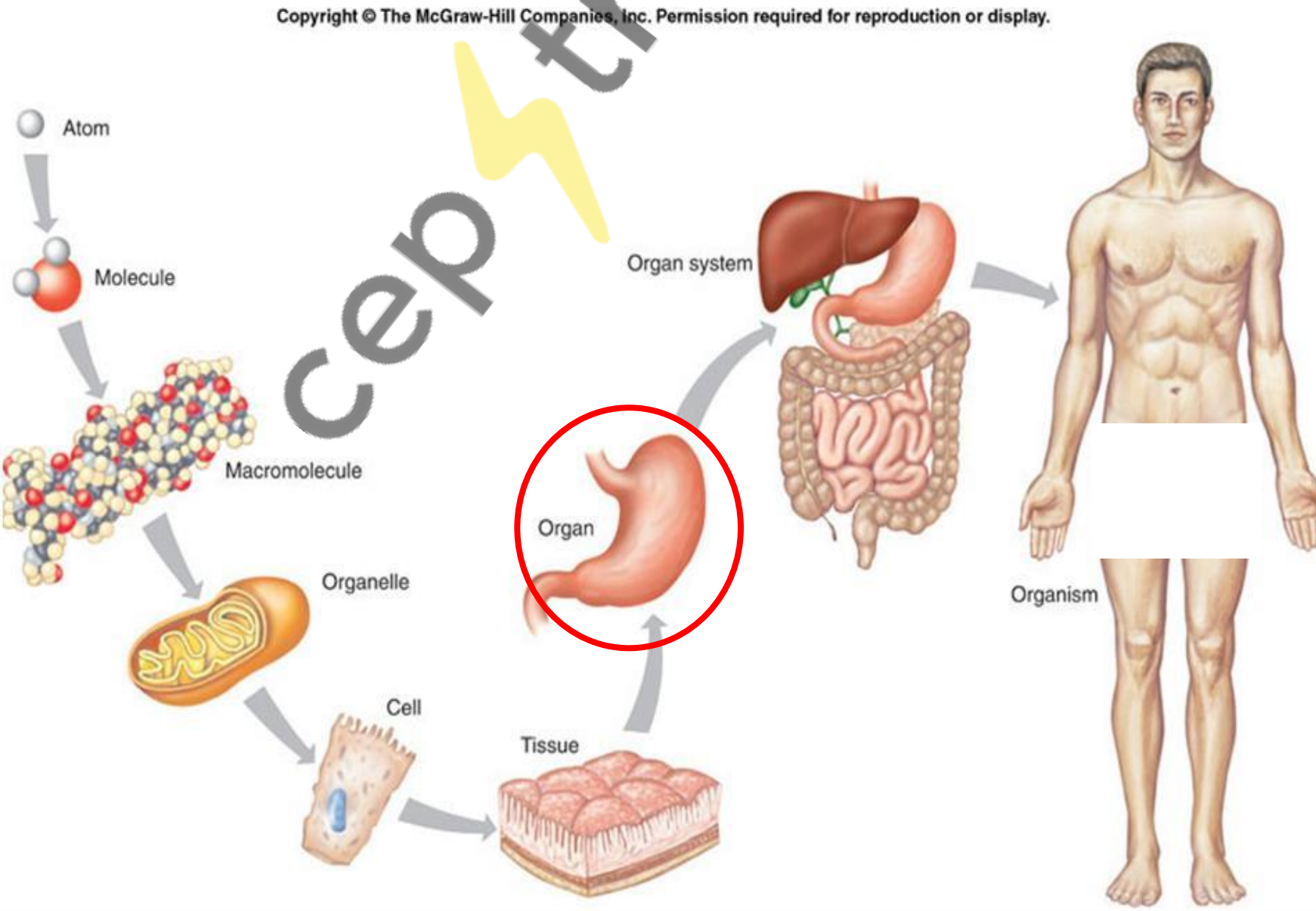
Various body systems present in a human body

1. Respiratory system
2. Cardiovascular system (heart and circulatory system)
3. Digestive system
4. Nervous system
5. Lymphatic system (or Immune system)
6. Endocrine system
7. Integumentary system (or Exocrine system)
8. Musculoskeletal system
9. Urinary system (or Excretory system or Renal system)
10. Reproductive system



Eukaryote – Homo sapiens (Human)

Fig. 1.03



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Number of organs present in a human body

- **79 organs** have been identified in the human body.
- **Skin** is the **largest and longest organ** according to its size and weight.
- **Pineal gland** near the center of the brain, in a groove between the hemispheres, is the **smallest organ**.
- **Liver** is the **largest internal** organ of the body.
- The **heart** is the **first major functional organ to develop** and starts to beat and pump blood at about three weeks into embryogenesis.
- The **last major organ** to develop and mature before birth are the **lungs**.
- The **brain** is the **last to reach its final mature** form and develops even after birth.

SENSORY ORGANS are as follows:-

- **Eyes**
- **Ears**
- **Nose**
- **Tongue**
- **Skin**

Number of organs present in a human body

News > Science

Mesentery: New organ discovered inside human body by scientists (and now there are 79 of them)

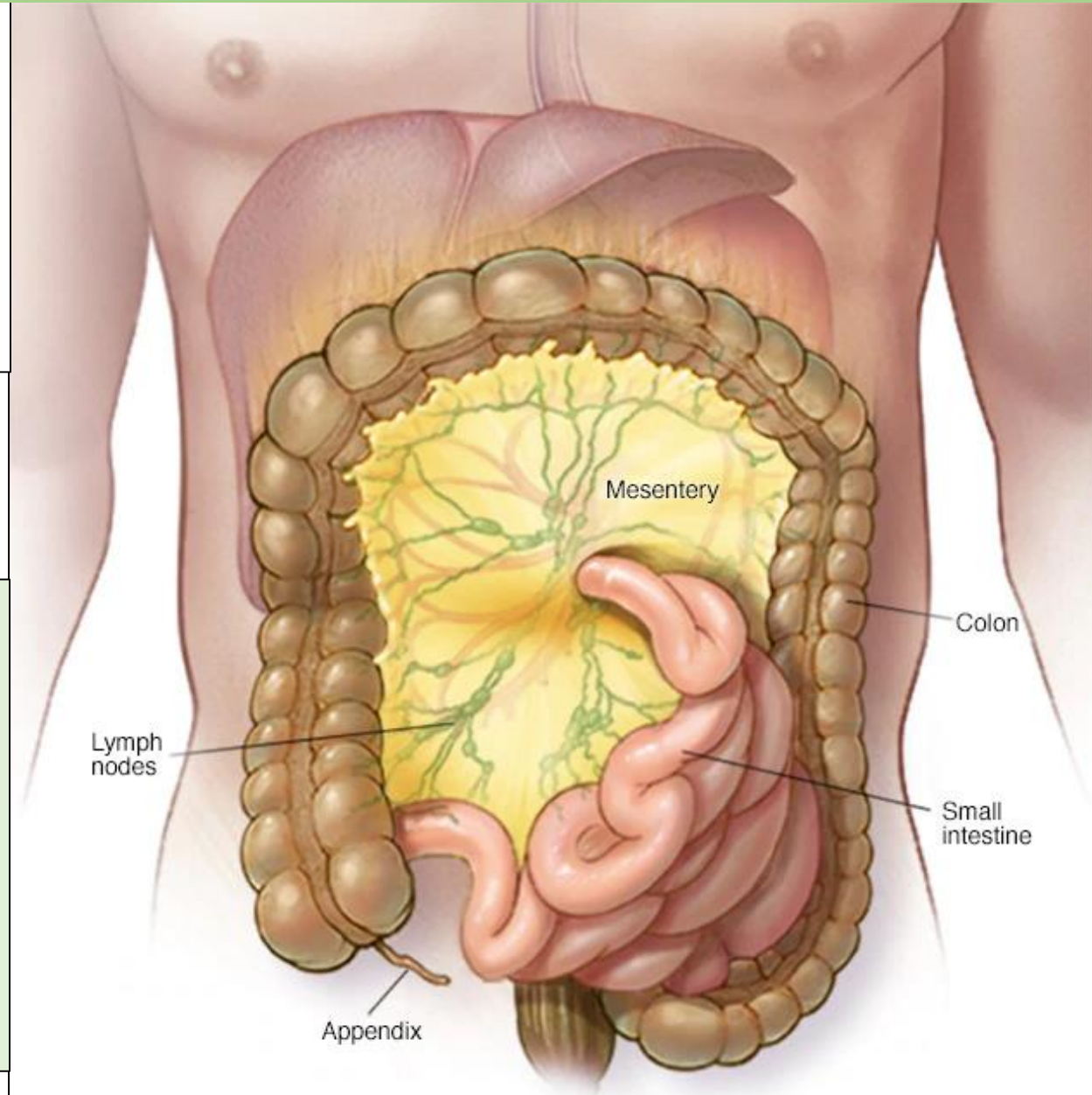
The finding opens up 'a whole new area of science'

Tom Embury-Dennis | @tomemburyd | Tuesday 3 January 2017 17:27 |
325 shares | 57 comments



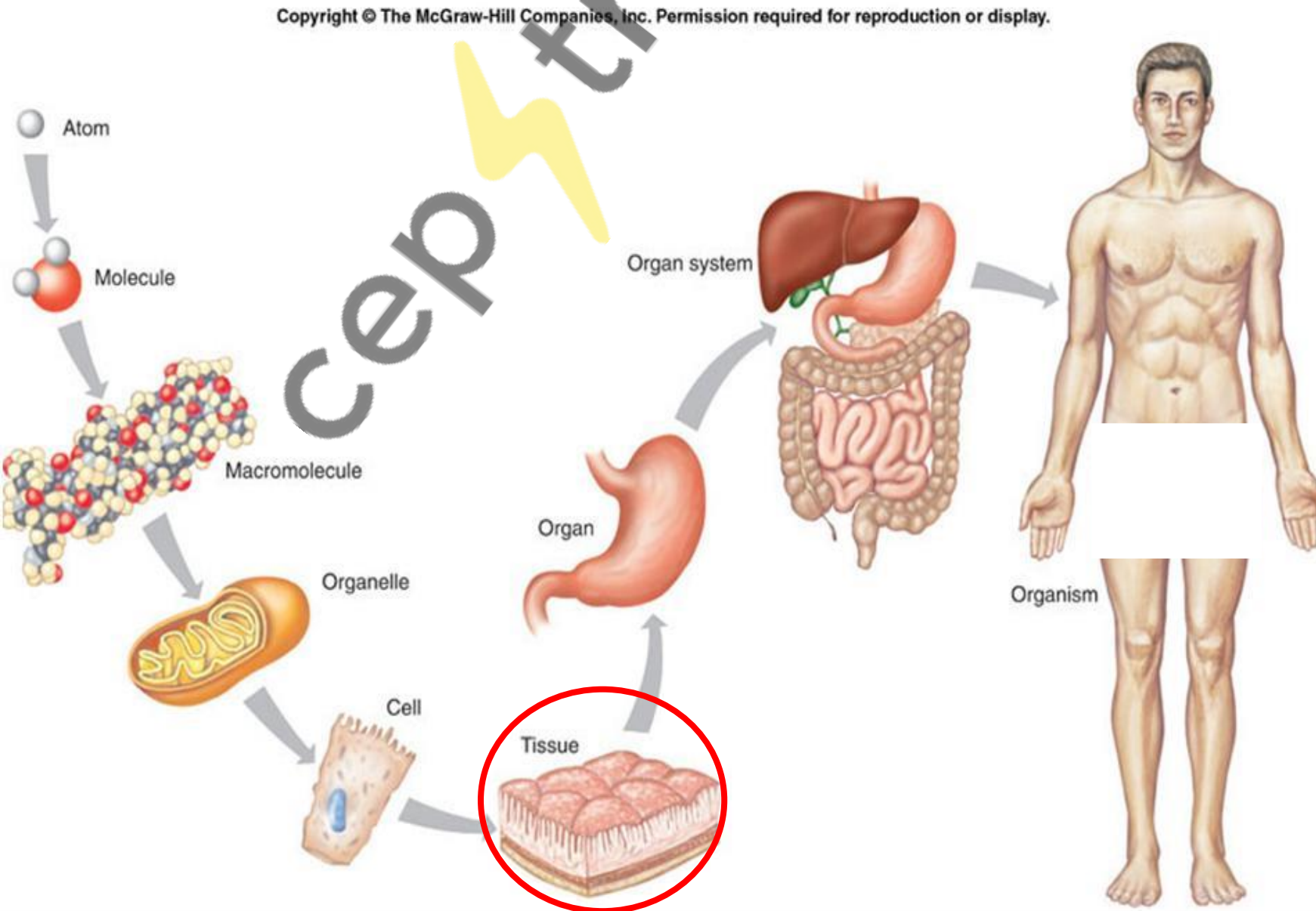
Click to follow
The Independent

- ❑ Large single continuous stretch of tissue that support and position all the digestive organs in the abdomen.
- ❑ The organ is responsible for transporting blood and lymphatic fluid between the intestine and the rest of the body.



Eukaryote – Homo sapiens (Human)

Fig. 1.03



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Different types of tissues present in a human body

Human body tissue consists of groups of cells with a similar structure working together for a specific function. There are **four** main types of tissues in a body.

Epithelial tissue (epithelium)

- is made of layers (sheets) of cells that cover the surfaces of the body that come into contact with the exterior world, lines internal cavities and passageways, and form glands.

Connective tissue

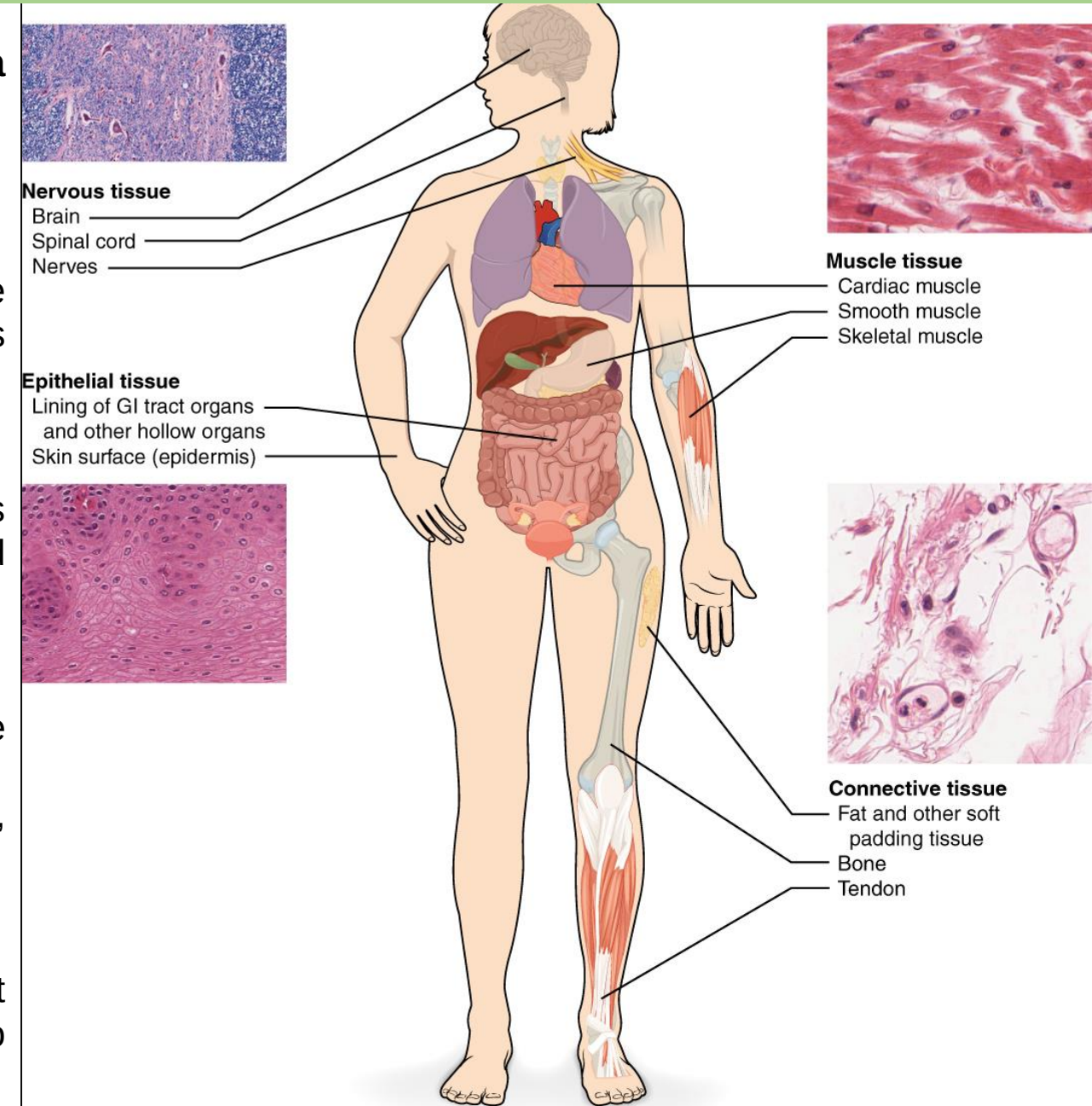
- binds the cells and organs of the body together and performs many functions, especially in the protection, support, and integration of the body.

Muscle tissue

- is excitable, responds to stimulation and contracts to provide movement.
- occurs as three major types: skeletal (voluntary) muscles, smooth muscles, and the cardiac muscle in the heart.

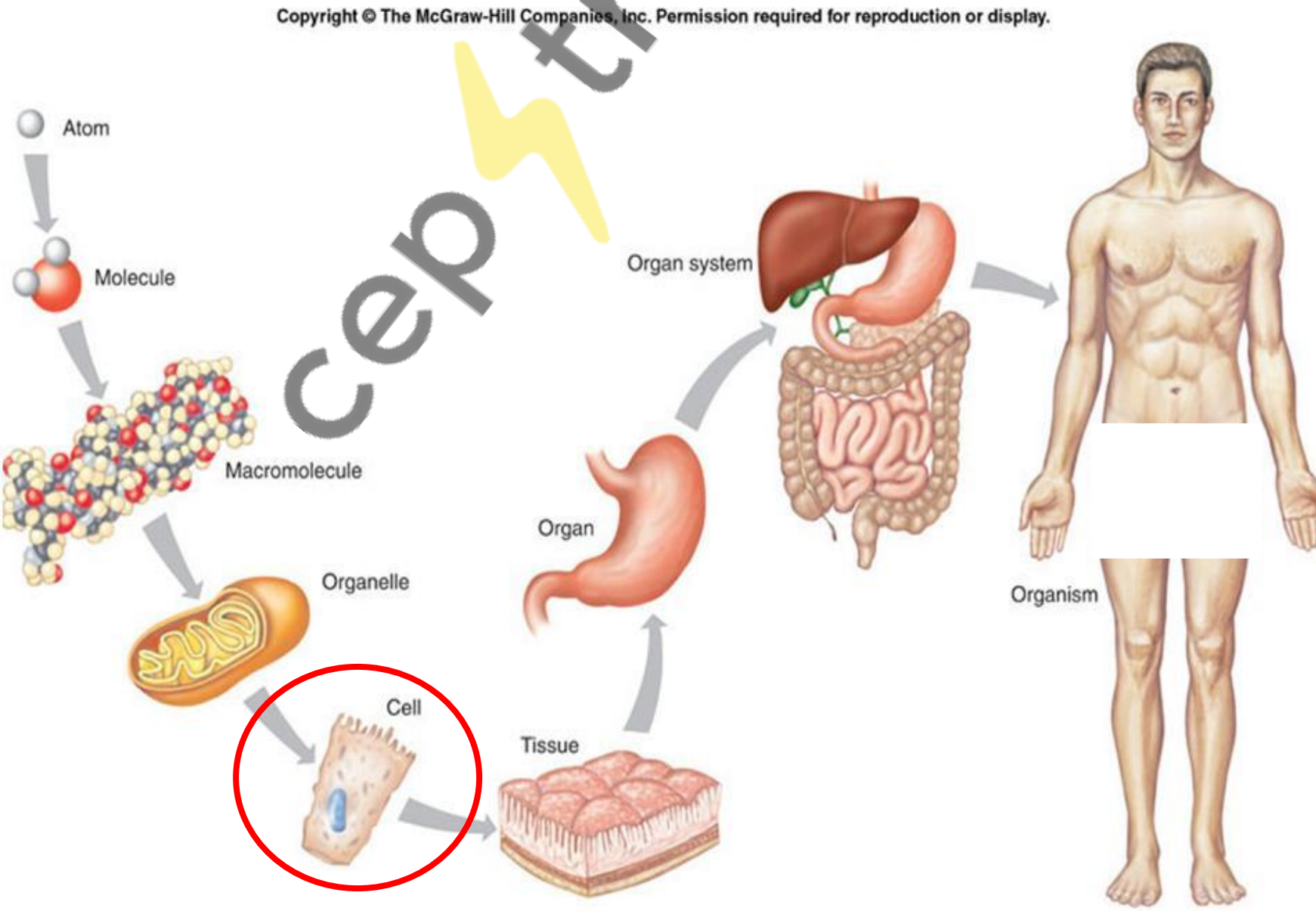
Nervous tissue

- is also excitable, allows the body to receive signals and transmit information as electric impulses from one region of the body to another.



Eukaryote – Homo sapiens (Human)

Fig. 1.03



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Number of cells / different cell types present in a human body

Total number of human cells present in a human body:

- ❑ There are 37.2 Trillion Cells in Your Body
- ❑ 1 trillion = 1,000,000,000,000 (1×10^{12}); so 37,200,000,000,000

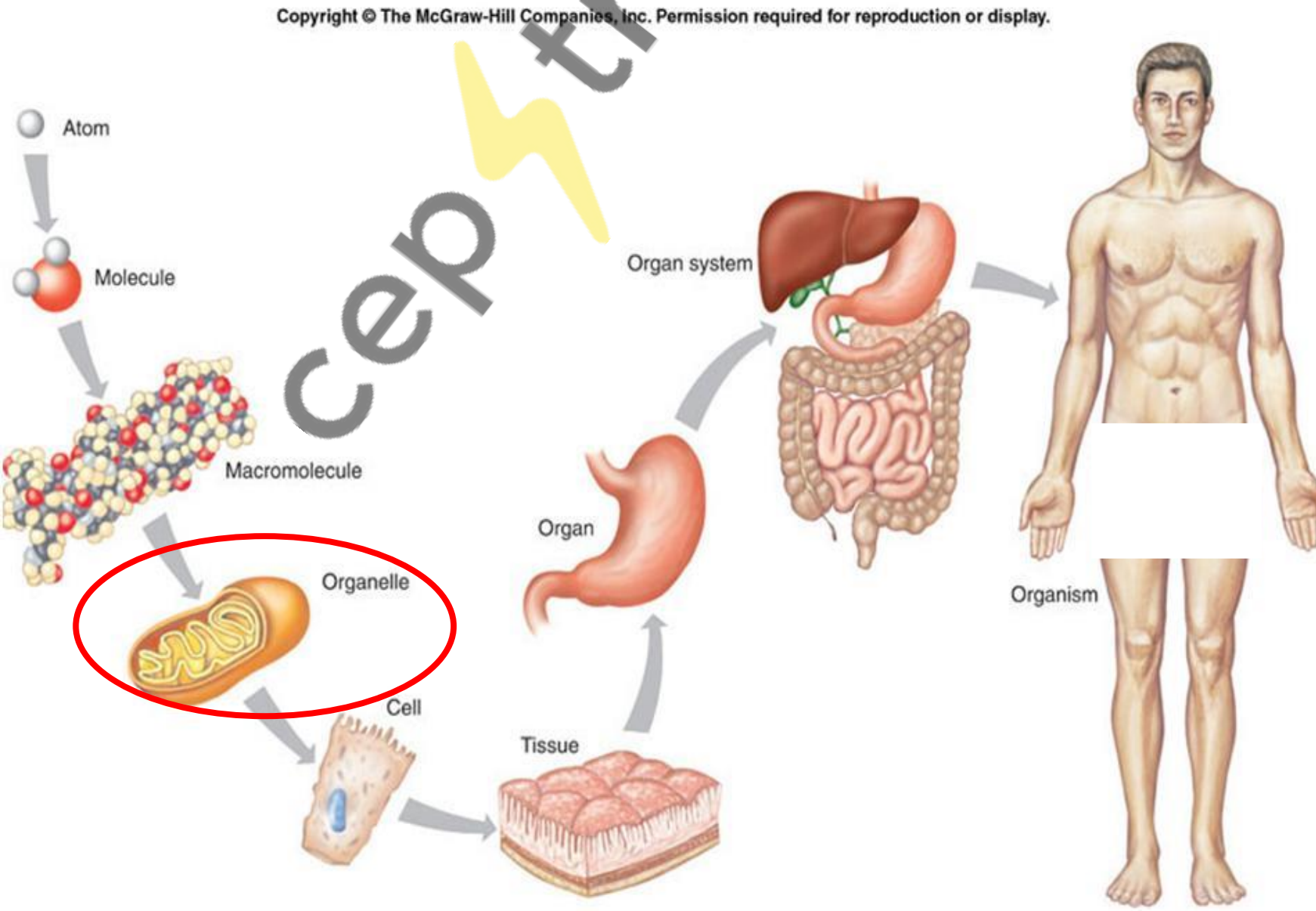
➤ Note: This does not include any of the millions of microbes living inside our body

Total number of different human cell types present in a human body:

- ❑ 200-220 cell types
- ❑ (neuron, cardiomyocyte, beta-cell, hepatocyte, fibroblast, keratinocyte, RBC, T lymphocyte, B lymphocyte, natural killer cell, etc.)

Eukaryote – Homo sapiens (Human)

Fig. 1.03



- ☐ The human body is the structure of a human being.
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Various organelles in a human cell and its major function(s)

1. **Cell membrane** – (is a membrane that separates the interior of all cells from the outside environment (the extracellular space) and protects the cell from its environment; acts as a barrier for cell contents. It consists of double phospholipid layer and monolayer of protein scattered around phospholipid layer. The other components in plasma membrane are cholesterol and glycoproteins).
2. **Cytosol** – (fluid that contains organelles; represents the material outside the nucleus and inside the cell membrane)
3. **Cytoskeleton** – (a network of protein structures that extend throughout the cytoplasm; provides the cell with an internal framework (e.g. microfilaments and microtubules))
4. **Nucleus** – (control center of the cell; contains genetic material; It is the center of the cell because it contains genetic material (DNA). It consists of three main regions: the nuclear membrane, the nucleolus and chromatin.

Nuclear membrane: Nuclear membrane serves as a barrier of nucleus. It consists of a double phospholipid membrane and contains nuclear pores that allow for the exchange of material with the rest of the cell.

Chromatin: It is composed of DNA and protein scattered throughout the nucleus. Chromatin condenses to form chromosomes when the cell divides)

5. **Nucleolus** – (functions as site of ribosome production; ribosomes then migrate to the cytoplasm through nuclear pores)
6. **Ribosome** (little dots) – sites of protein synthesis in the cell.

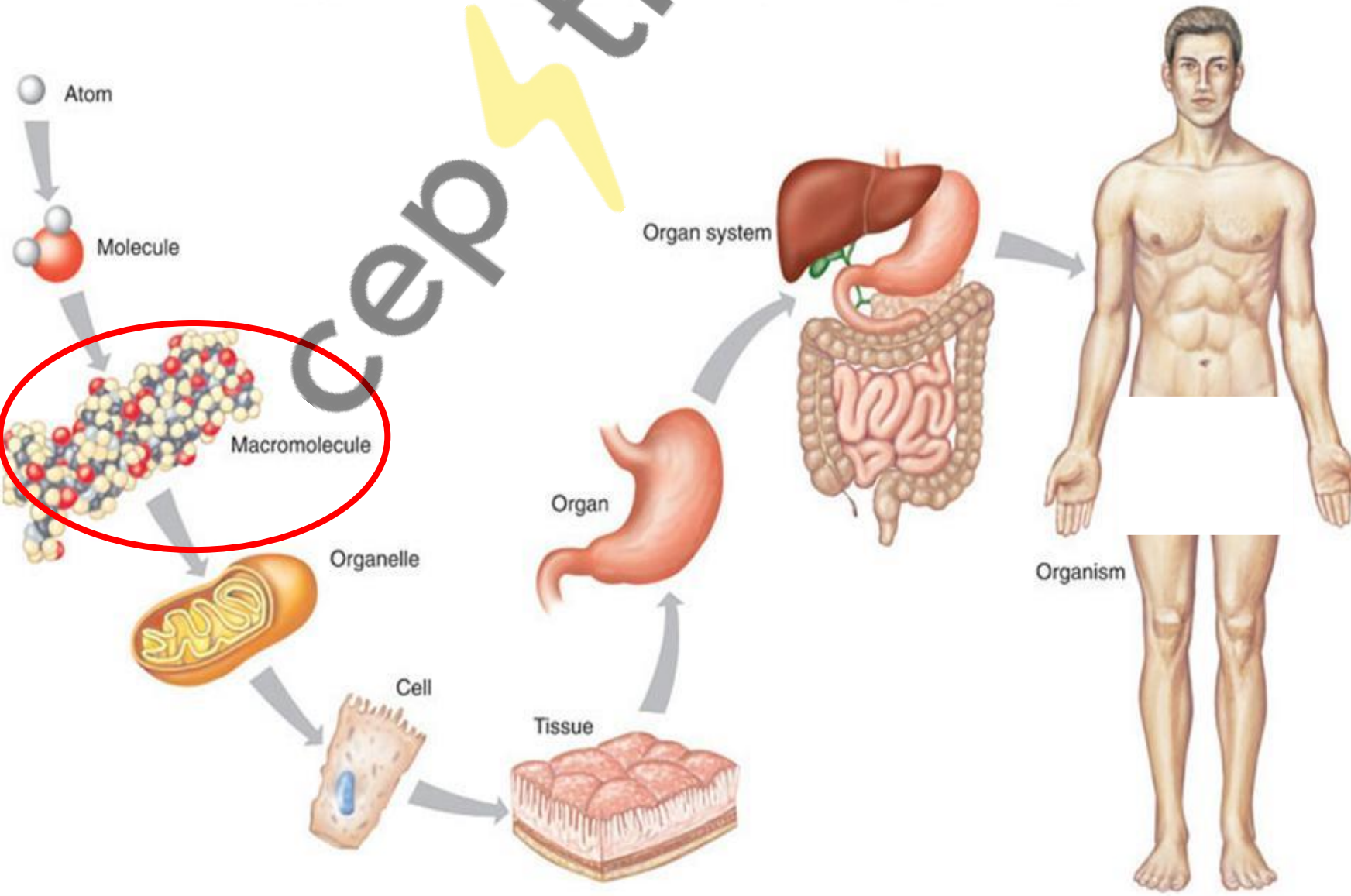
Various organelles in a human cell and its major function(s)

7. **Rough endoplasmic reticulum** – (carry ribosomes that represent sites of protein synthesis)
8. **Smooth endoplasmic reticulum** – (function in cholesterol synthesis and breakdown, fat metabolism, and detoxification of drugs)
9. **Smooth endoplasmic reticulum** – (function in cholesterol synthesis and breakdown, fat metabolism, and detoxification of drugs)
10. **Golgi apparatus** (or "Golgi body") – (modifies and packages proteins, secrete vesicles, plasma membrane components and lysosomes)
11. **Mitochondrion** – (powerhouse of the cell; generate ATP for cellular activities)
12. **Lysosomes** – (suicidal bag of the cell; contain enzymes that digest non-usable materials within the cell)
13. **Peroxisomes** – (detoxify harmful substances and break down free radicals; involved in catabolism of very long chain fatty acids,)
14. **Centrioles** – (involved in the organization of mitotic spindle fibers during cell division and in the completion of cytokinesis)
15. **Centrosomes** – (serves as the main microtubule organizing center (MTOC) of the animal cell, as well as a regulator of cell-cycle progression; are composed of two centrioles arranged at right-angles to each other)
16. **Vesicles** – (involved in metabolism, transport, buoyancy control, and temporary storage of food and enzymes)

Eukaryote – Homo sapiens (Human)

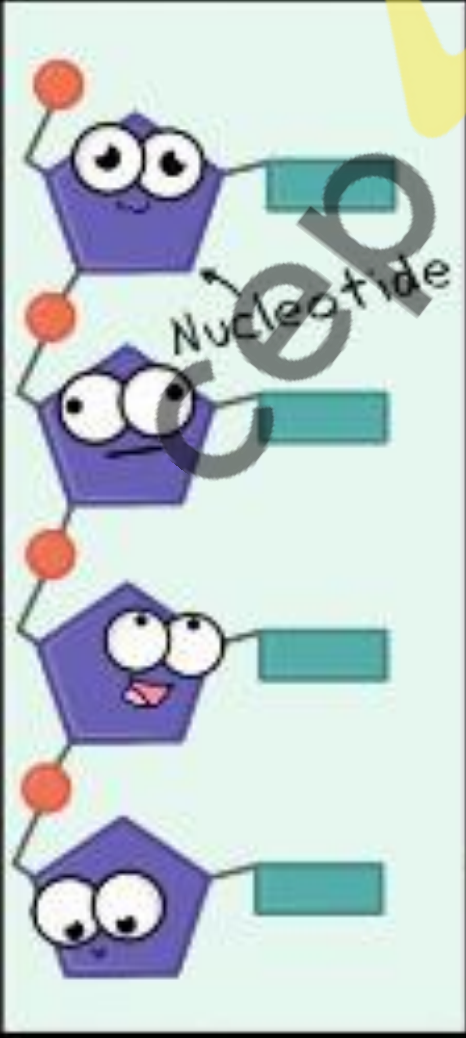
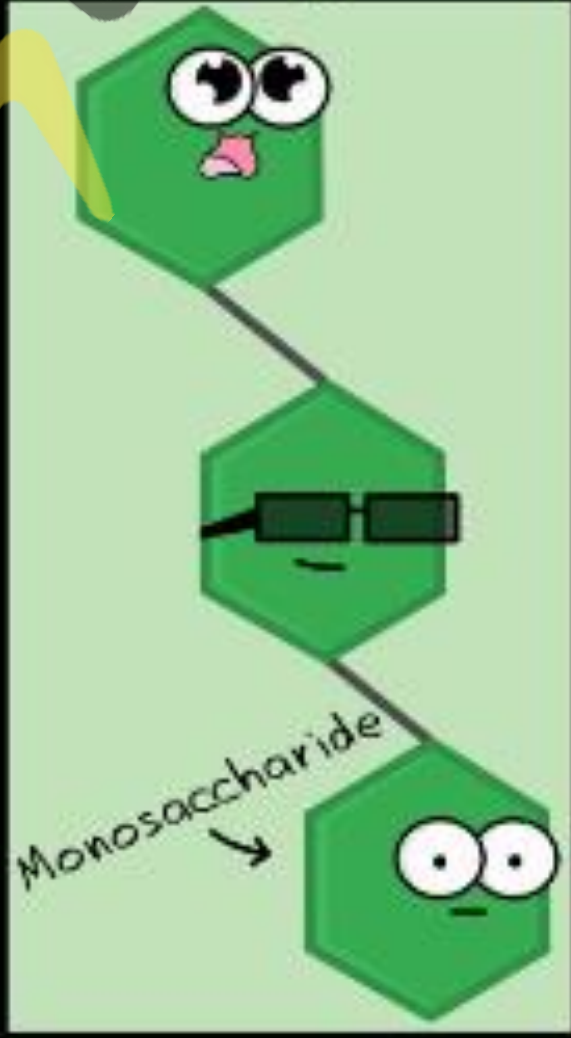
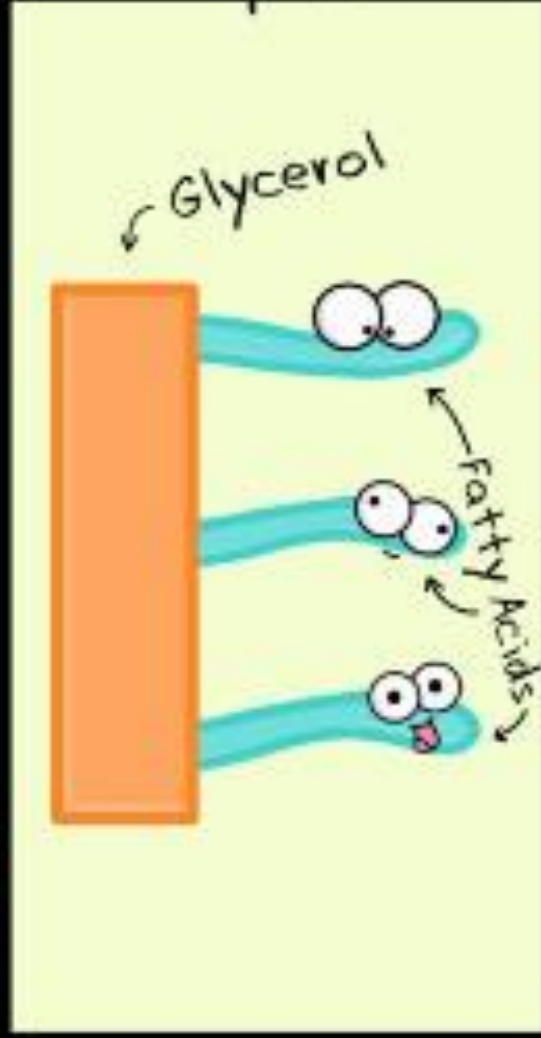
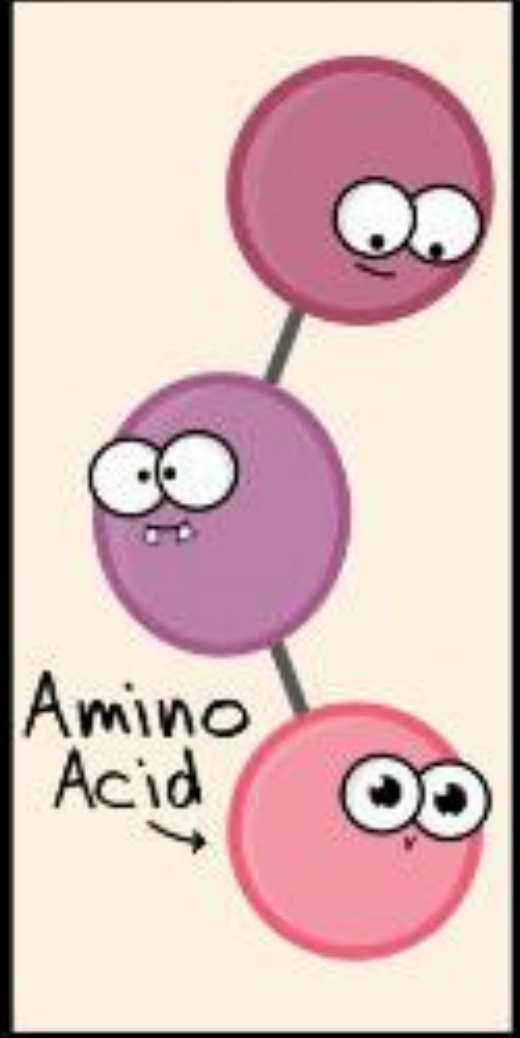
Fig. 1.03

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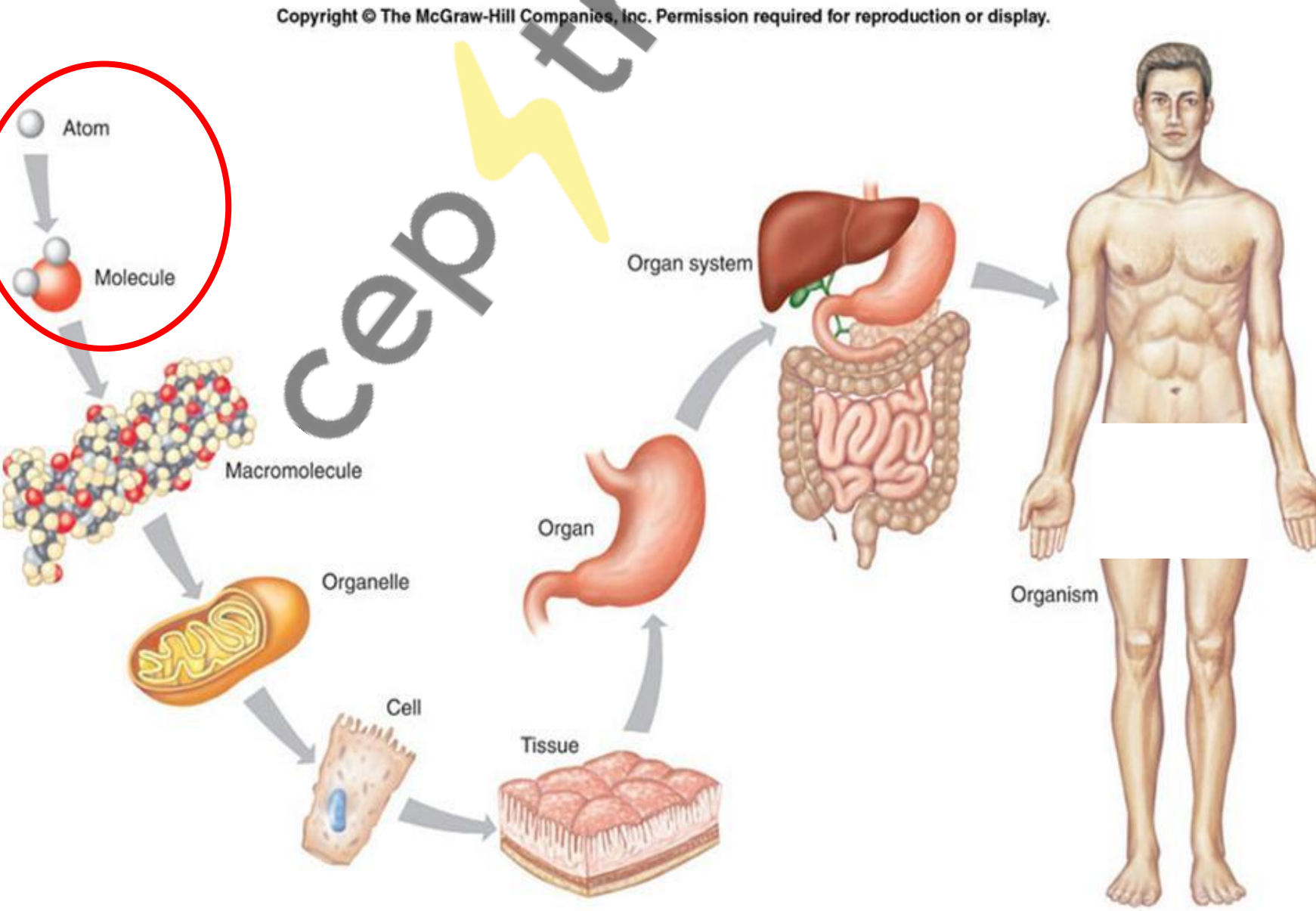
- ☐ The human body is the structure of a human being.
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Macromolecules

Nucleic Acid	Carbohydrate	Lipid	Protein
 Nucleotide	 Monosaccharide	 Glycerol Fatty Acids	 Amino Acid

Eukaryote – Homo sapiens (Human)

Fig. 1.03



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Number of molecules and atoms in human body

❑ How many molecules are there in the human body?

❑ <https://socratic.org/questions/how-many-molecules-are-in-the-human-body>

➤ The human body consists of about **2×10^{25} molecules**, and more than 99% of them are water.

❑ How many atoms are there in the human body?

❑ <https://www.thoughtco.com/how-many-atoms-are-in-human-body-603872>

❑ https://education.jlab.org/qa/mathatom_04.html

➤ There are approximately **7×10^{27} atoms** in the average human body. This is the estimate for a **70 kg** adult human male. Generally, a smaller person would contain fewer atoms; a larger person would contain more atoms.

➤ According to an estimate made by engineers at Washington University, there are around **10^{14} atoms in a typical human cell**.

➤ **Carbon, hydrogen, nitrogen, oxygen, calcium and phosphorus** together account for **99 percent** of the atoms in a person.

Different Levels of organization

Atom:

An atom is the smallest particle of an element or a molecule.
[carbon (C), Hydrogen (H), Oxygen (O), Nitrogen (N), etc.]

Molecule:

A molecule is a particle composed of two or more joined atoms by chemical bonds.
(carbon dioxide (CO₂), water (H₂O))

Macromolecule:

A macromolecule is a very large molecule.
(carbohydrates, lipids, proteins, and nucleic acids)

Organelle:

An organelle is a specialized subunit within in a cell which performs a particular function.
(cell membrane, cytoplasm, nucleus, mitochondria, etc.)

Cell:

The cell is a structural and functional unit of a living organism.
(neuron, cardiomyocyte, beta-cell, hepatocyte, fibroblast, keratinocyte, etc.)

Different Levels of organization

Tissue:

A tissue is a group of similar cells that perform a specialized function (epithelial, connective, muscle and nervous).

Organ:

An organ is a structure consisting of a group of tissues that perform a specialized function (skin, heart, brain, etc.).

Body System:

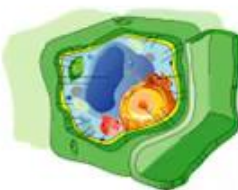
A body system is a group of organs that act together to perform a specialized function.
In total, there are 10 body systems.

Human body:

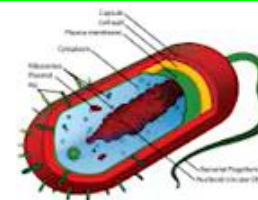
A living organism is the most complex level of organization. It consists of all the systems arranged in a discrete manner so as to facilitate functioning of the various organ systems in a synchronous manner.



EUKARYOTIC



PROKARYOTIC





Thank you for your attention

BT101

Introductory Biology

(Lecture 2)

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Guwahati

Introduction



Evolution of life:

☐ Origin of Life

- ☐ Darwin's concepts of evolution
- ☐ Biodiversity

Cell, the structural and functional unit of life:

- ☐ Three domains of life
- ☐ Cell types, cell organelles and structure
- ☐ Basic biomolecules of cell

Introduction



- ❑ Organisms are divided into **two** main classes, initially defined by whether they contain a **nucleus**.
- ❑ Prokaryotes (bacteria) lack a nuclear envelope; eukaryotic cells have a nucleus in which the genetic material is separated from the cytoplasm.
- ❑ Prokaryotic cells are generally smaller and simpler than eukaryotic cells; in addition to the absence of a nucleus, their genomes are less complex and they do not contain cytoplasmic organelles or a cytoskeleton.

Introduction



Table 1.1 Prokaryotic and Eukaryotic Cells

Characteristic	Prokaryote	Eukaryote
Nucleus	Absent	Present
Diameter of a typical cell	$\approx 1\mu\text{m}$	$10\text{--}100\mu\text{m}$
Cytoskeleton	Absent	Present
Cytoplasmic organelles	Absent	Present
<u>DNA</u> content (base pairs)	1×10^6 to 5×10^6	1.5×10^7 to 5×10^9
Chromosomes	Single circular <u>DNA</u> molecule	Multiple linear <u>DNA</u> molecules

<https://www.youtube.com/watch?v=zZtcMBTQaS4>
<https://www.youtube.com/watch?v=Pxujitlv8wc>
<https://www.youtube.com/watch?v=q9fthbK08Uc>

Introduction

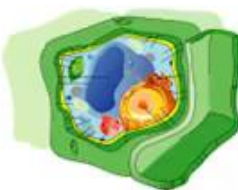


- ❑ **REPRODUCTION:** Asexual (binary fission) vs Sexual Reproduction
- ❑ **RIBOSOMES** (central dogma of molecular biology)
- ❑ **CHROMOSOMES** (circular vs linear; single vs multiple)
- ❑ Extrachromosomal material (plasmid)
- ❑ **MITOCHONDRIA vs CHLOROPLASTS**
- ❑ **EXAMPLES** of Prokaryotes and Eukaryotes
- ❑ **EXAMPLES** of Unicellular Eukaryotes

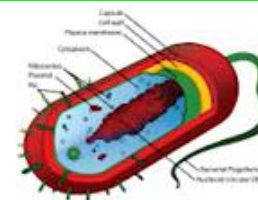
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<https://www.youtube.com/watch?v=Pxujitlv8wc>
<https://www.youtube.com/watch?v=q9fthbK08Uc>



EUKARYOTIC



PROKARYOTIC



Thank you for your attention

BT101

Introductory Biology

(Lecture 3)

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Introduction



Evolution of life:

☐ Origin of Life

- ☐ Darwin's concepts of evolution
- ☐ Biodiversity

Cell, the structural and functional unit of life:

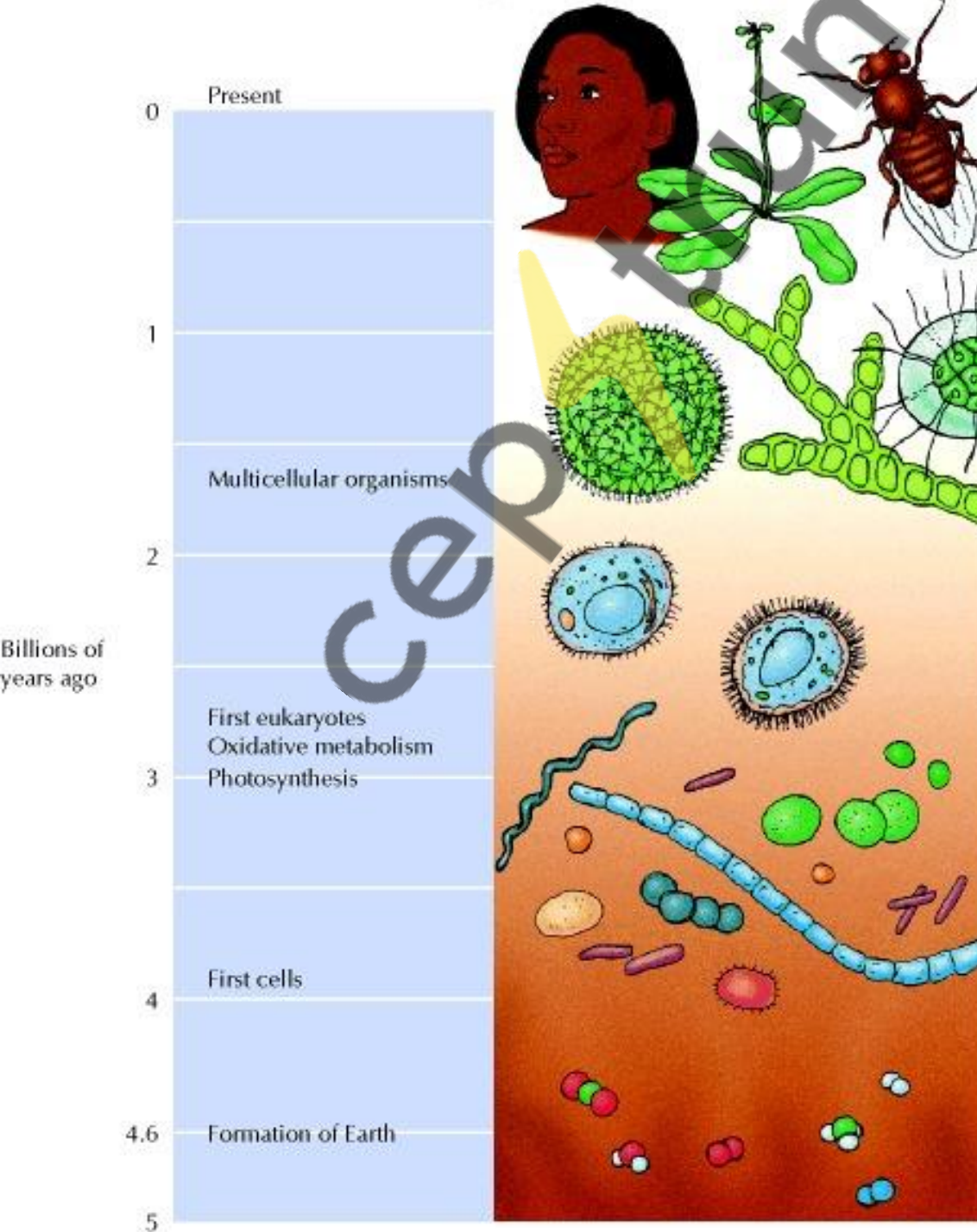
- ☐ Three domains of life
- ☐ Cell types, cell organelles and structure
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Introduction



- ❑ Cells are divided into **two** main classes, initially defined by whether they contain a nucleus.
- ❑ Prokaryotic cells (bacteria) lack a nuclear envelope; eukaryotic cells have a nucleus in which the genetic material is separated from the cytoplasm.
- ❑ Prokaryotic cells are generally smaller and simpler than eukaryotic cells; in addition to the absence of a nucleus, their genomes are less complex and they do not contain cytoplasmic organelles or a cytoskeleton.
- ❑ **In spite of these differences, the same basic molecular mechanisms govern the lives of both prokaryotes and eukaryotes, indicating that all present-day cells are descended from a single primordial ancestor.**
- ❑ **How did this first cell develop?**
- ❑ **And how did the complexity and diversity exhibited by present-day cells evolve?**

Introduction



Time scale of evolution.

The scale indicates the approximate times at which some of the major events in the evolution of cells are thought to have occurred.

Important Links



- ❑ <https://www.youtube.com/watch?v=8oSqXAwLsZc&t=158s>
- ❑ <https://www.youtube.com/watch?v=eCWwKrJyKxA>
- ❑ <https://www.youtube.com/watch?v=bBjD4A7R2xU>

Thank you for your attention

BT101

Introductory Biology

(Lecture 4)

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Introduction



- ❑ **Evolution** is the change in the inherited characteristics of biological populations over successive generations.

OR Simply

- ❑ **Evolution** is a process of gradual change that takes place over many generations, during which species of animals, plants, or insects slowly change some of their physical characteristics.
- ❑ **Evolutionary process** gives rise to diversity at every level of biological organization, including species, individual organism and molecules such as DNA and proteins.

Origin of Life



Origin of life on earth is also called **Protobiogenesis**.

The life came into existence from non living substances of primordial earth around 3.8 billion years ago.

Origin of life can be explained by following theories :

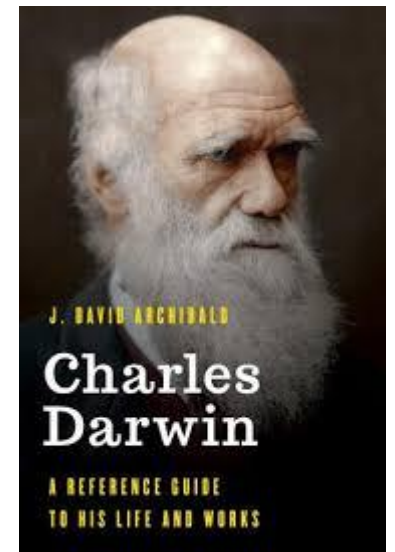
1. Theory of special creation
2. Theory of spontaneous generation
3. Theory of chemical evolution
4. Theory of panspermia/ cosmozoic theory

1. Theory of Special Creation



- I. All living organisms (different species or types of plants & animals) that we see today were/are created by God. (Father F. Saurez, 1548-1617)
- II. The diversity of life forms was always the same since the time of creation & will remain the same in future also.
- III. According to him Earth is around 4000 years old.

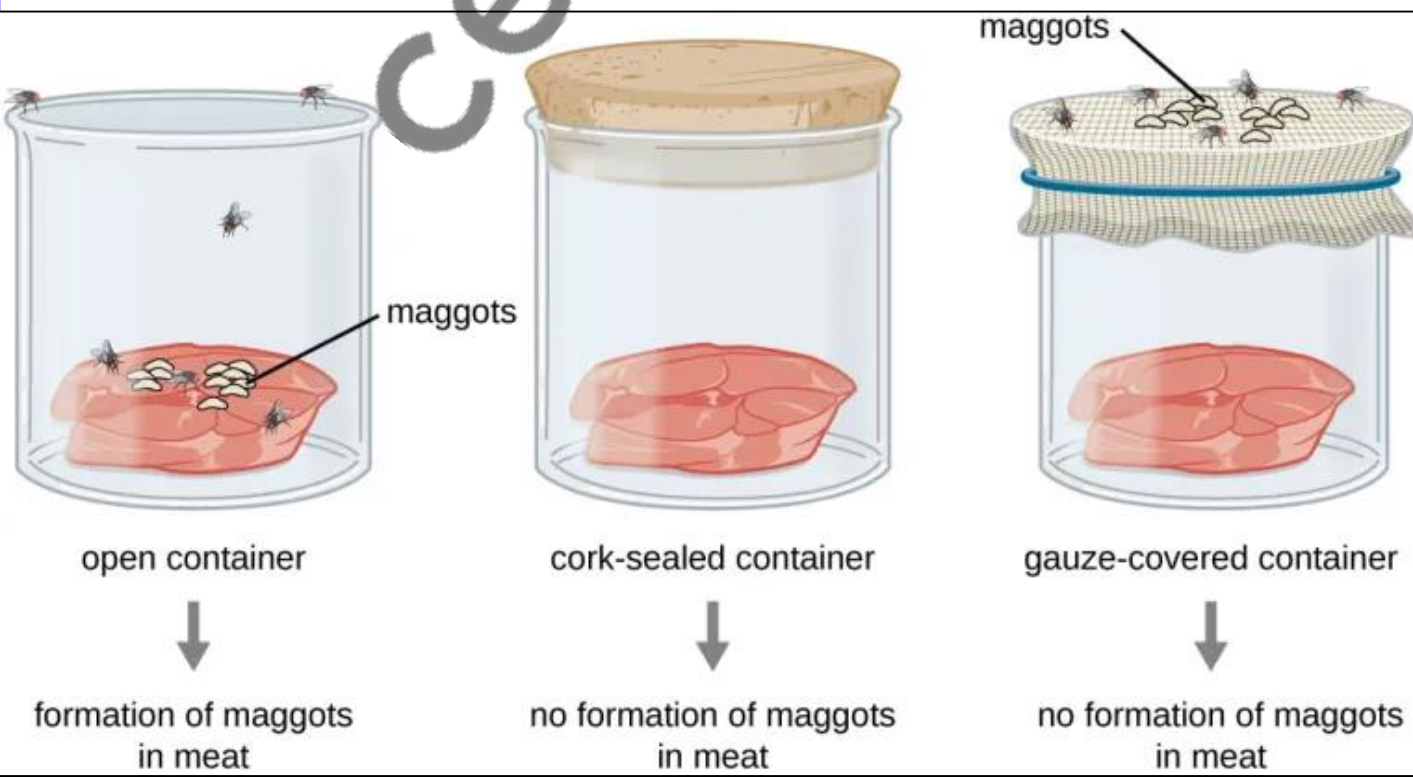
However, According to Charles Darwin (1809-1882), the existing life forms share similarities to varying degree among themselves and also with those life forms that existed millions of years ago.



2. Theory of Spontaneous Generation

- ❑ Spontaneous generation is an obsolete theory that states **life can arise from nonliving matter**.
- ❑ The Greek philosopher **Aristotle** first proposed this spontaneous generation theory.
- ❑ According to this theory, **organisms** do not descend from other organisms or from a parent. It only requires that certain conditions in their **environment** be fulfilled in order for creation to occur.
- ❑ Spontaneous generation explains the generation of complex organisms.
- ❑ Some examples are dust creating fleas, maggots arising from rotting meat, and bread or wheat left in a dark corner producing mice, etc.

https://www.youtube.com/watch?v=J_57hCdUg4w



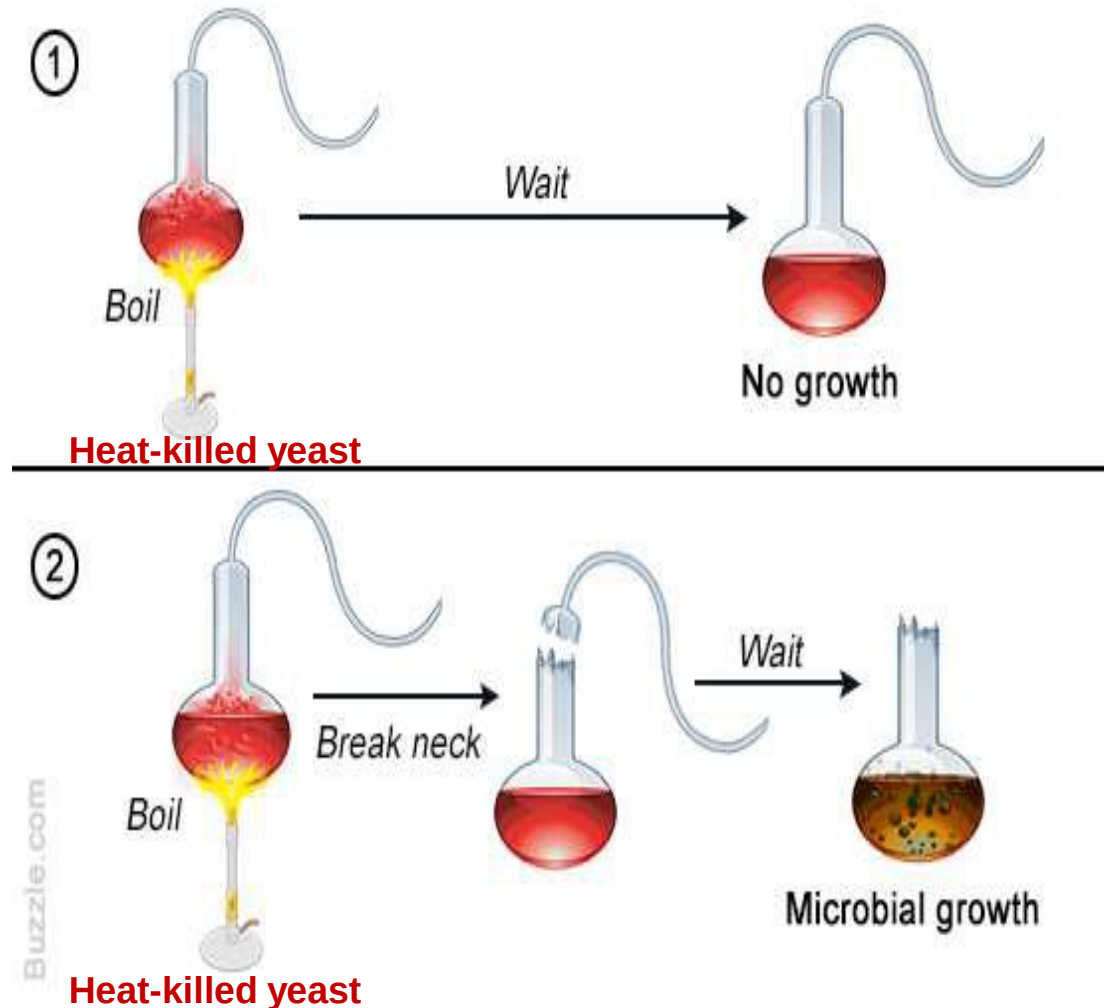
- ❑ Several scientists, including Francesco Redi, John Needham, Lazzaro Spallanzani, and Louis Pasteur, performed different experiments/research studies to disprove this theory.
- ❑ Francesco Redi showed that maggots arise from the eggs of flies rather than directly from rotting.
- ❑ Later, Louis Pasteur did experiments with flasks with twisted necks (swan-neck flasks) and proved sterilized broths in swan neck flasks remained sterile.
- ❑ Unless microbes are introduced from outside from air, the broths remained sterile, and there was no growth of microorganisms.
- ❑ Pasteur's experiments disproved spontaneous generation theory by proving **"life only comes from life"**.

2. Theory of Spontaneous Generation

But **Louis Pasteur (1822-1895)** demonstrated that life can arise only from the pre-existing life.



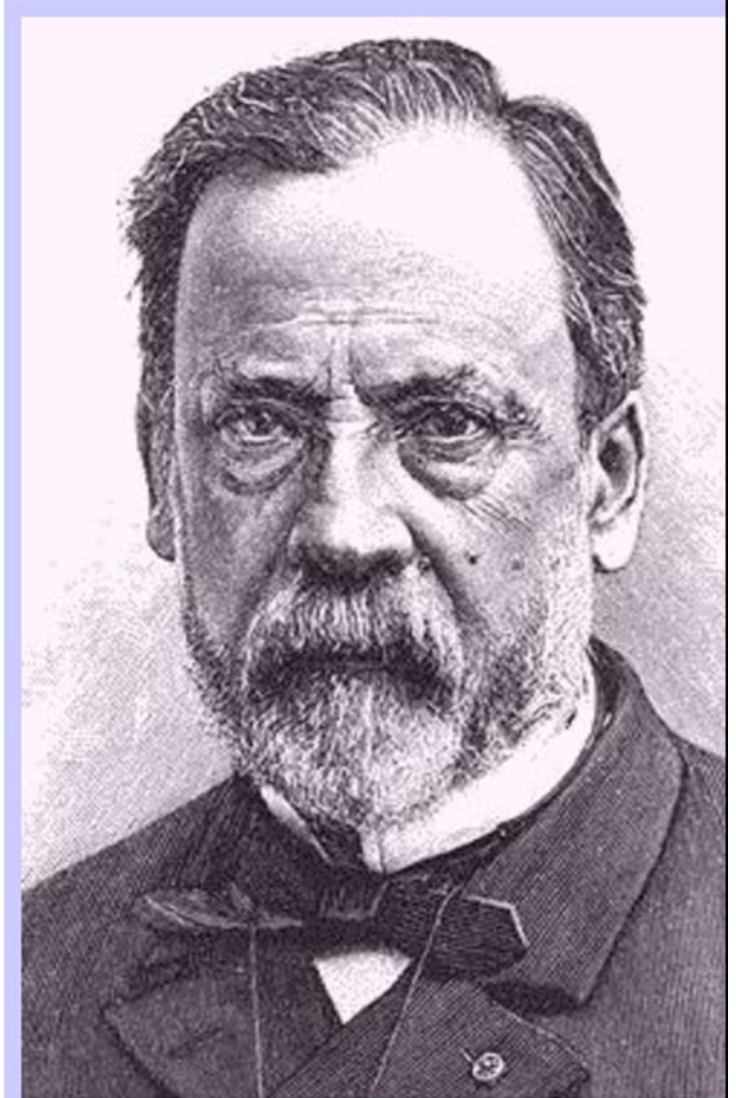
He showed that **No life** arose from the heat-killed yeast broth that was kept in **pre-sterilized closed flask (1)**, while **new organisms** arose from the heat-killed yeast that was kept in the flask left open in the air (2).



2. Theory of Spontaneous Generation

In a simple experiment using a sterilized flask with a bent neck, he showed that plain air cannot initiate the growth of microorganisms. A culture can grow in the flask only if germs enter it. "There is no known circumstance in which it can be confirmed that microscopic beings came into the world without germs, without parents similar to themselves," he concluded, in 1864 (4).

Louis Pasteur demonstrated that life comes only from life. One can only wonder what the history of biological science would be if this principle had been taken as fundamental. Perhaps today we would still be unsure how life on Earth began. But we would approach the question differently. We would assume that life here had to be seeded somehow. We would investigate possible mechanisms for this seeding. We would look for evidence that bacteria, the simplest known form of life, can survive in space, for example. We would look for means by which they could travel across interstellar space and survive for the millions of years such trips might take.



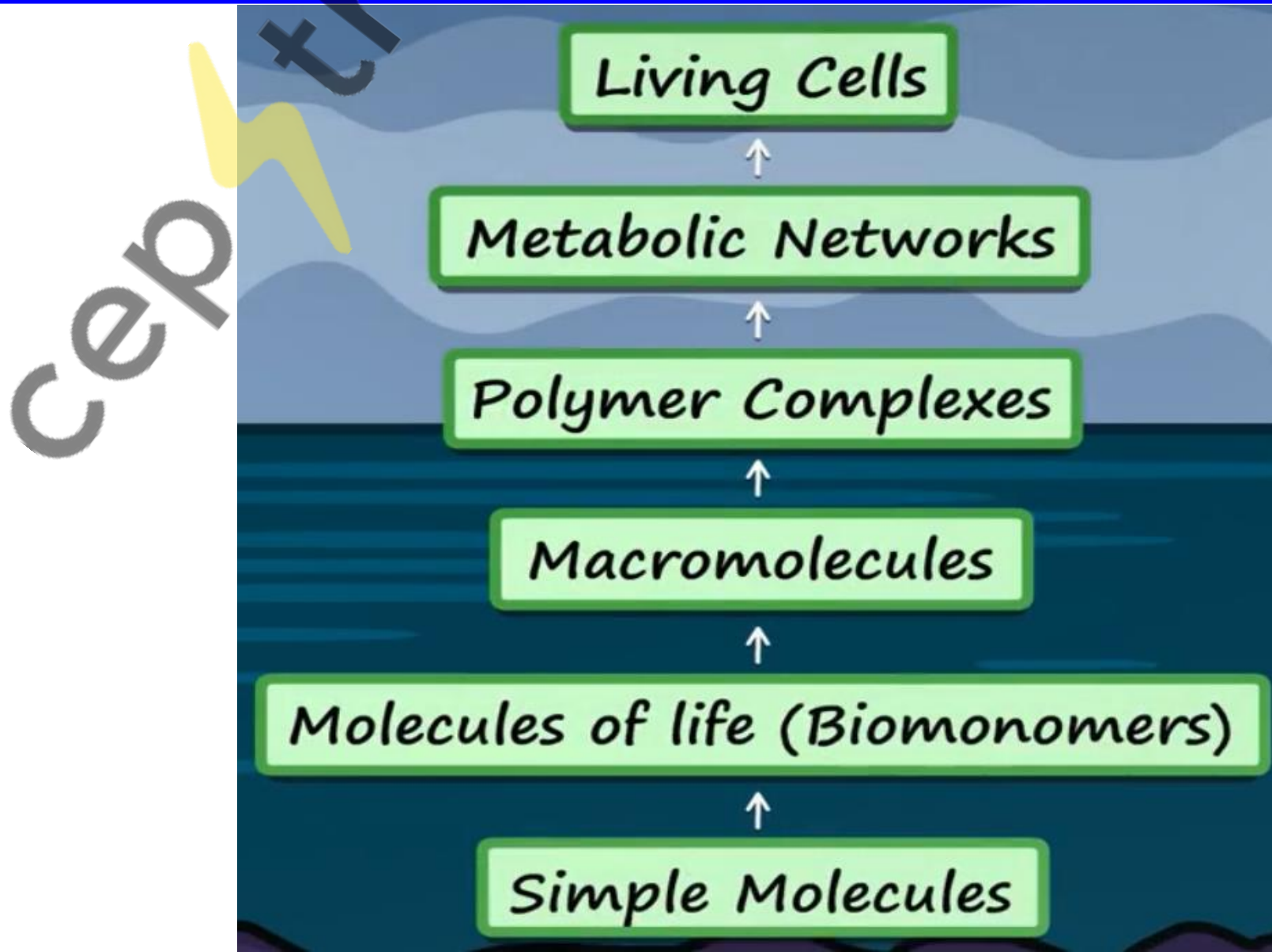
Louis Pasteur

3. Theory of Chemical Evolution



- ❑ In the 1920s British scientist **J.B.S. Haldane** and Russian biochemist **Aleksandr Oparin** independently set forth similar ideas concerning the conditions required for the origin of life on Earth.
- ❑ Both believed that **organic molecules could be formed from abiogenic materials in the presence of an external energy source** (e.g., ultraviolet radiation) and that the primitive atmosphere was reducing (having very low amounts of free oxygen) and contained ammonia and water vapour, among other gases.
- ❑ Both also suspected that the first life-forms appeared in the warm, primitive ocean and were heterotrophic (obtaining preformed nutrients from the compounds in existence on early Earth) rather than autotrophic (generating food and nutrients from sunlight or inorganic materials).
- ❑ Haldane, unfamiliar with Oparin's coacervates, believed that **simple organic molecules formed first and in the presence of ultraviolet light became increasingly complex, ultimately forming cells.**
- ❑ Haldane and Oparin's ideas formed the foundation for much of the research on **abiogenesis** that took place in later decades.

3. Theory of Chemical Evolution



3. Theory of Chemical Evolution



Theory of Chemical Evolution was proposed by Oparin & Haldane, 1920.

- They suggested that, under the strong reducing conditions in the atmosphere of early earth, inorganic molecules would spontaneously form organic molecules (simple sugars and amino acids).
- They proposed that first life form could have come from pre-existing non-living organic molecules (like RNA, proteins, etc.) and that formation of life was **preceded** by Chemical Evolution.

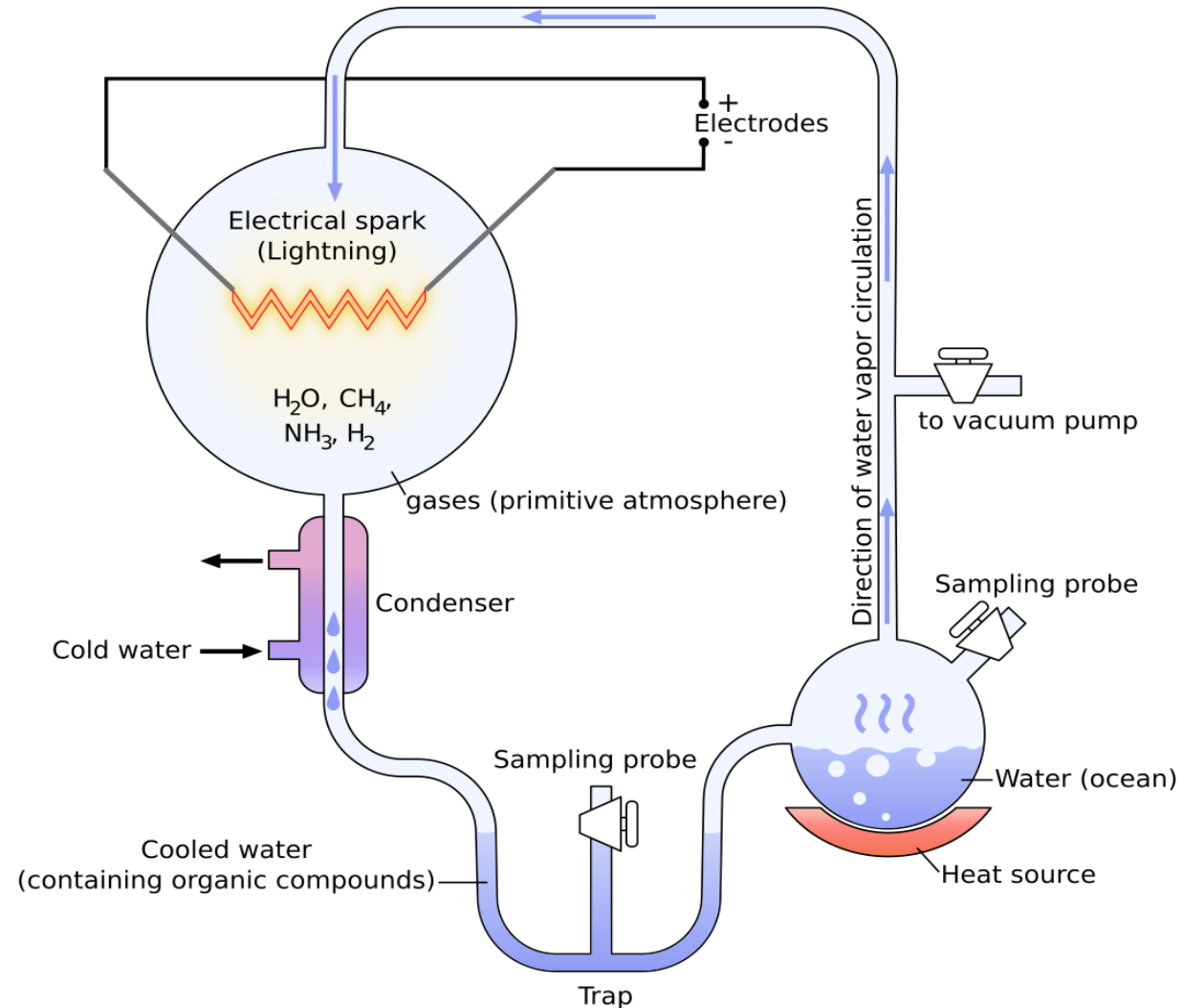
Experimental proof for Chemical Evolution of life



- ❑ It was first suggested in the 1920s that simple organic molecules could form and spontaneously polymerize into macromolecules under the conditions thought to exist in primitive Earth's atmosphere.
- ❑ At the time life arose, the atmosphere of Earth is thought to have contained little or no free oxygen, instead consisting principally of CO_2 and N_2 in addition to smaller amounts of gases such as H_2 , H_2S , and CO . Such an atmosphere provides reducing conditions in which organic molecules, given a source of energy such as sunlight or electrical discharge, can form spontaneously.
- ❑ The spontaneous formation of organic molecules was first demonstrated experimentally in the 1950s, when Stanley Miller (then a graduate student) showed that the discharge of electric sparks into a mixture of H_2 , CH_4 , and NH_3 , in the presence of water, led to the formation of a variety of organic molecules, including several amino acids (Figure).
- ❑ Although Miller's experiments did not precisely reproduce the conditions of primitive Earth, they clearly demonstrated the plausibility of the spontaneous synthesis of organic molecules, providing the basic materials from which the first living organisms arose.
- ❑ The next step in evolution was the formation of macromolecules. The monomeric building blocks of macromolecules have been demonstrated to polymerize spontaneously under plausible prebiotic conditions. Heating dry mixtures of amino acids, for example, results in their polymerization to form polypeptides.

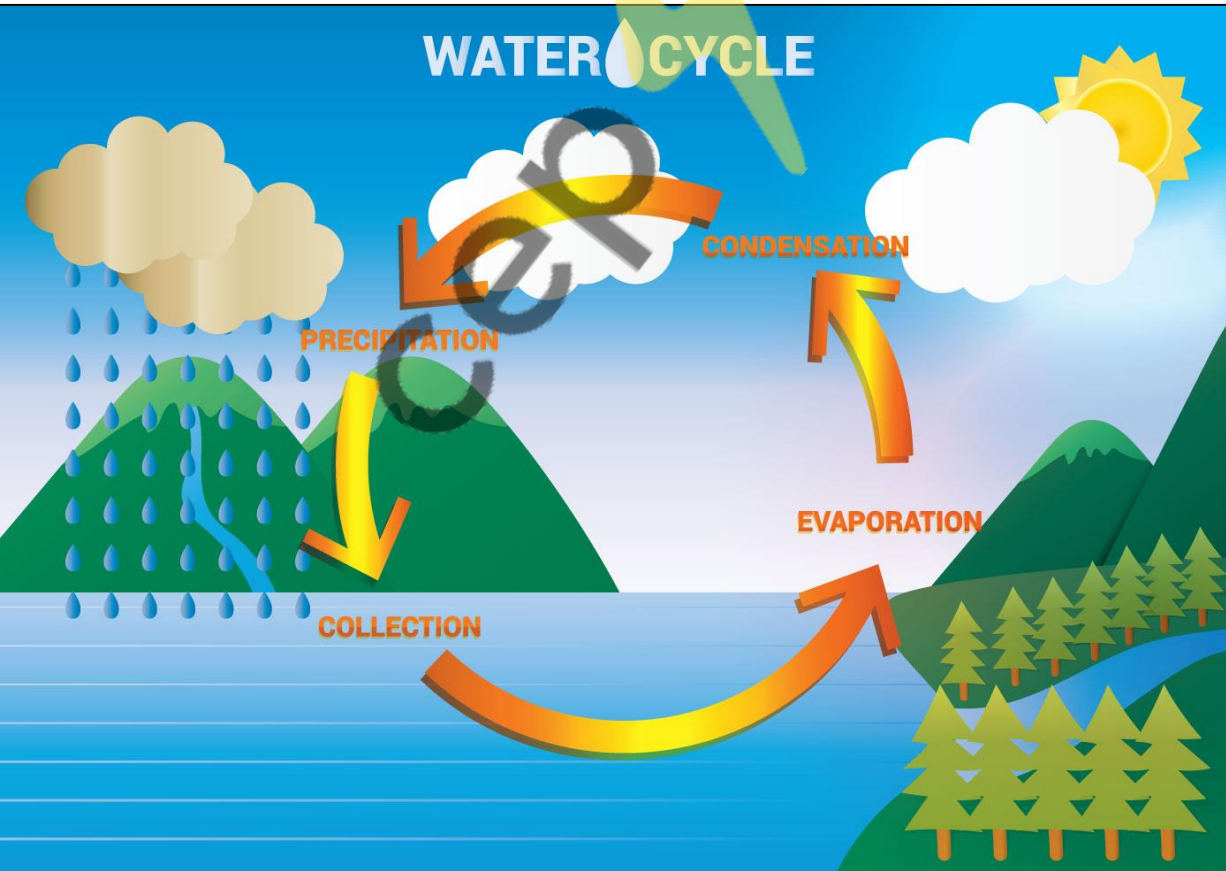
Experimental proof for Chemical Evolution of life

- Stanley Miller & Urey (1952) created the conditions similar to the primitive atmosphere in the laboratory by using glass apparatus & tubes.
- They created electric discharge using electrodes in a flask containing methane (CH_4), ammonia (NH_3), hydrogen (H_2) and water vapor at 800°C .
- The water containing chamber was heated to provide water vapor.
- After a few days, they observed the formation of amino acids that serve as the essential building blocks of protein.
- In similar experiments, many other scientists observed the formation of sugar, nitrogen bases, lipids, amino acids & even pigments.
- Chemical Evolution of life was more or less accepted.



(Miller-Urey experimental setup)

Water Cycle



Stages of Water Cycle

- ❑ The complete water cycle is carried into four stages which are as follows: **Evaporation, Condensation, Precipitation and Collection.**

Evaporation

- ❑ This is the initial stage of the water cycle.
- ❑ The process by which water from its liquid state changes to vapour, a gaseous state, is termed as evaporation. During the water cycle, water in the water bodies get heated up and evaporates in the form of vapour, mixes with the air and disappears.

Condensation

- ❑ When the evaporated water vapour loses its thermal energy, it becomes liquid through the process of condensation. Formation of clouds are examples of condensation.

Precipitation

- ❑ Rain, snow, sleet, or hail are all examples of Precipitation. After the condensation, atmospheric water vapour forms sufficiently large water droplets and falls back to the earth with the help of gravity.

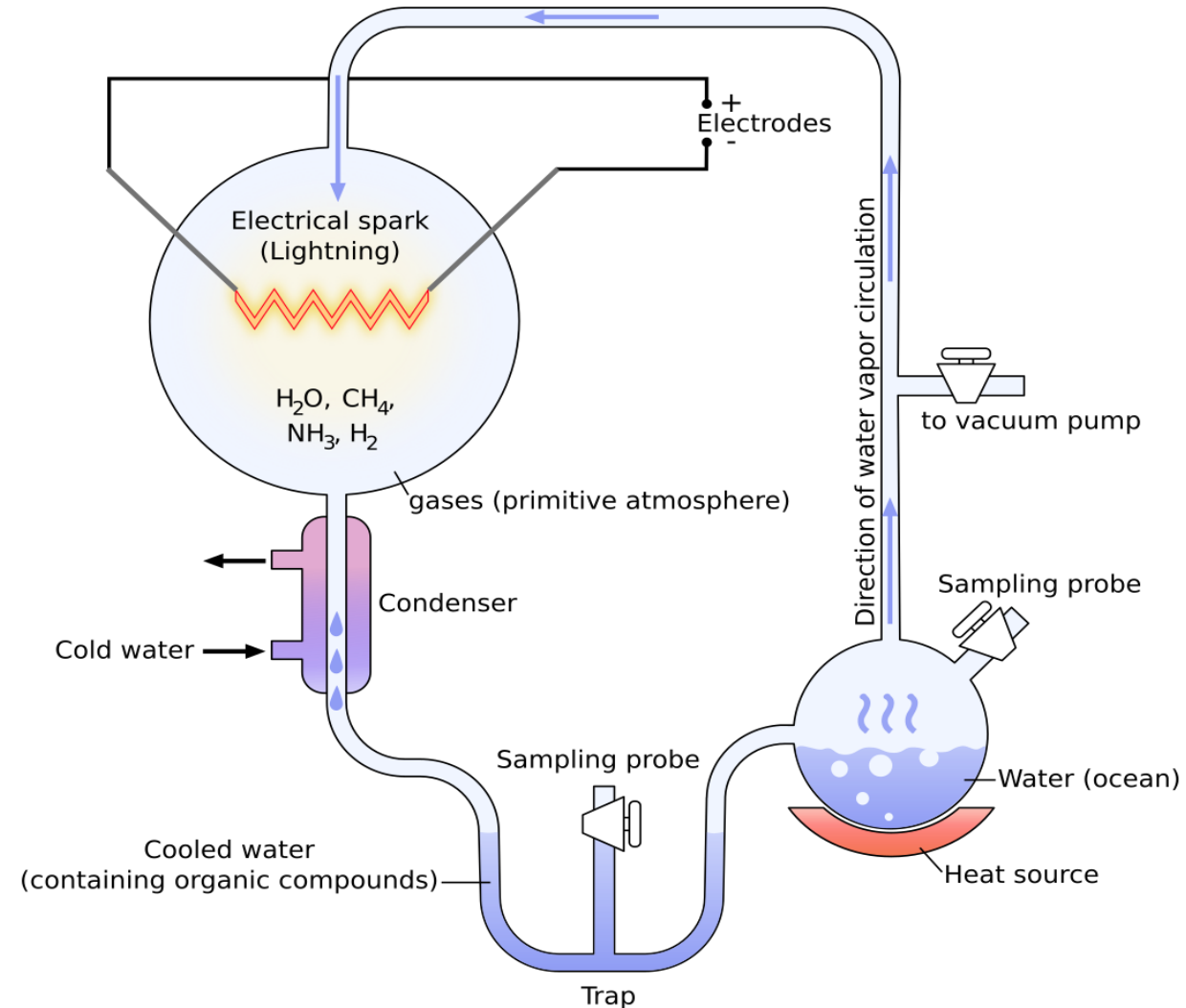
Deposition or Collection

- ❑ This is the final stage of the water cycle. Deposition occurs when evaporated water vapour falls back to earth as precipitation. This water may fall back into the different water bodies, including oceans, rivers, ponds, lakes and even end up on the land, which in turn becomes a part of the groundwater.

Overall, the water cycle process describes how water is balanced in the atmosphere. It also plays an important role in ensuring the availability of water for all living organisms and also it has a great impact on our environment.

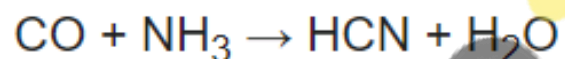
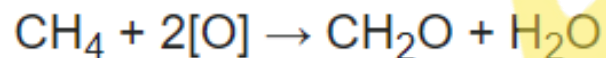
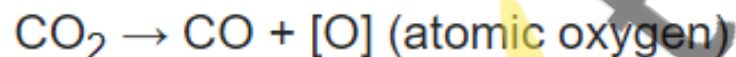
Experimental proof for Chemical Evolution of life

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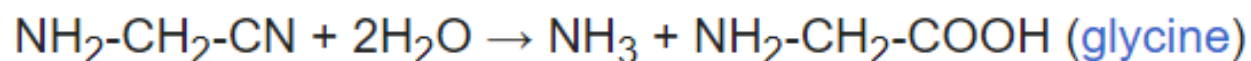
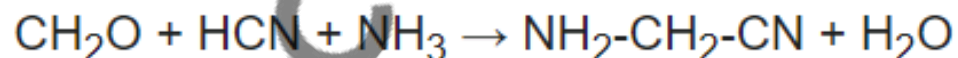


(Miller-Urey experimental setup)

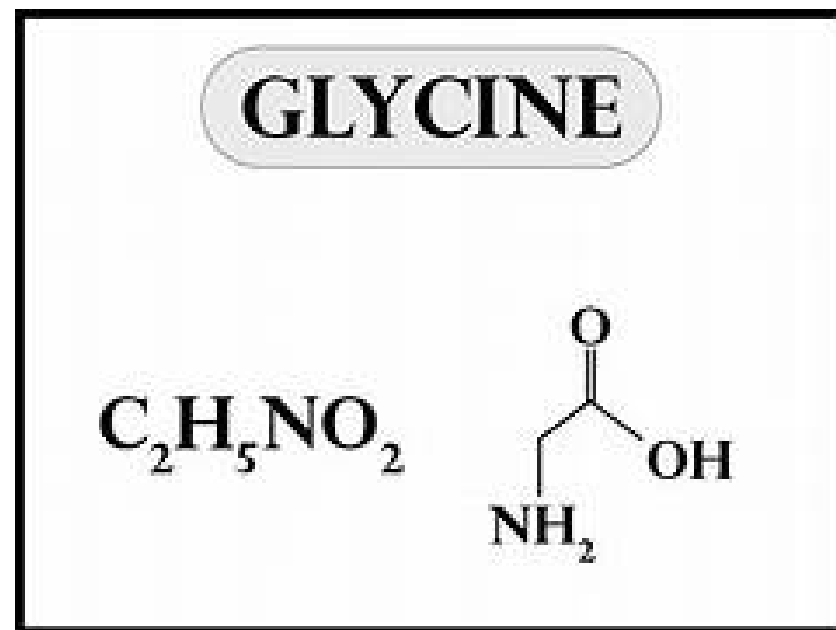
Experimental proof for Chemical Evolution of life



The formaldehyde, ammonia, and HCN then react by **Strecker synthesis** to form amino acids and other biomolecules:



- ❑ Furthermore, water and formaldehyde can react to produce various sugars like ribose.
- ❑ The experiments showed that simple organic compounds of building blocks of proteins and other macromolecules can be formed from gases with the addition of energy.

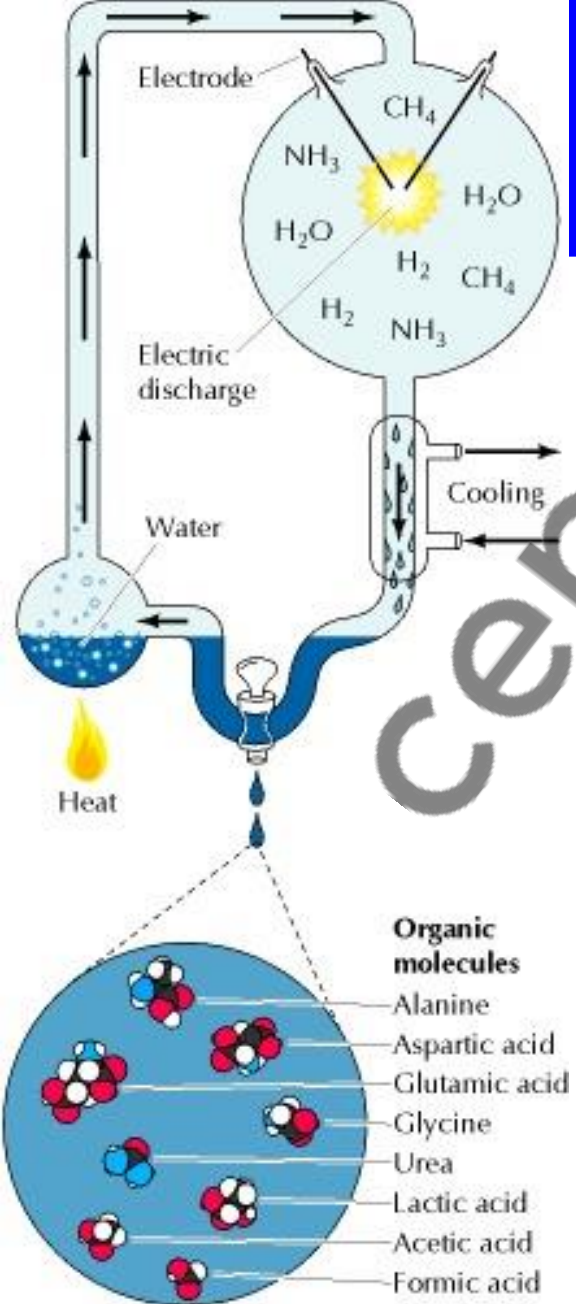


(Miller-Urey experiment)

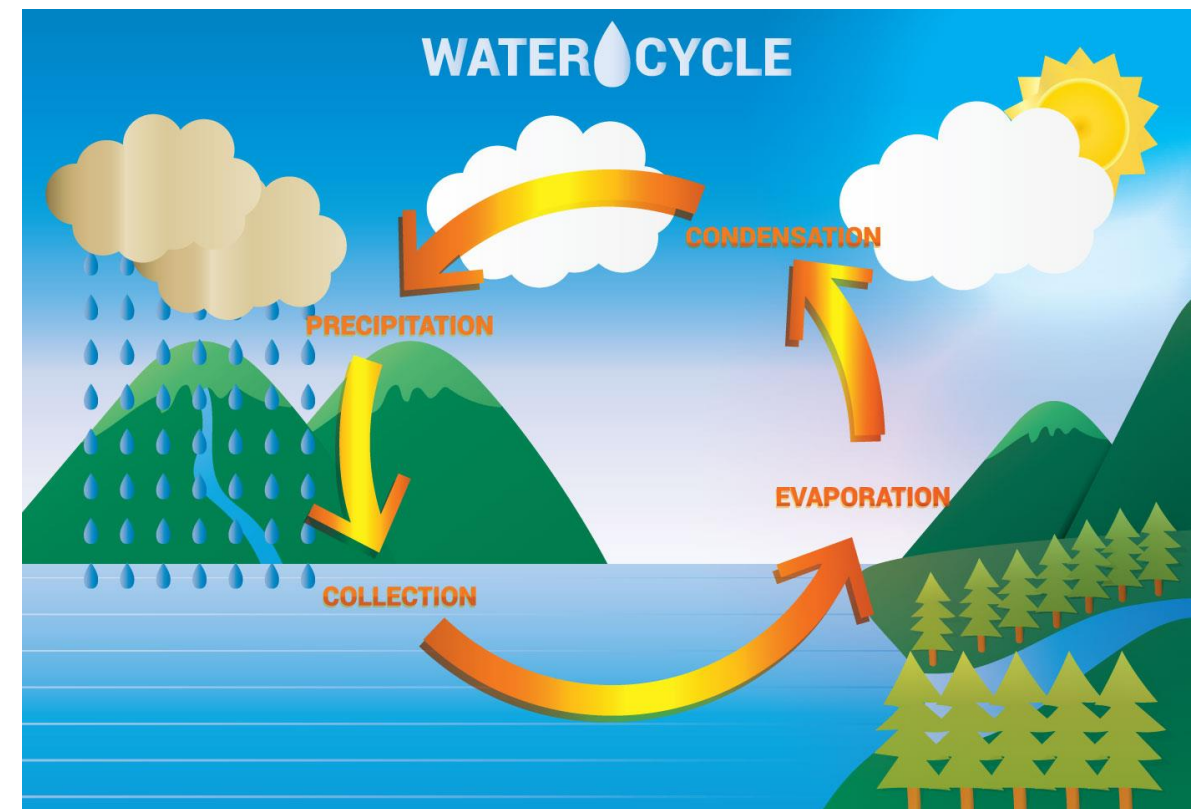
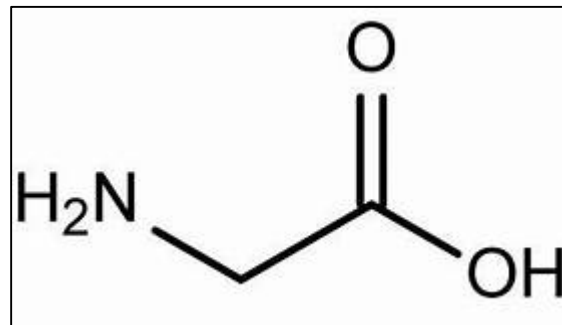
Experimental proof for Chemical Evolution of life

Figure: Spontaneous formation of organic molecules

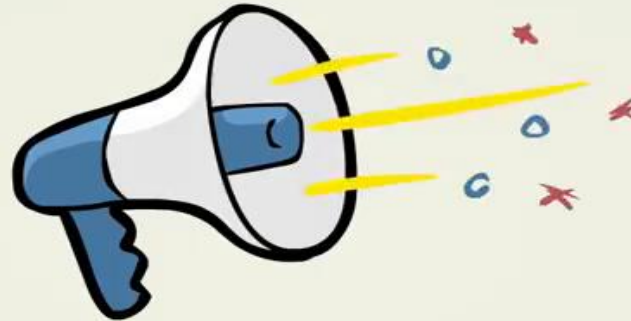
- ❑ Water vapor was refluxed through an atmosphere consisting of CH_4 , NH_3 , and H_2 , into which electric sparks were discharged.
- ❑ Analysis of the reaction products revealed the formation of a variety of organic molecules, including the amino acids alanine, aspartic acid, glutamic acid, and glycine.



Glycine



(Miller-Urey experimental setup)



Stated Clearly

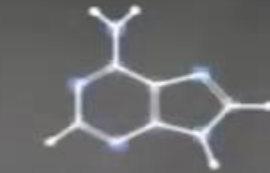
4. Theory of Panspermia or Cosmozoic theory



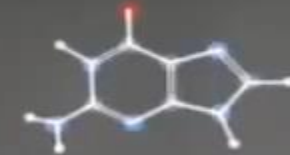
- ❑ Some scientist believe that life has come on the earth from other planets or outer space (through meteoroids and/or asteroids and/or comets) in the form microscopic life (spores or microorganisms).
- ❑ This is called 'Cosmozoa' or 'Pansperms' (seeds everywhere) which were preserved inside the meteorites.
- ❑ These meteorites struck into barren earth to release the Cosmozoa and they might have evolved into present day life forms.



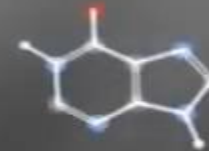
- ❑ **Meteoroid:** A meteoroid, especially one that has hit Earth's surface.
- ❑ **Asteroid:** A rocky object that orbits the sun and has an average size between a meteoroid and a planet.



ADENINE



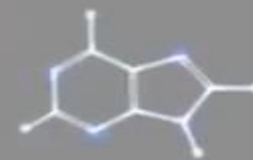
GUANINE



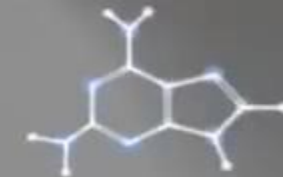
HYPOXANTHINE



XANTHINE



PURINE



DIAMINOPURINE

Conclusion



Scientists usually propose a four-stage process of formation for the first life:

- ☐ formation of small organic molecules (amino acids, nucleic acid bases,...),
- ☐ and these combine to make larger biomolecules (proteins, nucleic acids, lipids, ...),
- ☐ which self-organized, by a variety of interactions, into a semi-alive system
- ☐ that gradually transformed into a more sophisticated form, a living organism.

Important Links to see Videos



- ❑ <https://www.youtube.com/watch?v=NNijmxsKGbc>
- ❑ <https://www.youtube.com/watch?v=1g7AKVZ3HC4>
- ❑ <https://www.youtube.com/watch?v=TK1E3heBSil>
- ❑ <https://www.youtube.com/watch?v=mRzxTzKIsp8>

Radioactive dating:

- ❑ https://www.youtube.com/watch?v=phZeE7Att_s&t=117s
- ❑ <https://www.youtube.com/watch?v=oe45GegJUvM&t=1s>
- ❑ https://www.youtube.com/watch?v=5_vKvJO7Kqw&t=4s
- ❑ https://www.youtube.com/watch?v=8QnsA_1pEd8&t=335s

Thank you for your attention

BT101

Introductory Biology

(Lecture 5)

Dr. Rajkumar P Thummer
Assistant Professor
Department of Biosciences and Bioengineering
IIT Guwahati
Guwahati

Introduction



Evolution of life:

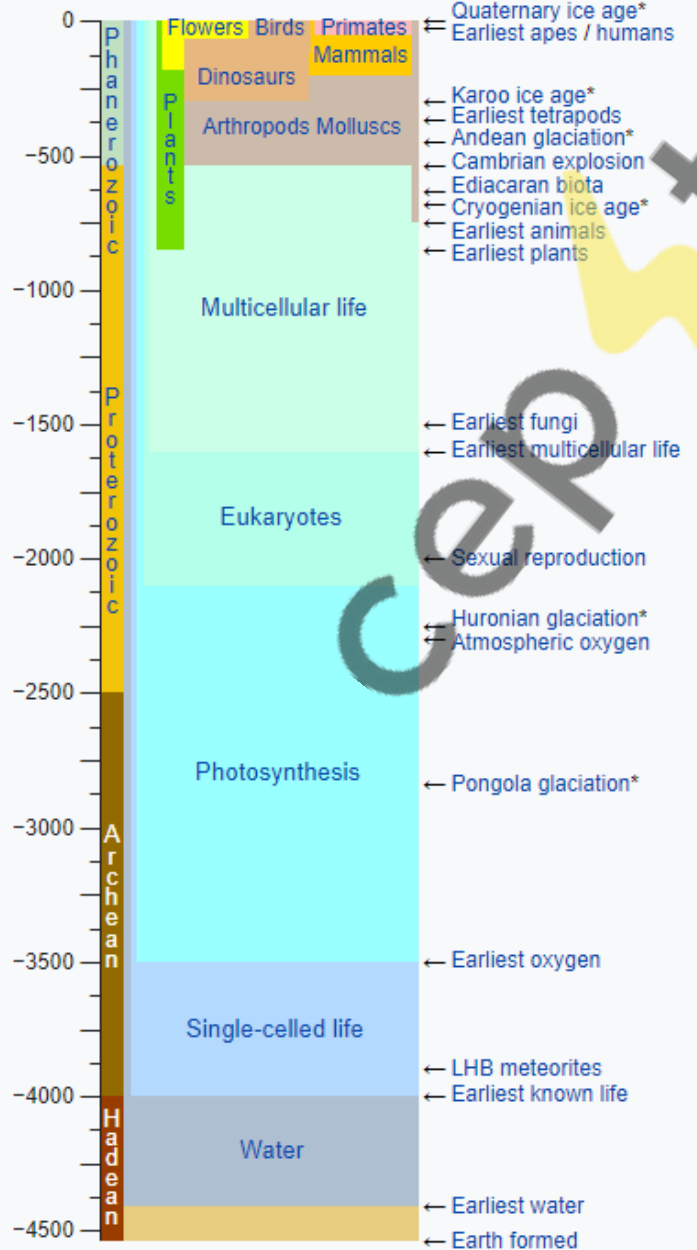
- ☐ Origin of Life;
- ☐ Darwin's concepts of evolution
- ☐ Biodiversity

Cell, the structural and functional unit of life:

- ☐ **Three domains of life**
- ☐ Cell types, cell organelles and structure
- ☐ Basic biomolecules of cell

Life timeline

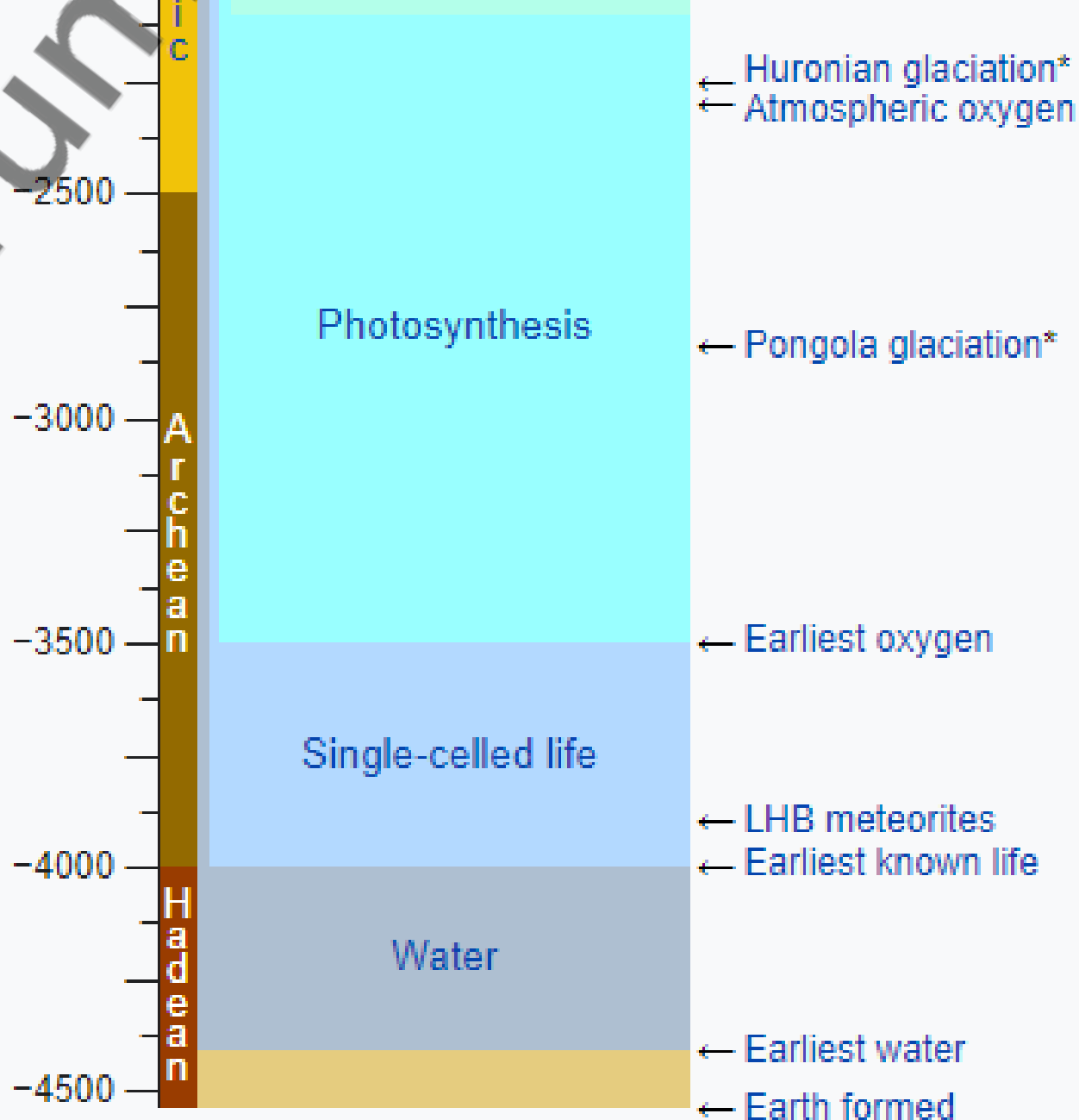
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(million years ago)

*Ice Ages

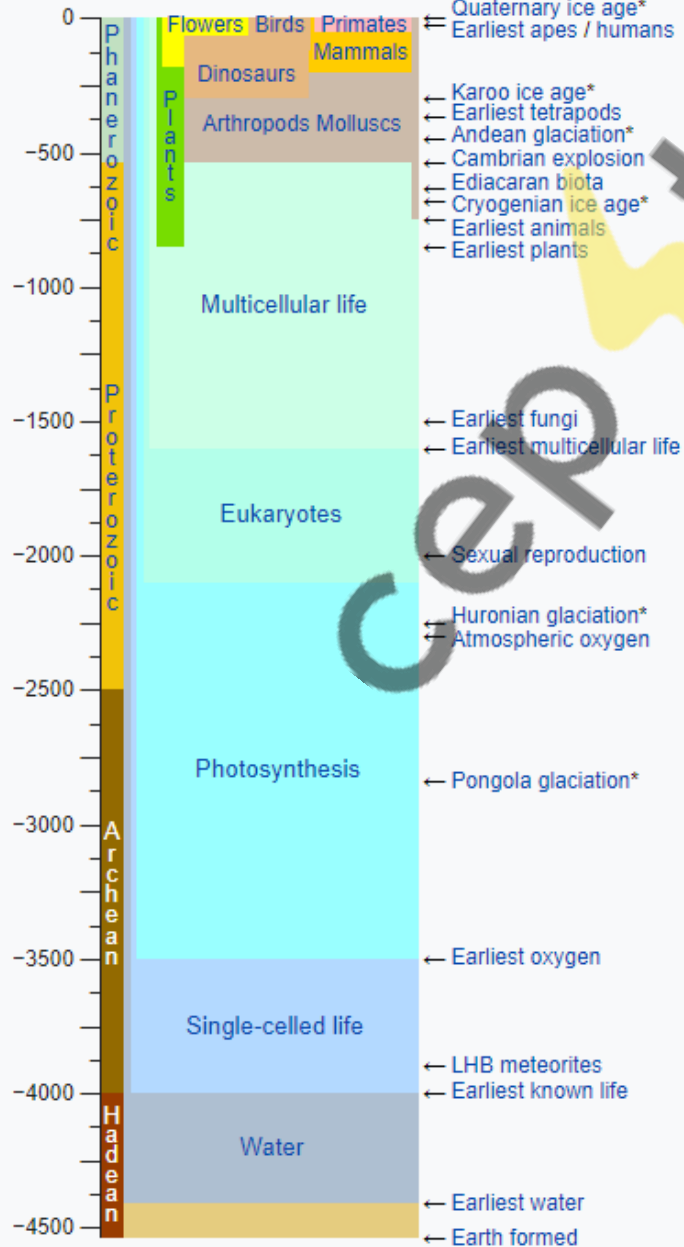
(million years ago)



*Ice Ages

Life timeline

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 Quaternary ice age*
 Earliest apes / humans

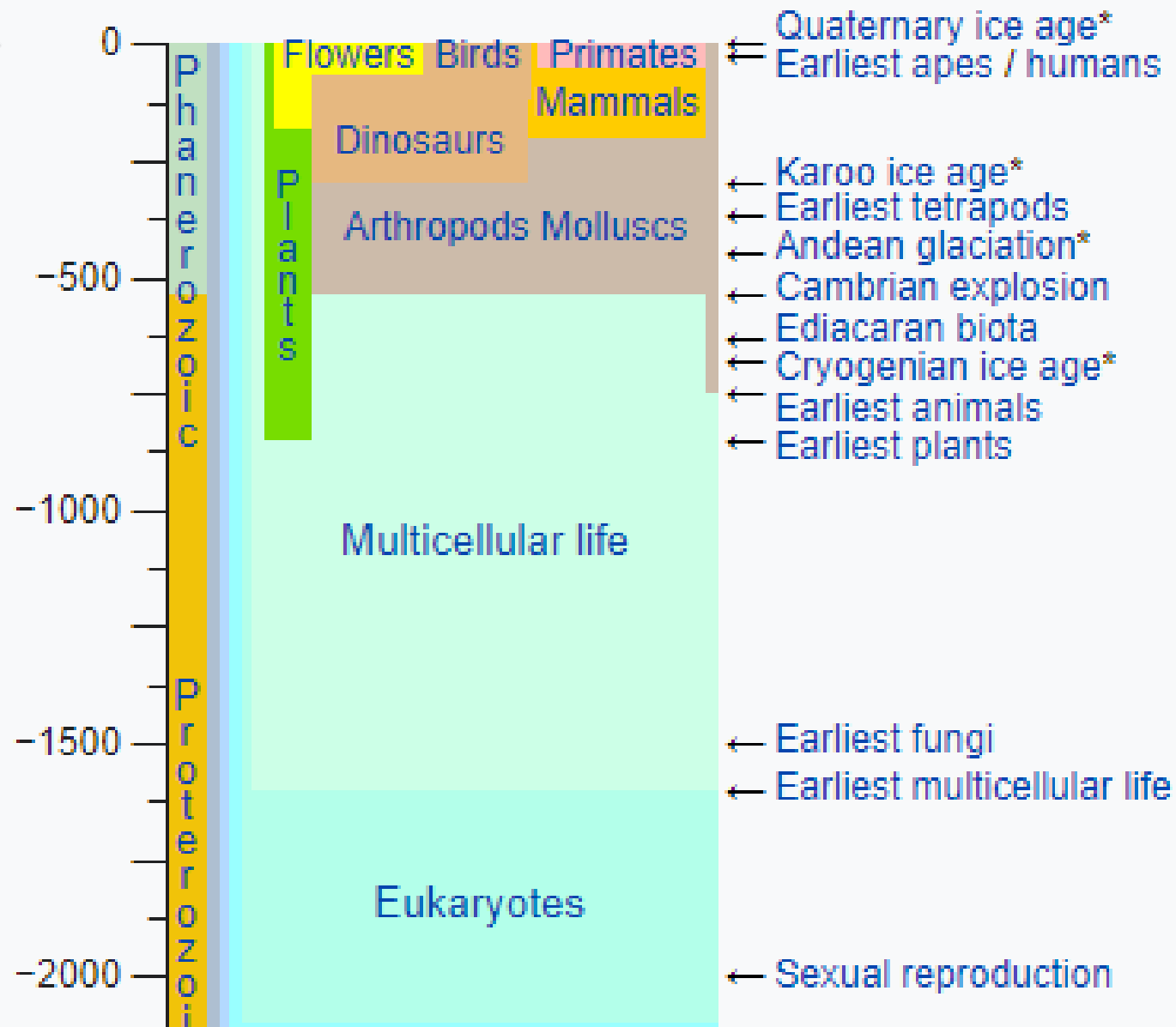


(million years ago)

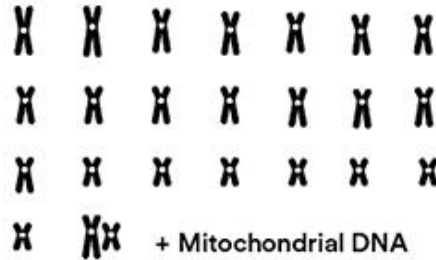
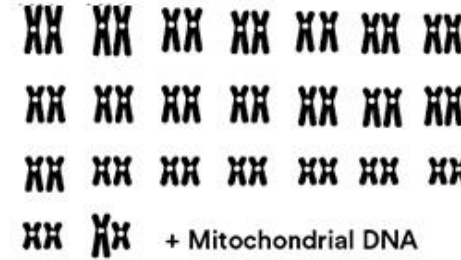
*Ice Ages

Life timeline

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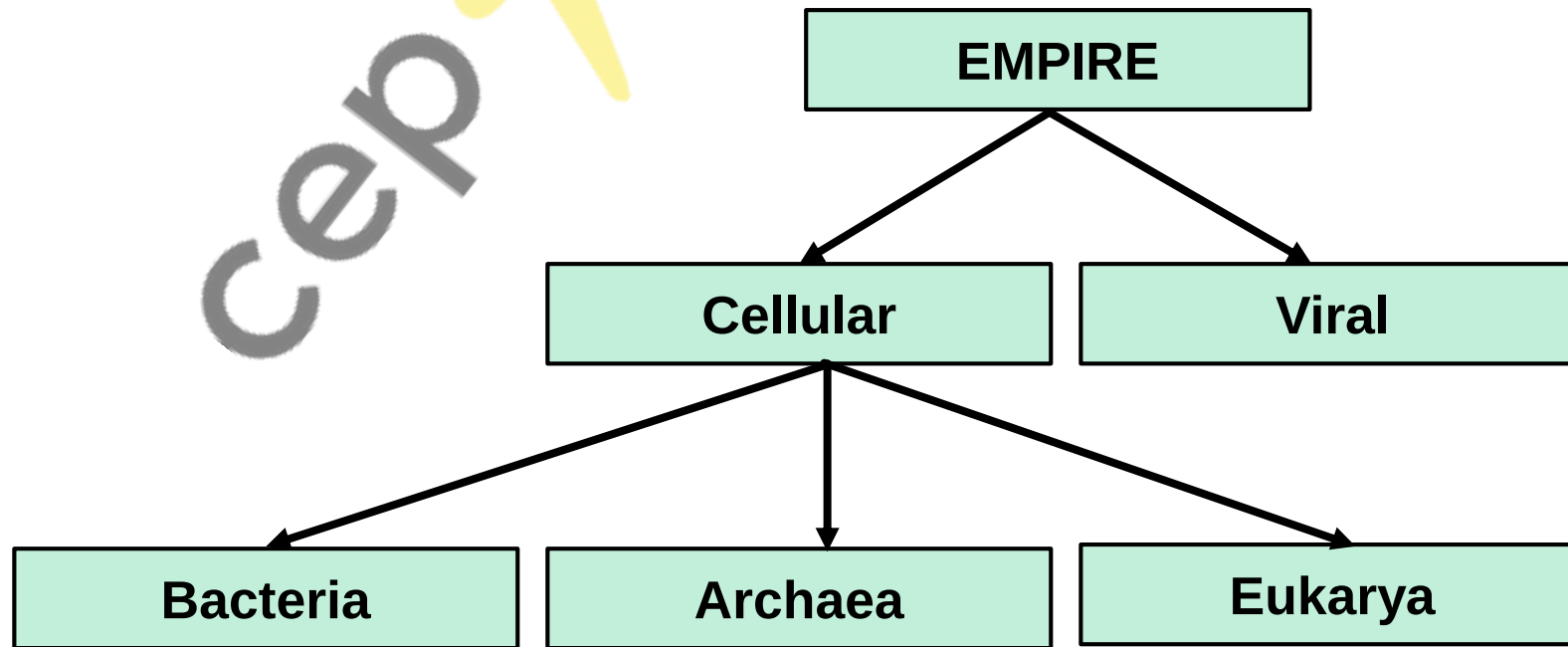


Human Genome

	Reference Genome	A Person's Genome
What is it?	 + Mitochondrial DNA	 + Mitochondrial DNA
How many chromosomes?	24 (22 + X + Y)	46 (23 PAIRS)
How many letters?	~3.2 bn	~6.4 bn
How to think about it?	- The Human Genome Project and its goal of a first draft of "the human genome" - Serves as a standard for comparison - A "consensus" genome sequence	- The genome of a person - The genome within a person's cells - The whole genome sequence of an individual

- ❑ That human genome—the real, physical human genome—is diploid; in other words, it has two pairs of each chromosome—one from each parent. Everyone has a chromosome 1 from mom and another from dad; and a chromosome 2 from mom and another from dad; and so forth. The total is 46 chromosomes, or two pairs of 23. Thus, 6.4 billion letters (or bases) long.
- ❑ The human genome contains about 3.2 billion base pairs of DNA. However, within the genome there are only about 20,000 genes that code for proteins, comprising only 1.5% of the genome.

Two empires



Large and in charge

Giant viruses are so huge they can even have their own parasites – the biggest are almost the same size as some bacteria



Human immunodeficiency virus (HIV)

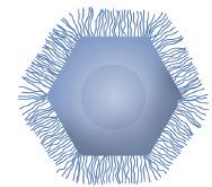


~100nm

Genes: 9

Discovered: Human blood

Mimivirus

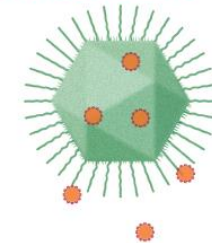


~400nm

Genes: 1018

Discovered: Water tower, Bradford, UK

Mamavirus



~400nm

Genes: 1023

Discovered: Water tower, Paris, France

Can be infected by **Sputnik virophage**

● **SPUTNIK VIROPHAGE**

~50nm Genes: 21

Megavirus



440nm

Genes: 1120

Discovered: Ocean off Chile

Pithovirus

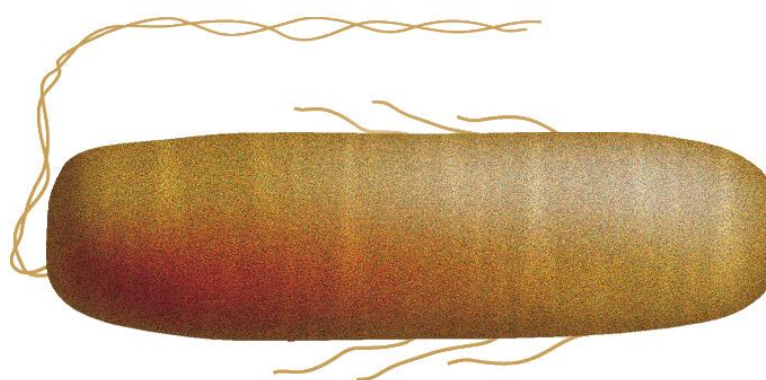


1500nm

Genes: 467

Discovered: 30,000-year-old
Siberian ice core

E. coli bacterium



~2000nm

Genes: 4288

Discovered: Human colon

500nm

- ❑ Viruses are amazing molecular machines that are much tinier than even the smallest cells.
- ❑ They are intracellular parasites.
- ❑ These selfish genetic elements typically encode some proteins essential for viral replication, but they never contain the full set of genes for the proteins and RNAs required for translation, membrane function, or metabolism.
- ❑ Therefore, viruses exploit cells to produce their components.

Scientists identify more than 140,000 virus species in the human gut

Study opens up new research avenues for understanding how viruses living in the gut affect human health

Prof Rob Knight, from University of California San Diego, told the BBC: "You're more microbe than you are human."

Originally it was thought our cells were outnumbered 10 to one.

"That's been refined much closer to one-to-one, so the current estimate is you're about 43% human if you're counting up all the cells," he says.

But genetically we're even more outgunned.

The human genome - the full set of genetic instructions for a human being - is made up of 20,000 instructions called genes.

But add all the genes in our microbiome together and the figure comes out between two and 20 million microbial genes.

Prof Sarkis Mazmanian, a microbiologist from Caltech, argues: "We don't have just one genome, the genes of our microbiome present essentially a second genome which augment the activity of our own.

"What makes us human is, in my opinion, the combination of our own DNA, plus the DNA of our gut microbes."

“ Eight percent of our DNA consists of remnants of ancient viruses, and another 40 percent is made up of repetitive strings of genetic letters that is also thought to have a viral origin.”

8 percent

Viral remnants make up 8 percent of the human genome, and a new study finds that these sequences are still active in healthy people.

01-Nov-2022

Reproduction



- ❑ Reproduction is a common characteristic of Living Things.
- ❑ In general terms, reproduction is defined as the biological process of giving birth to their young ones identical to their parents.
- ❑ There are two different types of reproduction:
 - **Sexual Reproduction**
 - When an organism reproduces sexually, it integrates genetic material from both parents to create a genetically unique organism.
 - **Asexual Reproduction** (Asexual reproduction occurs by binary fission, wherein a cell divides into two)
 - Binary fission is the most common mode of reproduction seen in bacteria.
 - Binary Fission: It is a common and simple method of asexual reproduction in bacteria. In this process of asexual reproduction, the parent cell divides into two, leading to two equal-sized daughter cells. Under favourable conditions, bacteria multiply rapidly and may double in ~20 minutes.

Three domains of life

The tree of life consists of three domains: Bacteria, Archaea and Eukarya

- ❑ All living organisms consist of elementary units called cells.
- ❑ All cells are believed to have evolved from a common progenitor because the structures and molecules in all cells have so many similarities.
- ❑ In recent years, detailed analysis of DNA sequences from a variety of Prokaryotic organisms revealed that they are of **two distinct** types: so-called “True” bacteria or **Eubacteria** and **Archaea**.
- ❑ Based on the assumption that the organisms with more similar genes evolved from a common progenitor more recently than those with more dissimilar genes. The researchers developed an Evolutionary Lineage Tree as shown in Figure 1.
- ❑ According to this tree, the **Archaea** and the **Eukaryotes** diverged from the **Eubacteria** and later they also diverged from each other.

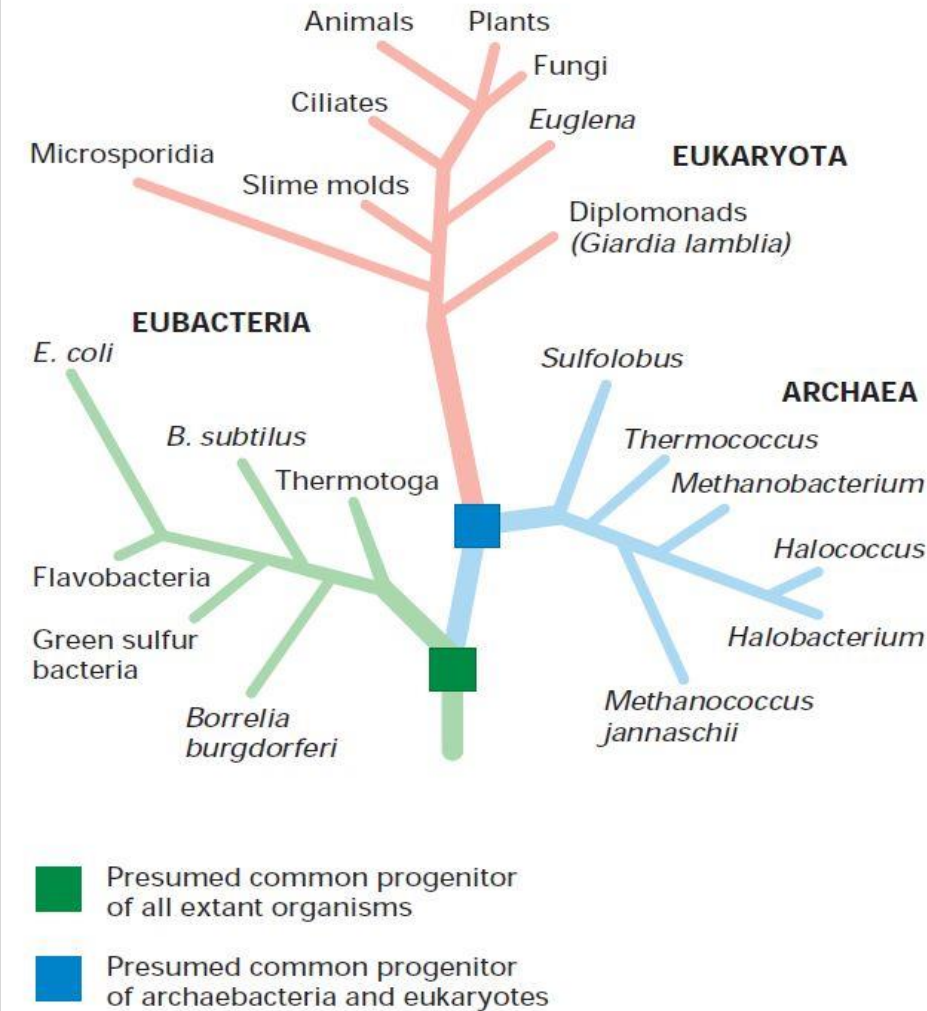
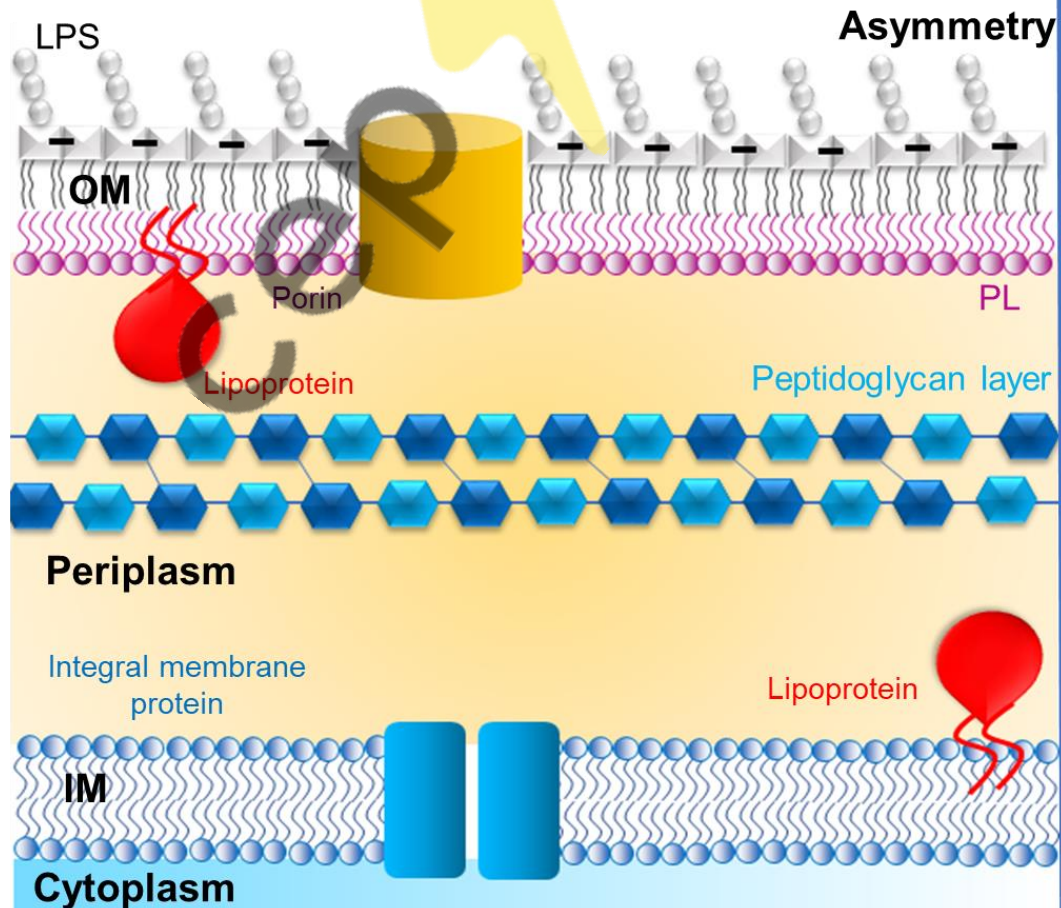
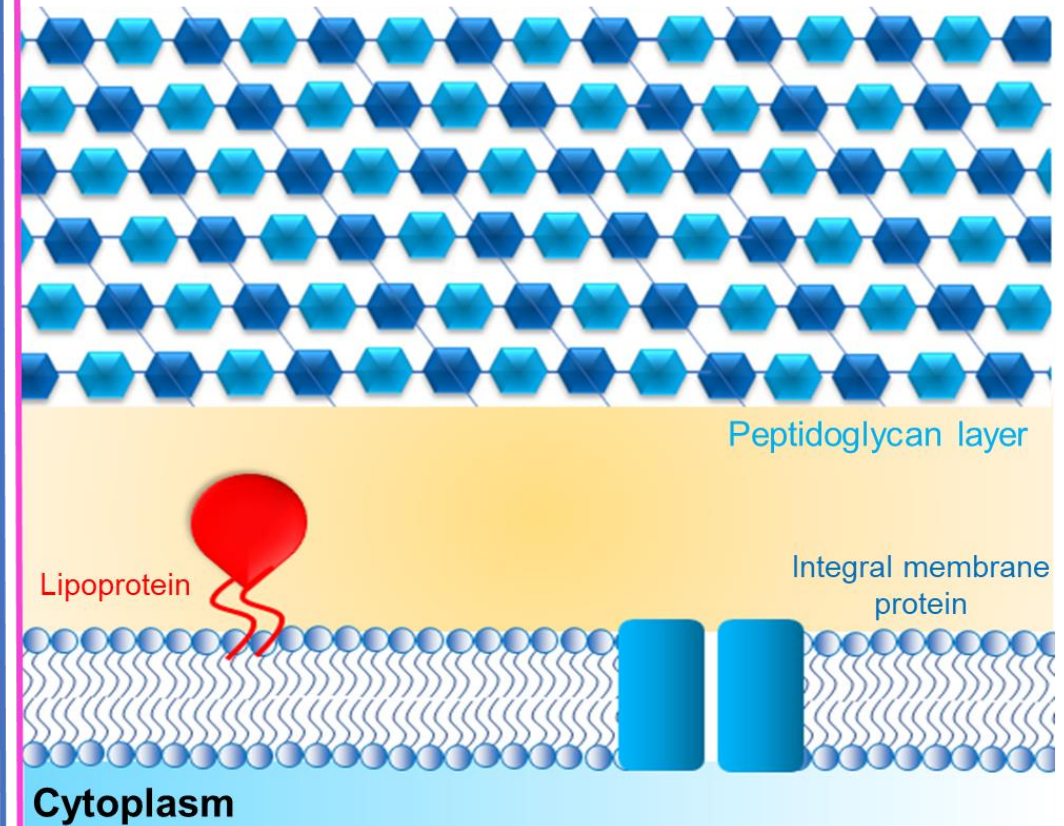


Fig. 1. All organisms from simple bacteria to complex mammals probably evolved from a common, single-celled progenitor. *Credit: Molecular Cell Biology - Lodish

Gram-negative bacteria

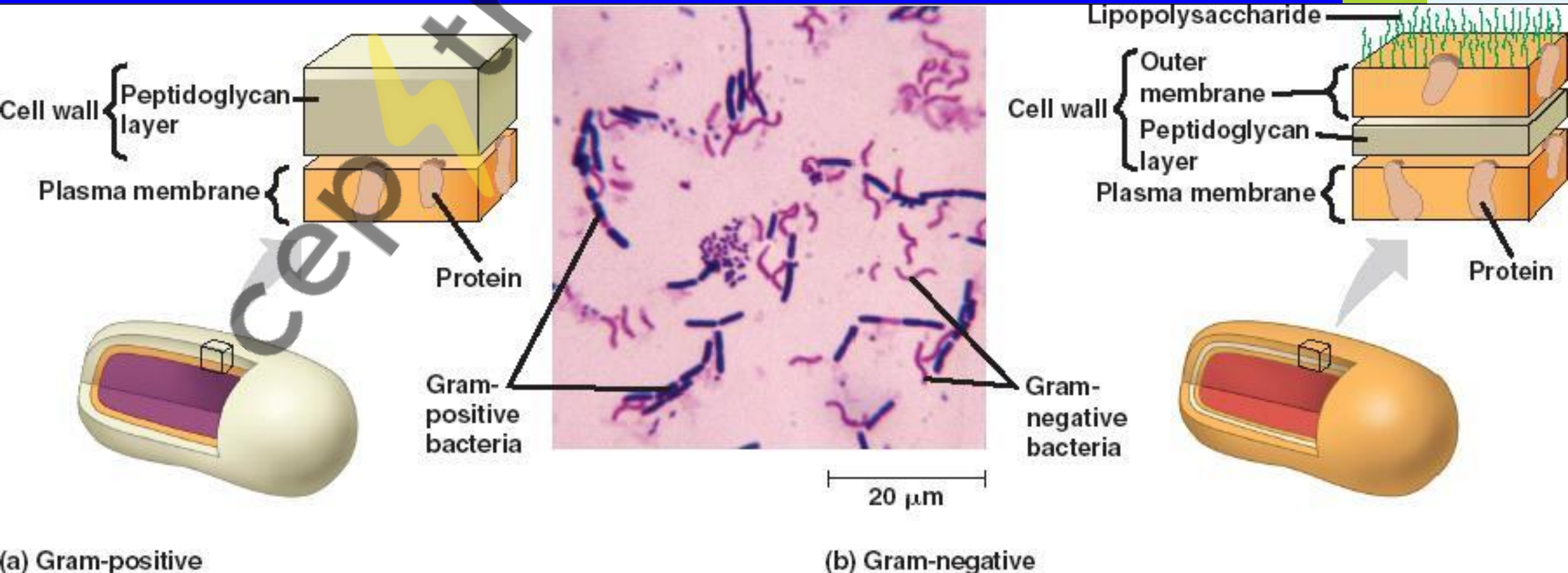


Gram-positive bacteria



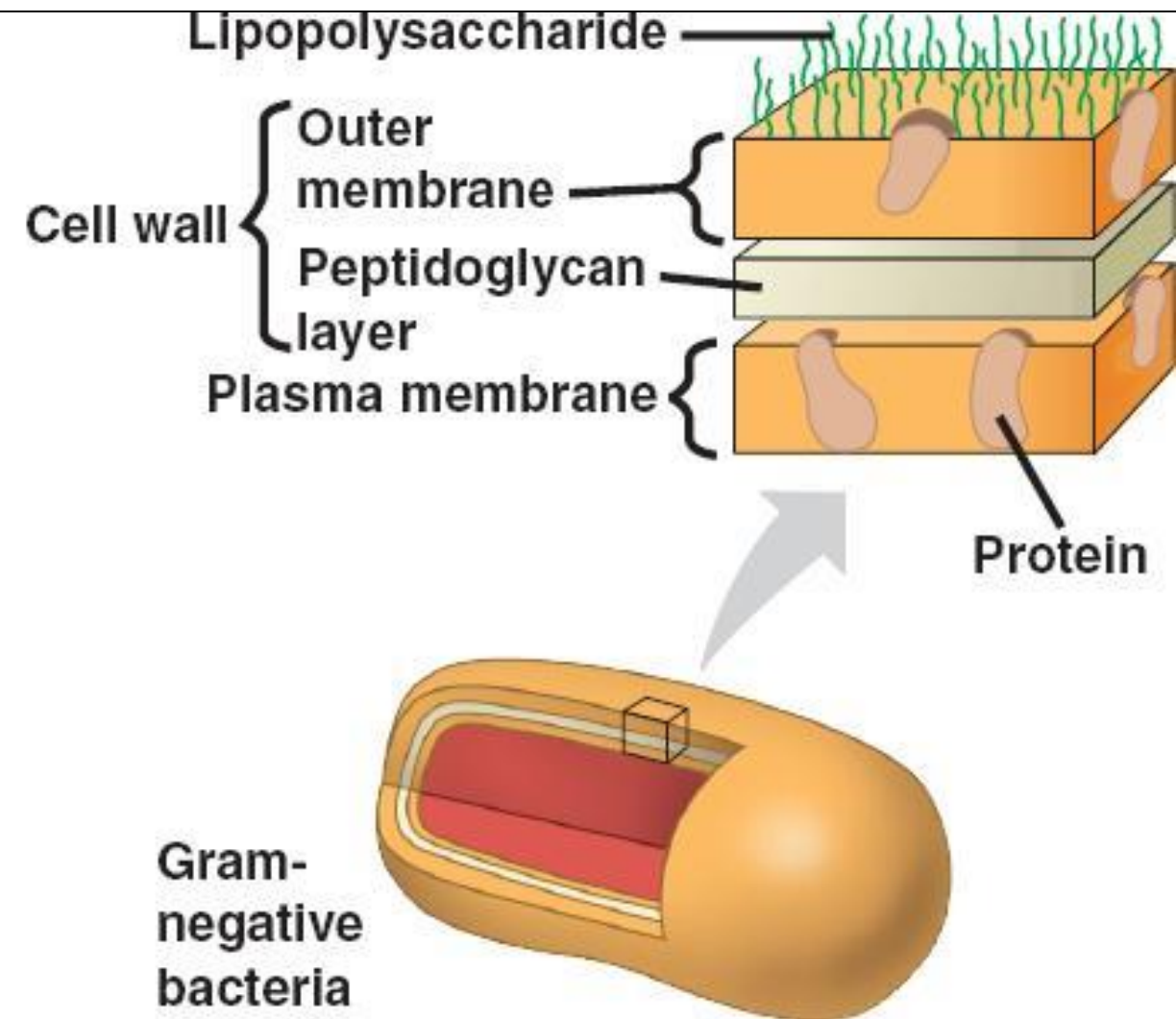
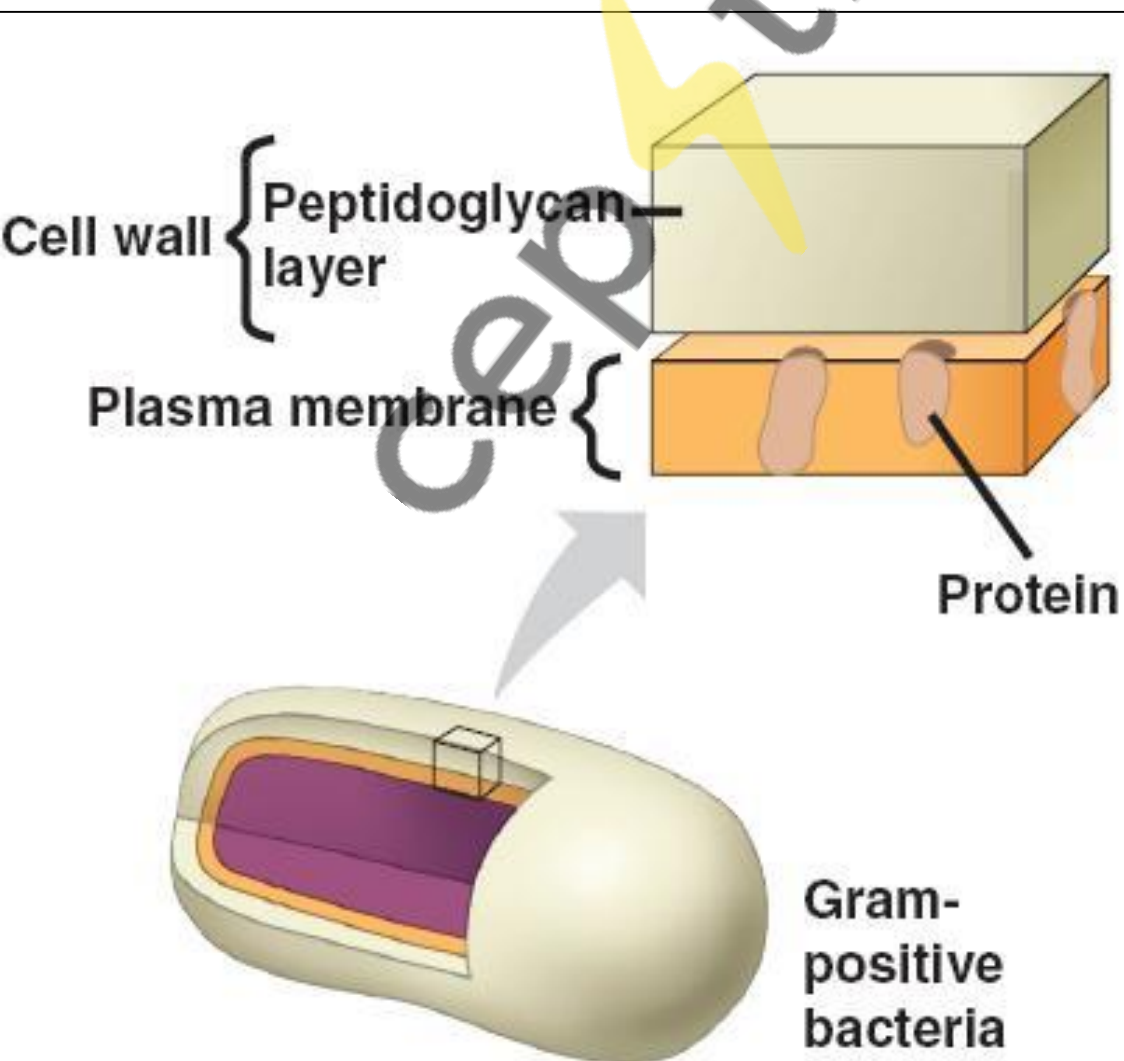
- ❑ Gram stain or Gram staining is a method of staining used to classify bacterial species into two large groups: **Gram-positive bacteria and Gram-negative bacteria.**
- ❑ Gram-positive bacteria have a thick mesh-like cell wall made of peptidoglycan (50–90% of cell envelope), and as a result are stained purple by crystal violet, whereas gram-negative bacteria have a thinner layer (10% of cell envelope), so do not retain the purple stain and are counter-stained pink by safranin.

Bacteria

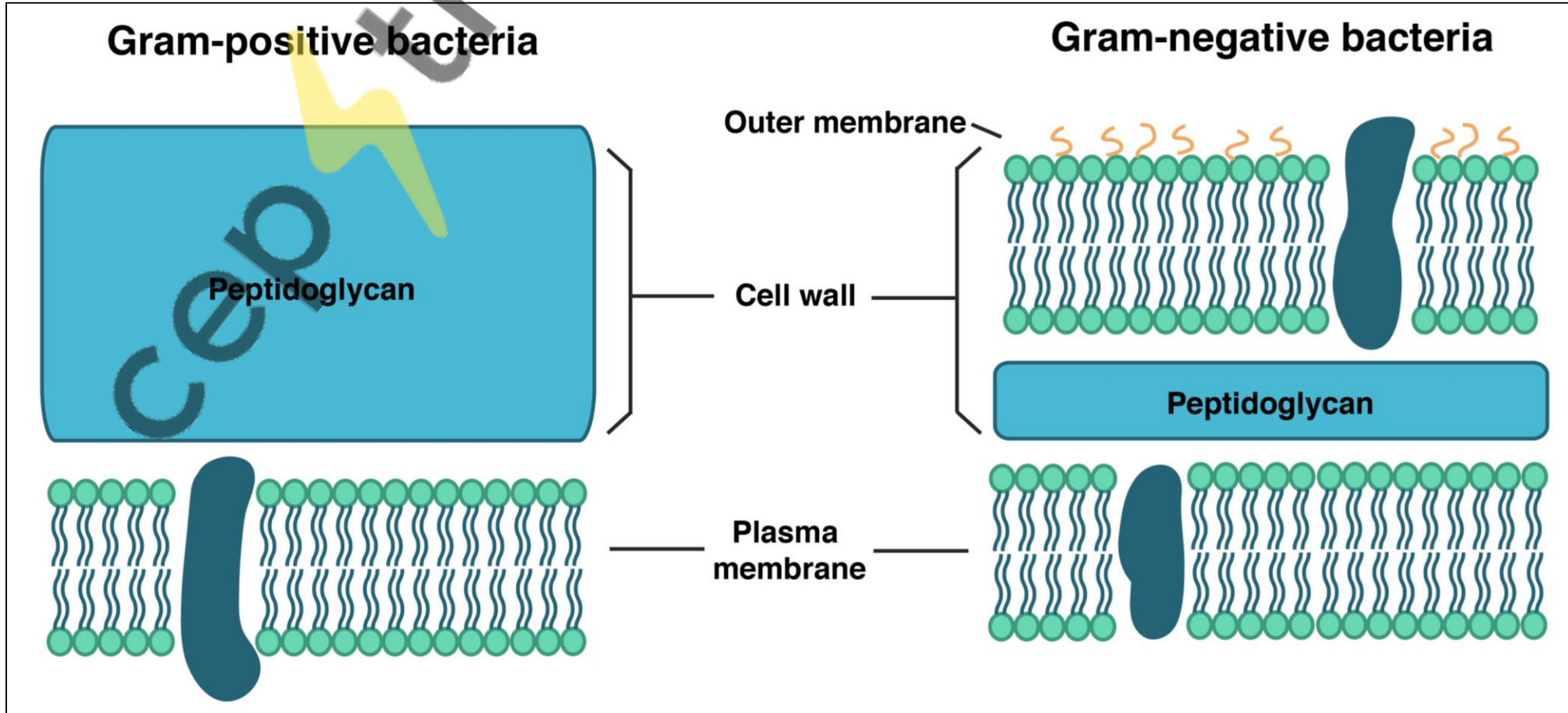


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Bacteria

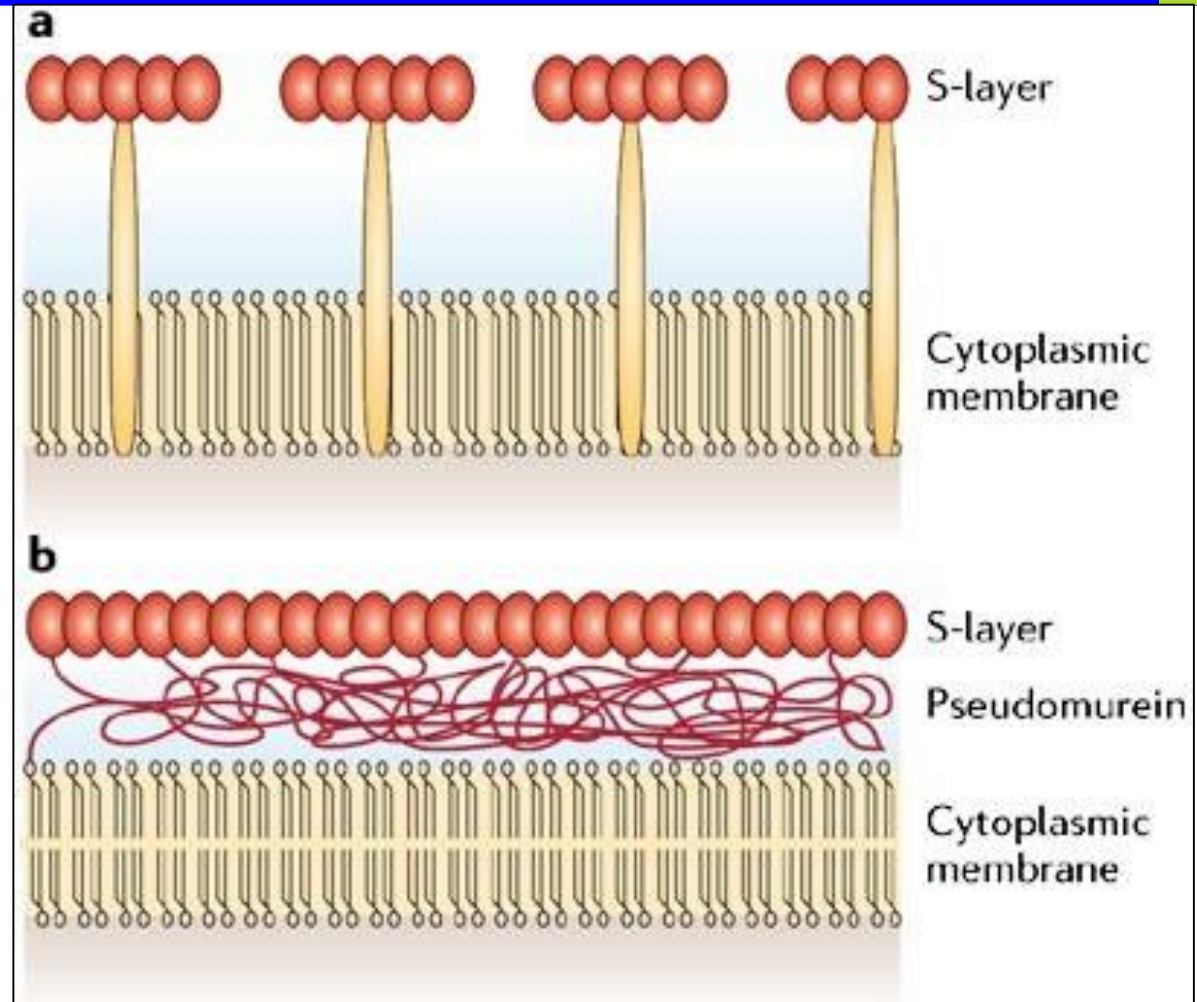
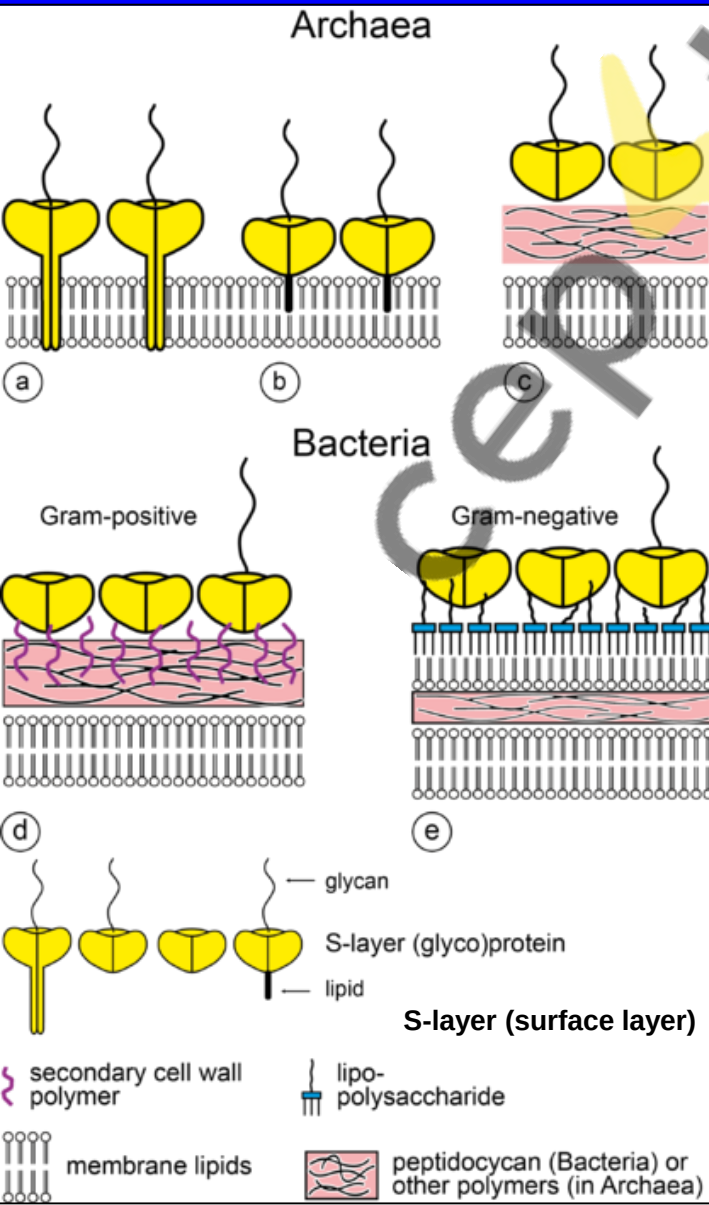


Bacteria

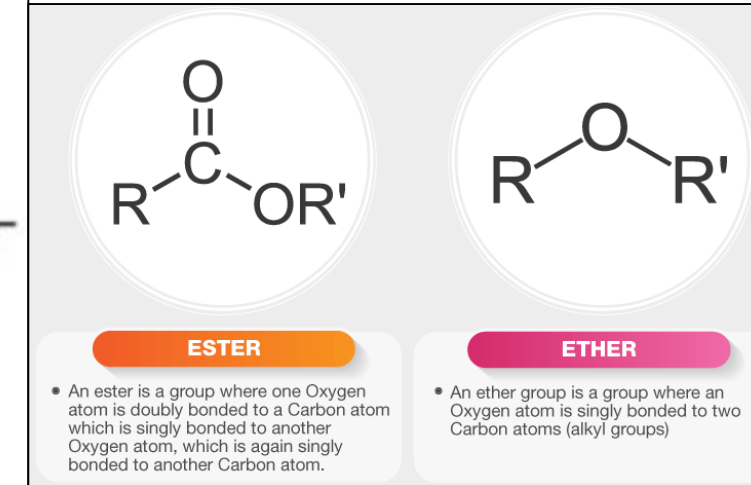
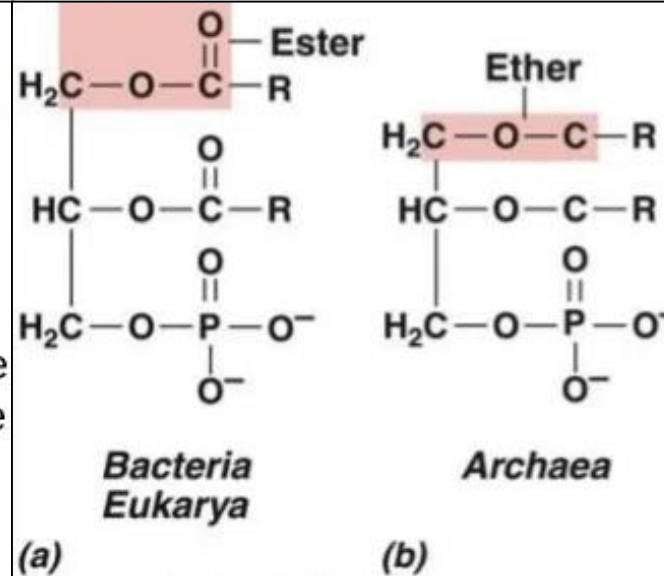
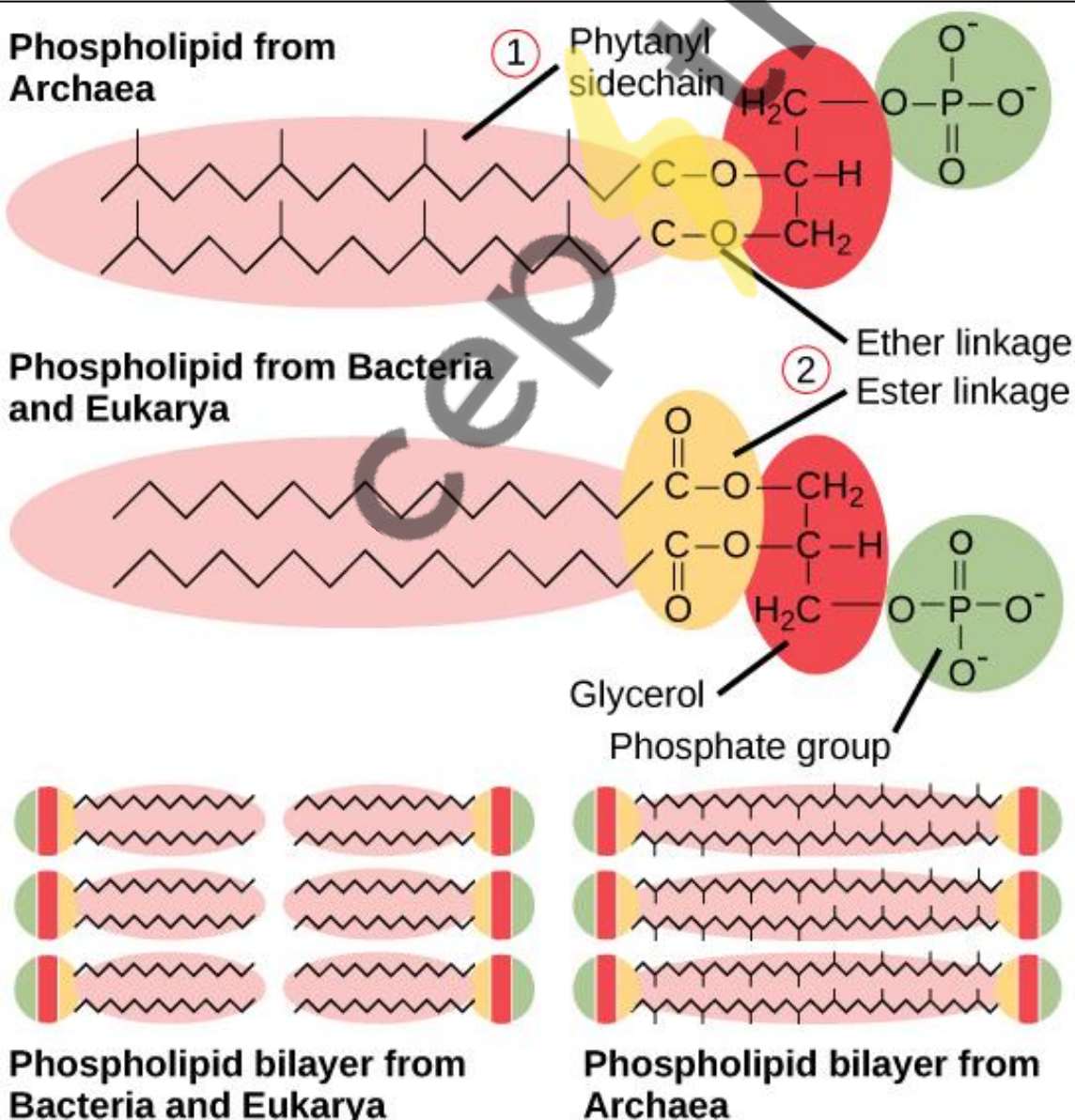


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Archaea vs Eubacteria (cell wall)



Archaea vs Eubacteria vs Eukarya (cell membrane)



- ❑ Archaeol is one of the main core membrane lipids of archaea, one of the three domains of life.
- ❑ Standard archaeol is 2,3-di-O-phytanyl-sn-glycerol, with two phytanyl chains binding to the position of sn-2 and sn-3 of glycerol by ether bonds.
- ❑ Archaeol is generally composed by linking two phytanyl chains to the sn-2 and sn-3 positions of a glycerol molecule. The highly branched side chains are speculated to account for the very low permeability of archaeol-based membrane, which may be one of the key adaptations of archaea to extreme environment.
- ❑ One of the key features that distinguishes archaea from bacteria and eukarya is their membrane lipids, where archaeol plays an important role.
- ❑ Because of this, archaeol is also broadly used as a biomarker for ancient archaea.

Important Links to see Videos



☐ **Bacteria:**

- ☐ <https://www.youtube.com/watch?v=PJmtkQpqJxE>
- ☐ <https://www.youtube.com/watch?v=N7dTM7nLw1Q>
- ☐ <https://www.youtube.com/watch?v=pgr-HeVNbOY&t=391s>

☐ **Three domains of life:**

- ☐ <https://www.youtube.com/watch?v=6LZACcEmQtE>

☐ **Archaea and the tree of life:**

- ☐ <https://www.youtube.com/watch?v=hw-ij3822DY&list=PL6P3nr9Jc5loak1Lld5QwCt6W0MIhGzQ9&t=435s>
- ☐ <https://www.youtube.com/watch?v=rd37jBXfM4k>
- ☐ <https://www.youtube.com/watch?v=7LdmcFoYQtU>

☐ **Virus:**

- ☐ <https://www.youtube.com/watch?v=FmX8au0xGIY&t=613s>
- ☐ <https://www.youtube.com/watch?v=YI3tsmFsrOg>

BT101

Introductory Biology

(Lecture 6)

Dr. Rajkumar P Thummer
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Introduction



Evolution of life:

- ☐ Origin of Life;
- ☐ Darwin's concepts of evolution
- ☐ **Biodiversity**

Cell, the structural and functional unit of life:

- ☐ Three domains of life
- ☐ Cell types, cell organelles and structure
- ☐ Basic biomolecules of cell

What is Biodiversity?



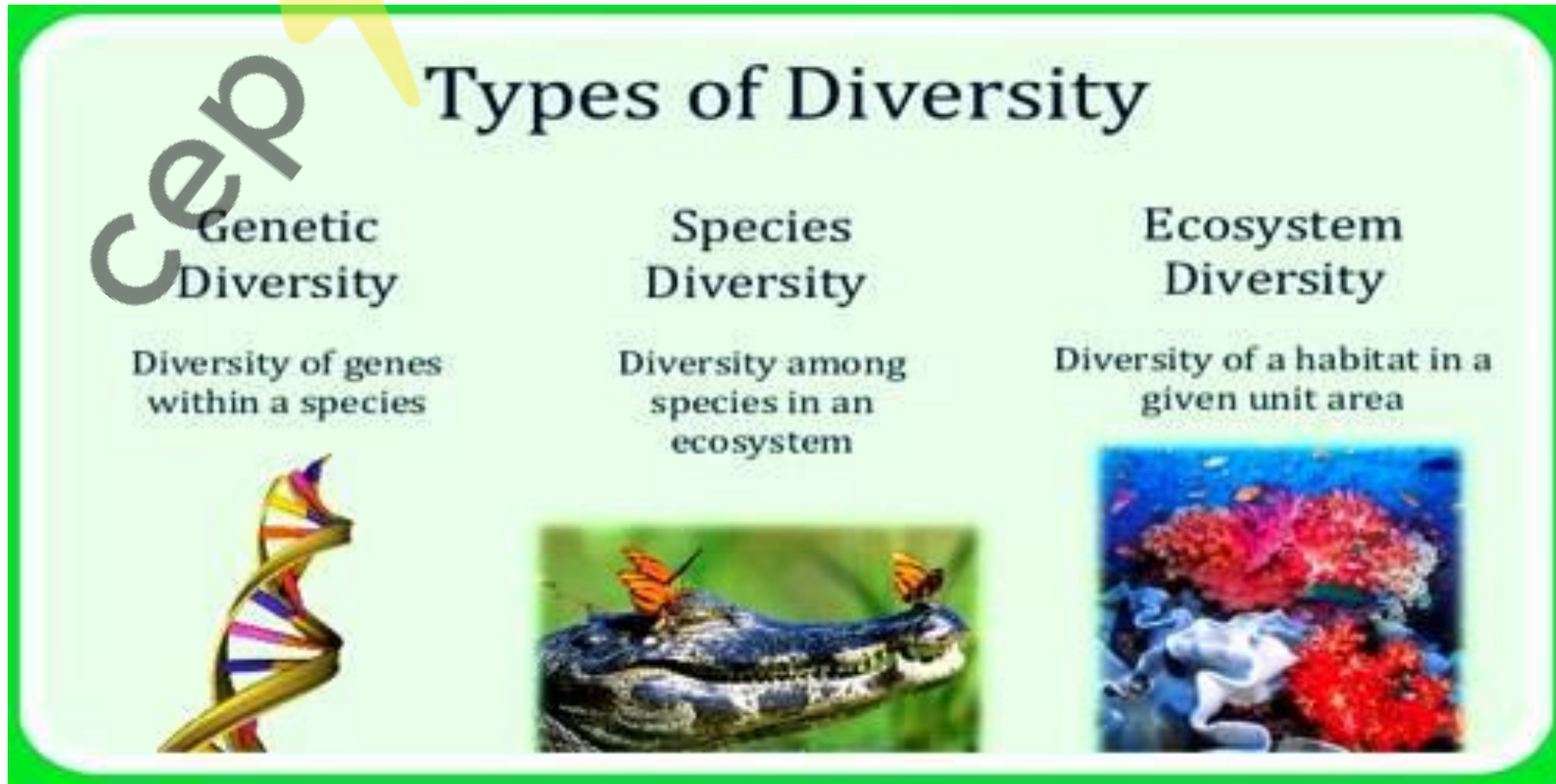
- ❑ The term Biodiversity was first coined by **Walter G. Rosen in 1985**.
- ❑ Biodiversity, is a short term for biological diversity, refers to all of the population, species and communities in a defined area.
- ❑ Biodiversity is the variation among living organisms from different sources including terrestrial, marine and desert ecosystems, and the ecological complexes of which they are a part.
- ❑ Biodiversity forms the foundation of the vast array of ecosystem services, that critically contribute to human well-being.
- ❑ It typically measures the variation at genetic, species, and ecosystem level.

Types of Biodiversity

1. Genetic diversity

2. Species diversity

3. Ecosystem diversity



1. Genetic Diversity

- Genetic diversity is the **variation in the genetic composition** of individuals in a **population, community or species**.
- **Allows individuals to adapt to different conditions. Thus, high genetic diversity increases the ability of the populations and species to survive in the major changes in their environment such as the Climate Change.**



2. Species Diversity

- Species diversity is the variety of species in a particular habitat or ecosystem.
- The richness of species in an ecosystem is usually referred to as Species Diversity.



The diversity of the smaller organisms (e.g. Phytoplankton, the microscopic marine algae) is less known than the larger organisms (e.g. Mammals such as Dolphins and Whales).

3. Ecosystem Diversity



- Ecosystem diversity describes the variation in all **Living and Non-living** things in a **particular Geographic or Ecological Region**.
- Ecosystems comprise unique combinations of animals, plants, micro-organisms and physical characteristics that define a location.
- For example, a small pond constitute an ecosystem and possesses a set of flora and fauna different from a river which is another type of ecosystem.

Ecosystem Diversity (Terrestrial and Aquatic)

Grass land ecosystem



Forest Ecosystem



Desert Ecosystem



Marine Ecosystem



Pond Ecosystem



River Ecosystem



Why is Biodiversity important?



- ☐ Oxygen to breathe
- ☐ Food to eat
- ☐ Clothes to wear
- ☐ House to stay

Why is Biodiversity important?



- ❑ Everything that exists in an ecosystem is part of the web of life, including humans.
- ❑ Plant, animal, and insect species interact and depend upon one another for what each offers, such as food, shelter, oxygen, and soil enrichment.
- ❑ "It is reckless (hasty) to suppose that biodiversity can be diminished indefinitely without threatening humanity itself." - **Edward O. Wilson** (Father of Biodiversity)
- ❑ At the ecosystem level, biodiversity provides the conditions and drives the processes that sustain the global economy- and our survival as a species.

Magnitude of Biodiversity

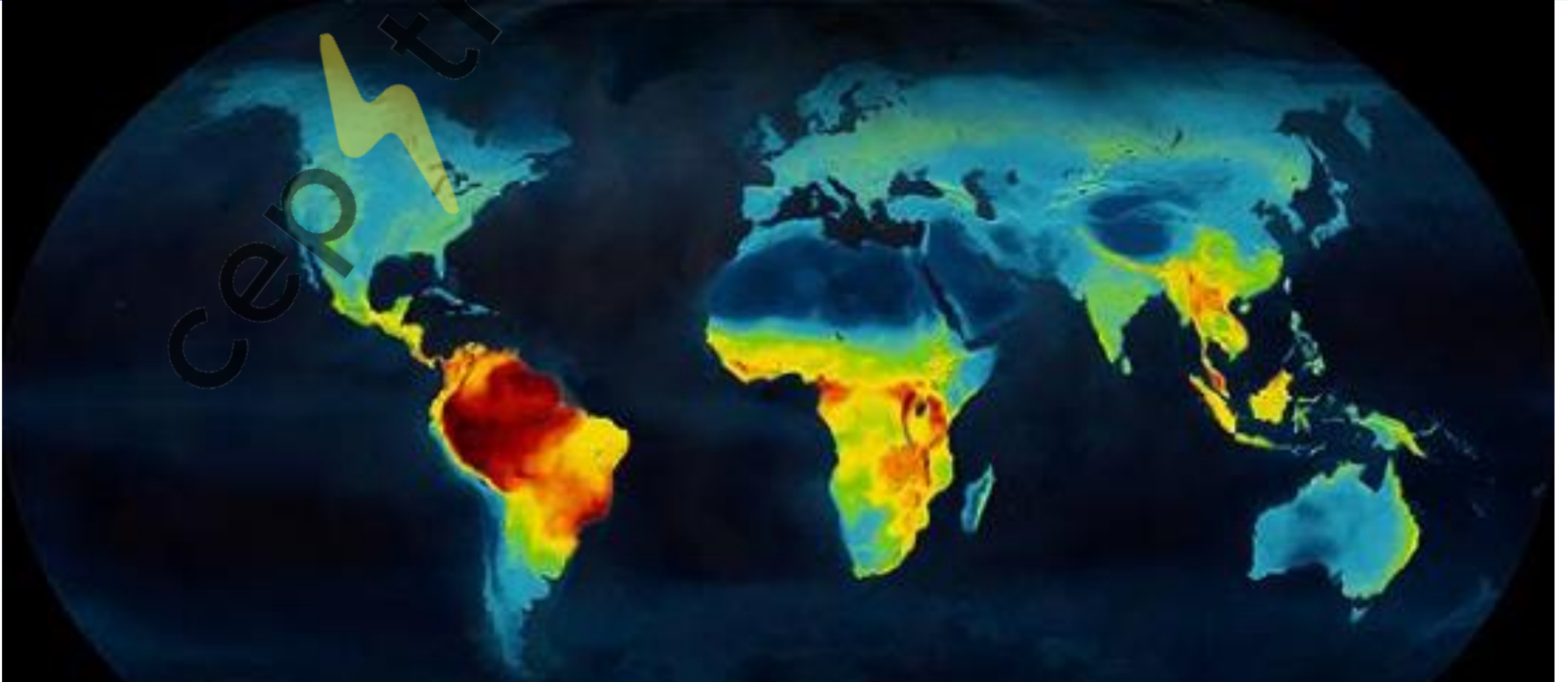


- ❑ The known and described number of species on earth is between 1.7 and 1.8 million.
- ❑ **Current estimates of the total number of species range from 5 to 30 million.**
- ❑ About 61% of these species are insects.

Table lists the Approximate number of species on earth.

GROUP	Number of Species
MAMMALS (Human, Bat, Whale etc.)	4650
BIRDS	9700
REPTILES (crawls, moves on its belly or on short legs. e.g. Snake, Corcodile, lizard, etc.)	7150
FISH	27000
AMPHIBIANS (Can live both on land and in water, Frog, etc.)	4780
INSECTS	1025000
HIGHER PLANTS	270000
ALGAE	40000
PROTOZOANS (Single celled organisms e.g. <i>Amoeba</i> , <i>Paramecium</i>)	40000

Magnitude of Biodiversity



Distribution of **living terrestrial vertebrate** species, highest concentration of diversity shown in red in equatorial regions, declining polewards (towards the blue end of the spectrum) (Mannion 2014)



Benefits of Biodiversity



❑ **Consumptive value:**

- Food/Drink
- Fuel
- Medicine: painkillers, antibiotics, anticancer drugs
- Crop varieties
- Industrial Material: paper, rubber

❑ **Non-Consumptive Value:**

- Recreation: visiting beach, whale watching, photography, surfing, scuba diving and boating, where resource is not removed from Ocean or Coast.
- Education and Research
- Traditional value

Threats to Biodiversity



Natural causes:

- Narrow Geographical Area
- Over Population
- Low Breeding rate
- Natural Disasters

Anthropogenic (by Human) causes:

- Pollution
- Hunting
- Global warming and climate change
- Agriculture

Threats to Biodiversity



Biodiversity is under serious threat as a result of **Human activities**.

The main dangers worldwide are

- i) population growth and resource consumption
- ii) climate change and global warming
- iii) habitat conversion and urbanization
- iv) overexploitation of natural resources and
- v) environmental degradation
- vi) etc.

Why is Biodiversity important?



Important Links to see Videos



- ❑ **Biodiversity:**
- ❑ https://www.youtube.com/watch?v=GK_vRtHJZu4&t=2s
- ❑ https://www.youtube.com/watch?v=ewJImPSV_XE (in hindi)

BT101

Introductory Biology

(Lecture 7)

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Introduction



Evolution of life:

- ❑ Origin of Life;
- ❑ Darwin's concepts of evolution;
- ❑ Biodiversity.

Cell, the structural and functional unit of life:

- ❑ Three domains of life;
- ❑ Cell types, cell organelles and structure;
- ❑ Basic biomolecules of cell.

Three domains of life

The tree of life consists of three domains: Bacteria, Archaea and Eukarya

- ❑ All living organisms consist of elementary units called cells.
- ❑ All cells are believed to have evolved from a common progenitor because the structures and molecules in all cells have so many similarities.
- ❑ In recent years, detailed analysis of DNA sequences from a variety of Prokaryotic organisms revealed that they are of **two distinct** types: so-called “True” bacteria or **Eubacteria** and **Archaea**.
- ❑ Based on the assumption that the organisms with more similar genes evolved from a common progenitor more recently than those with more dissimilar genes. The researchers developed an Evolutionary Lineage Tree as shown in Figure 1.
- ❑ According to this tree, the **Archaea** and the **Eukaryotes** diverged from the **Eubacteria** and later they also diverged from each other.

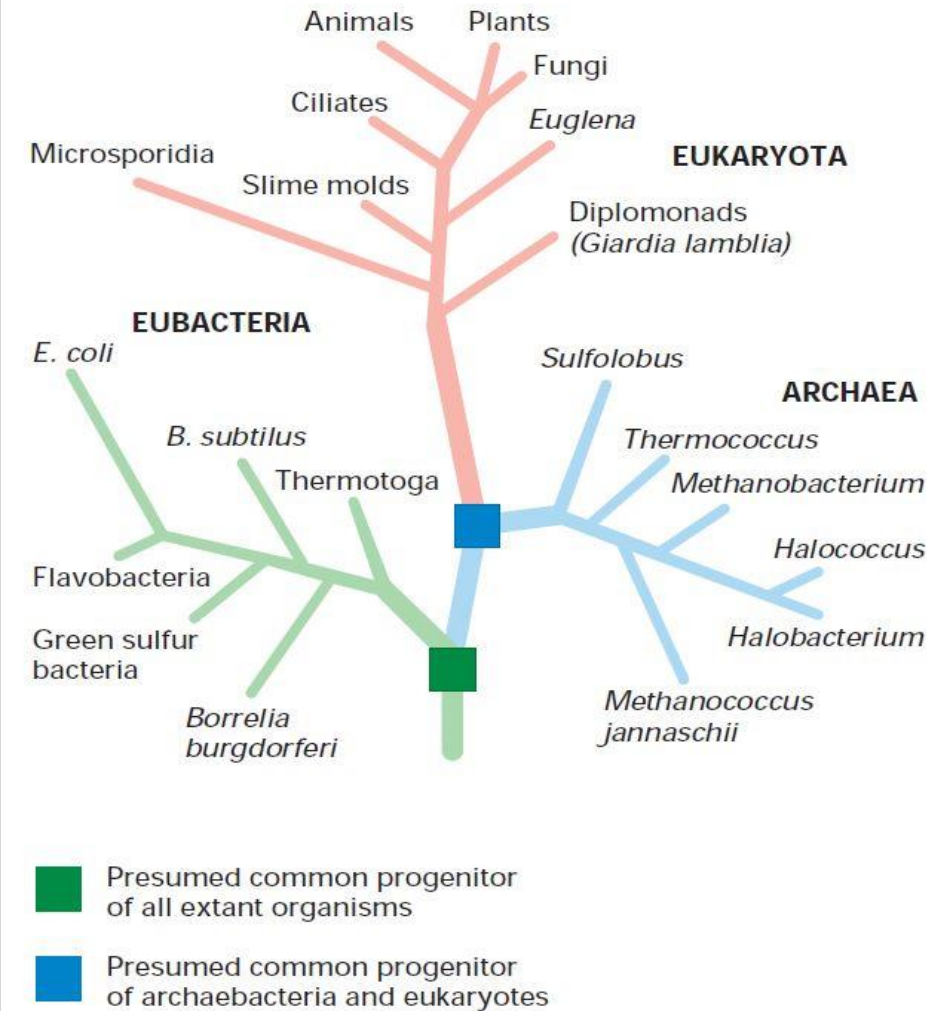


Fig. 1. All organisms from simple bacteria to complex mammals probably evolved from a common, single-celled progenitor. *Credit: Molecular Cell Biology - Lodish

The Cell

- The cell is the basic unit of life. Microorganisms such as Bacteria, Yeast and Amoeba exist as single cells. By contrast, an adult human is made up of around ~37.2 trillion cells (1 trillion = 10^{12}), which are mostly organized into tissues and organs.
- The first person to observe and record cells was Robert Hooke (1635–1703)
- Antonie Van Leeuwenhoek is known for his pioneering work in Microscopy.



Important characteristics of a cell

1. All cells store their hereditary information in the same chemical code: DNA.
2. All cells replicate their hereditary information by DNA template polymerization: Replication.
3. All cells transcribe portions of their hereditary information into an intermediary form: RNA.
4. All cells translate RNA into protein: Translation
5. All cells use proteins as catalysts.
6. Each protein is encoded by a specific gene (segment of DNA).
7. All cells are enclosed in a plasma membrane allowing the nutrients and waste materials exchange.

Introduction



Table 1.1 Prokaryotic and Eukaryotic Cells

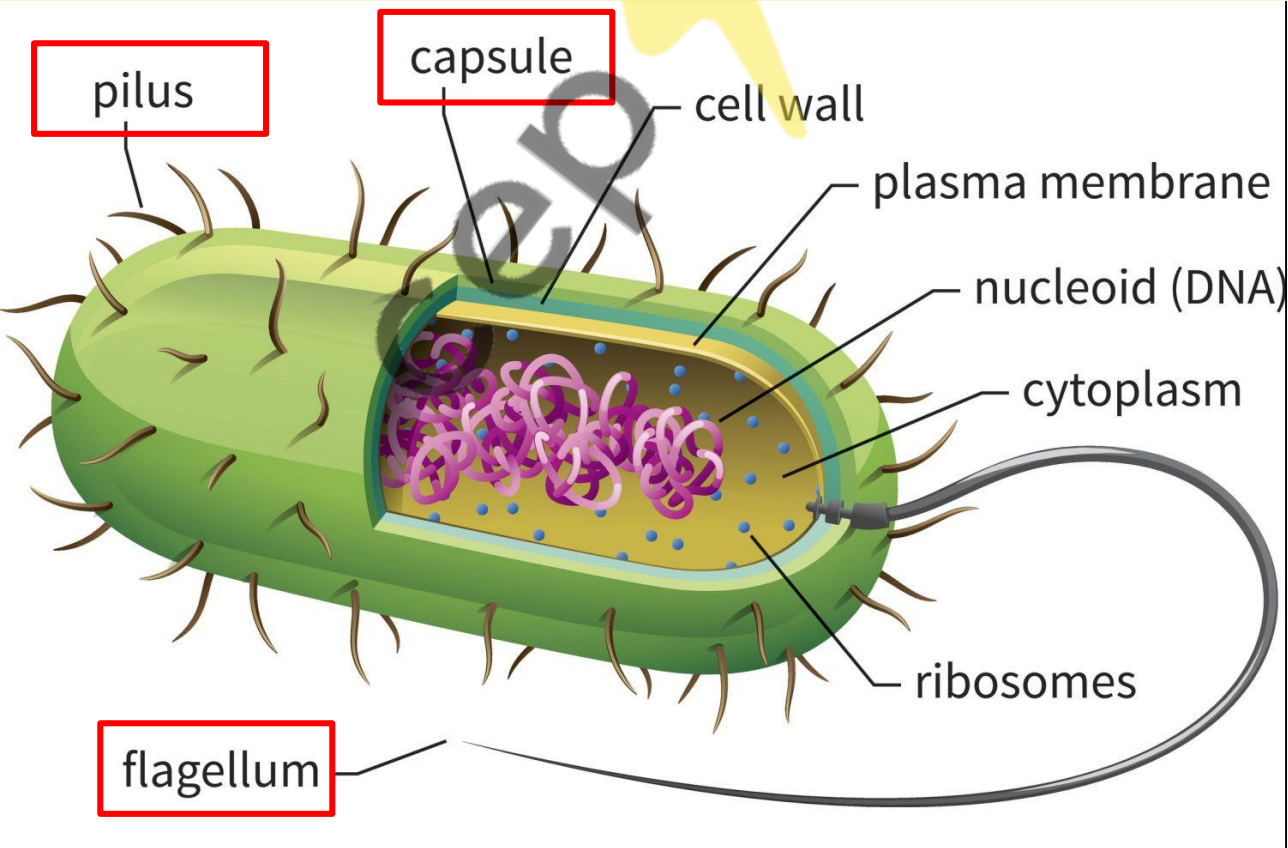
Characteristic	Prokaryote	Eukaryote
Nucleus	Absent	Present
Diameter of a typical cell	$\approx 1\mu\text{m}$	$10\text{--}100\mu\text{m}$
Cytoskeleton	Absent	Present
Cytoplasmic organelles	Absent	Present
<u>DNA</u> content (base pairs)	1×10^6 to 5×10^6	1.5×10^7 to 5×10^9
Chromosomes	Single circular <u>DNA</u> molecule	Multiple linear <u>DNA</u> molecules

<https://www.youtube.com/watch?v=zZtcMBTQaS4>

<https://www.youtube.com/watch?v=Pxujitlv8wc>

Prokaryotes

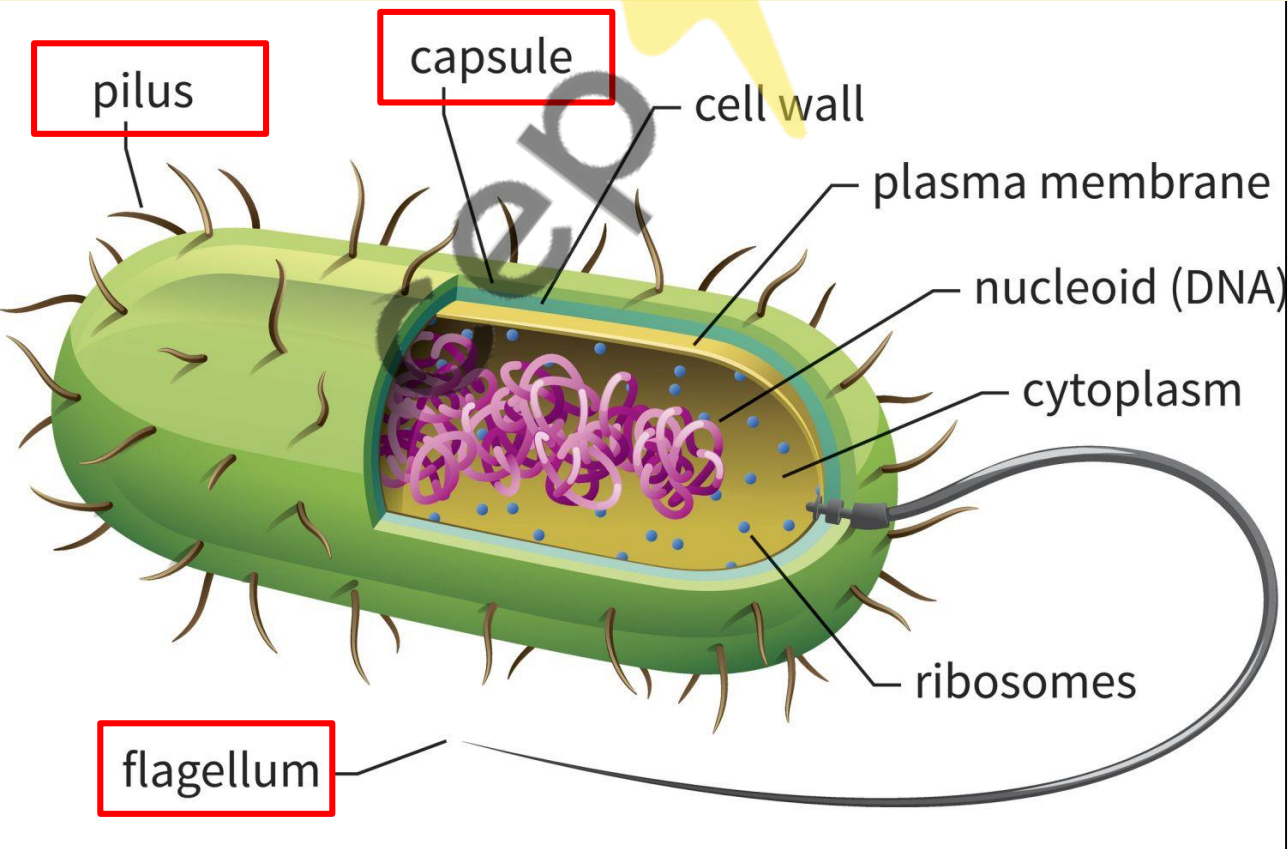
- ❑ It is a single-celled organism that lacks a membrane-bound nucleus (karyon), mitochondria or any other membrane-bound organelle.



- ❑ The **capsule** is a gelatinous layer covering the entire bacterium. It is composed of polysaccharide (long chains of carbohydrate molecules).
- ❑ The capsule—which can be found in both gram negative and gram-positive bacteria—is different from the second lipid membrane – bacterial outer membrane, which contains lipopolysaccharides and lipoproteins and is found only in gram-negative bacteria.
- ❑ The capsule is considered a virulence factor because it enhances the ability of bacteria to cause disease (e.g. prevents phagocytosis).
- ❑ The capsule can protect cells from engulfment by eukaryotic cells, such as macrophages.
- ❑ A capsule-specific antibody may be required for phagocytosis to occur. Capsules also contain water which protects the bacteria against desiccation.
- ❑ They also exclude bacterial viruses and most hydrophobic toxic materials such as detergents.
- ❑ Capsules also help cells adhere to surfaces.

Prokaryotes

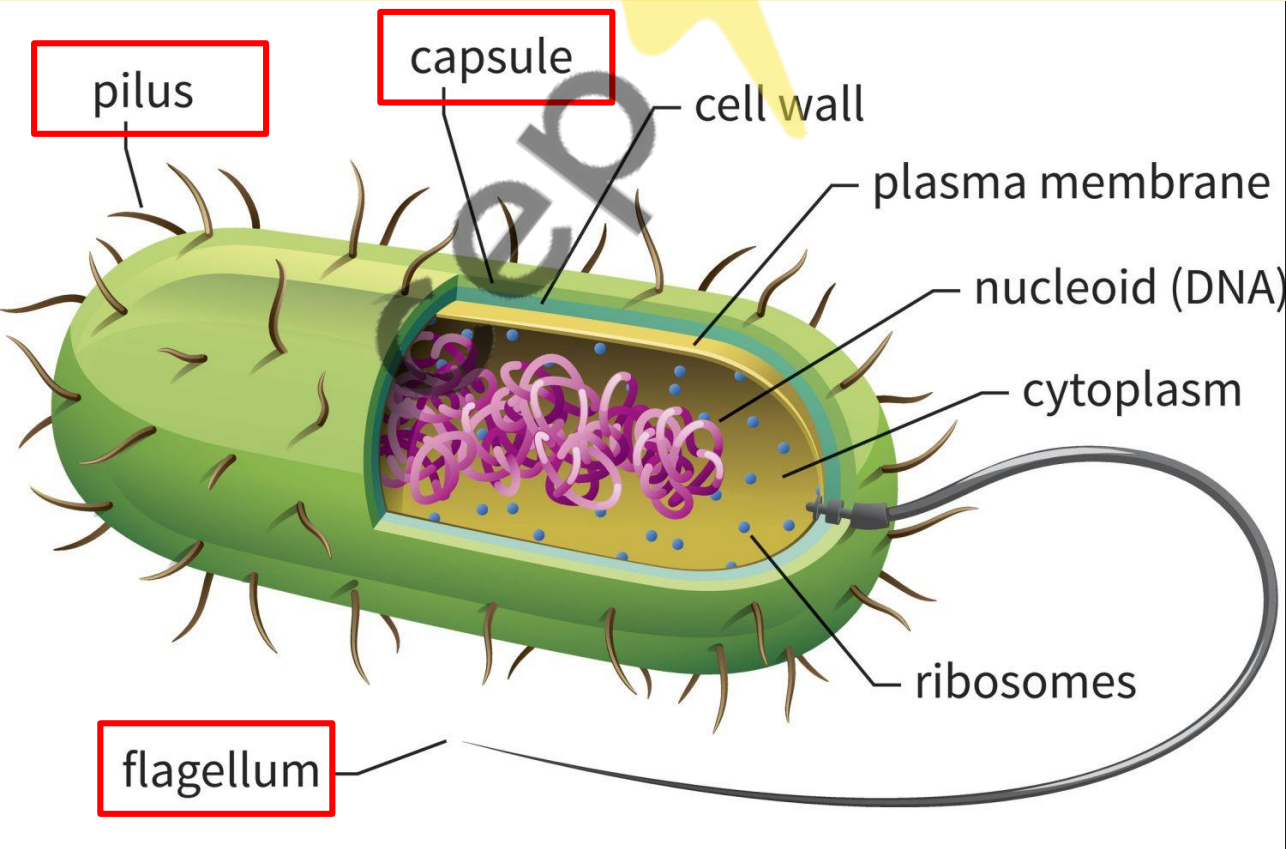
- ❑ It is a single-celled organism that lacks a membrane-bound nucleus (karyon), mitochondria or any other membrane-bound organelle.



- ❑ **Flagella** help move the bacteria toward nutrients and other attractants, a process called chemotaxis.
- ❑ The long filament, which acts as a propeller, is composed of many subunits of a single protein, flagellin, arranged in several intertwined chains.
- ❑ The energy for movement, the proton motive force, is provided by adenosine triphosphate (ATP), derived from the passage of ions across the membrane.

Prokaryotes

- ❑ It is a single-celled organism that lacks a membrane-bound nucleus (karyon), mitochondria or any other membrane-bound organelle.



- ❑ **Pili** are hair-like filaments that extend from the cell surface. They are shorter and straighter than flagella and are composed of subunits of pilin, a protein arranged in helical strands.
- ❑ Conjugative pili allow for the transfer of DNA between bacteria, in the process of bacterial conjugation. They are sometimes called "sex pili", in analogy to sexual reproduction, because they allow for the exchange of genes via the formation of "mating pairs".
- ❑ Pili are responsible for virulence in the pathogenic strains of many bacteria

Various organelles in a human cell and its major function(s)



1. **Cell membrane** – (is a membrane that separates the interior of all cells from the outside environment (the extracellular space) and protects the cell from its environment; acts as a barrier for cell contents. It consists of double phospholipid layer and monolayer of protein scattered around phospholipid layer. The other components in plasma membrane are cholesterol and glycoproteins).
2. **Cytosol** – (fluid that contains organelles; represents the material outside the nucleus and inside the cell membrane)
3. **Cytoskeleton** – (a network of protein structures that extend throughout the cytoplasm; provides the cell with an internal framework (e.g. microfilaments and microtubules))
4. **Nucleus** – (control center of the cell; contains genetic material; It is the center of the cell because it contains genetic material (DNA). It consists of three main regions: the nuclear membrane, the nucleolus and chromatin.
Nuclear membrane: Nuclear membrane serves as a barrier of nucleus. It consists of a double phospholipid membrane and contains nuclear pores that allow for the exchange of material with the rest of the cell.
Chromatin: It is composed of DNA and protein scattered throughout the nucleus. Chromatin condenses to form chromosomes when the cell divides)
5. **Nucleolus** – (functions as site of ribosome production; ribosomes then migrate to the cytoplasm through nuclear pores)
6. **Ribosome** (little dots) – sites of protein synthesis in the cell.

Various organelles in a human cell and its major function(s)



7. **Rough endoplasmic reticulum** – (carry ribosomes that represent sites of protein synthesis)
8. **Smooth endoplasmic reticulum** – (function in cholesterol synthesis and breakdown, fat metabolism, and detoxification of drugs)
9. **Golgi apparatus** (or "Golgi body") – (modifies and packages proteins, secrete vesicles, plasma membrane components and lysosomes)
10. **Mitochondrion** – (powerhouse of the cell; generate ATP for cellular activities)
11. **Lysosomes** – (suicidal bag of the cell; contain enzymes that digest non-usable materials within the cell)
12. **Peroxisomes** – (detoxify harmful substances and break down free radicals; involved in catabolism of very long chain fatty acids,)
13. **Centrioles** – (involved in the organization of mitotic spindle fibers during cell division and in the completion of cytokinesis)
14. **Centrosomes** – (serves as the main microtubule organizing center (MTOC) of the animal cell, as well as a regulator of cell-cycle progression; are composed of two centrioles arranged at right-angles to each other)
15. **Vesicles** – (involved in metabolism, transport, buoyancy control, and temporary storage of food and enzymes)

Important Links to see Videos



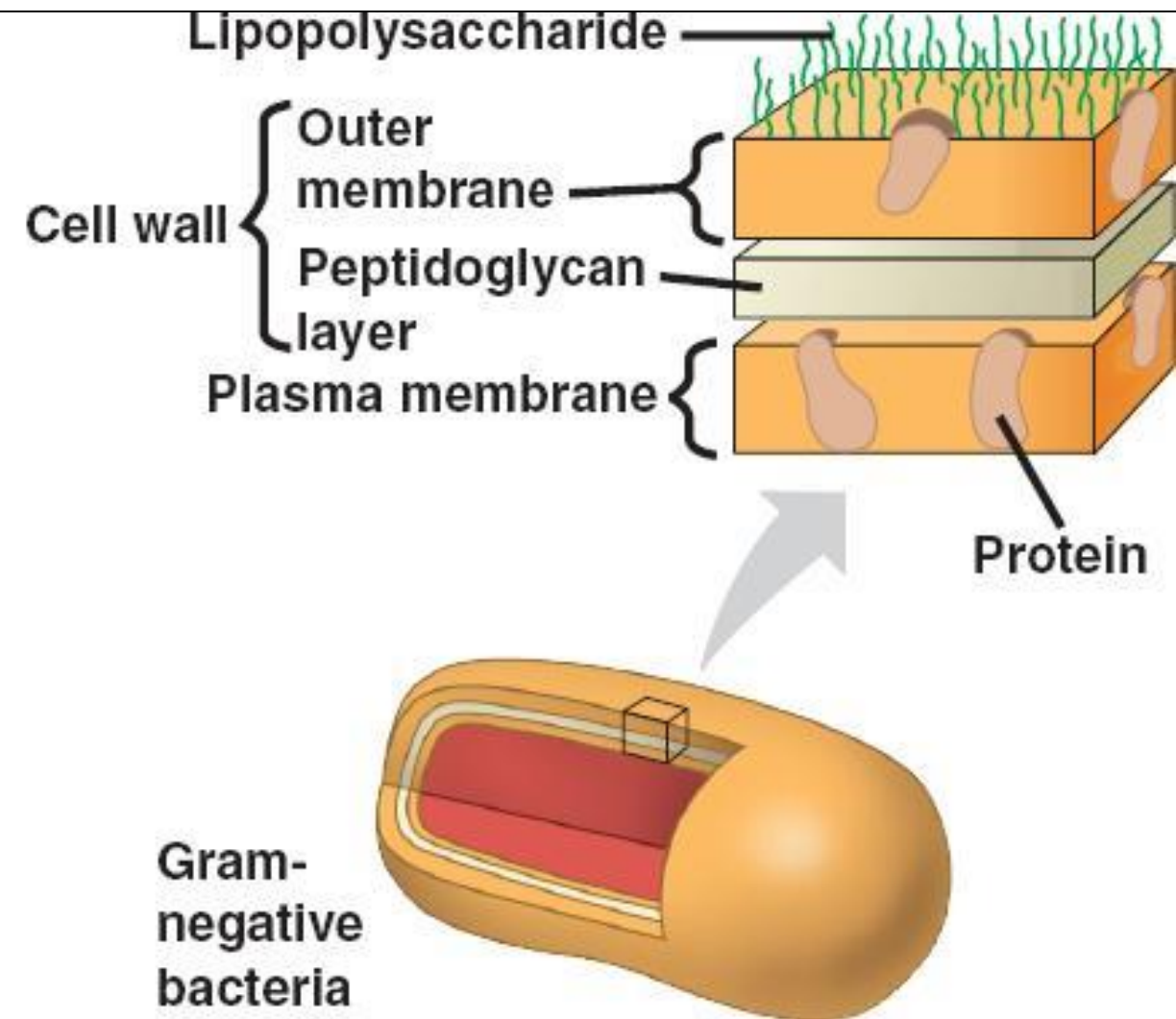
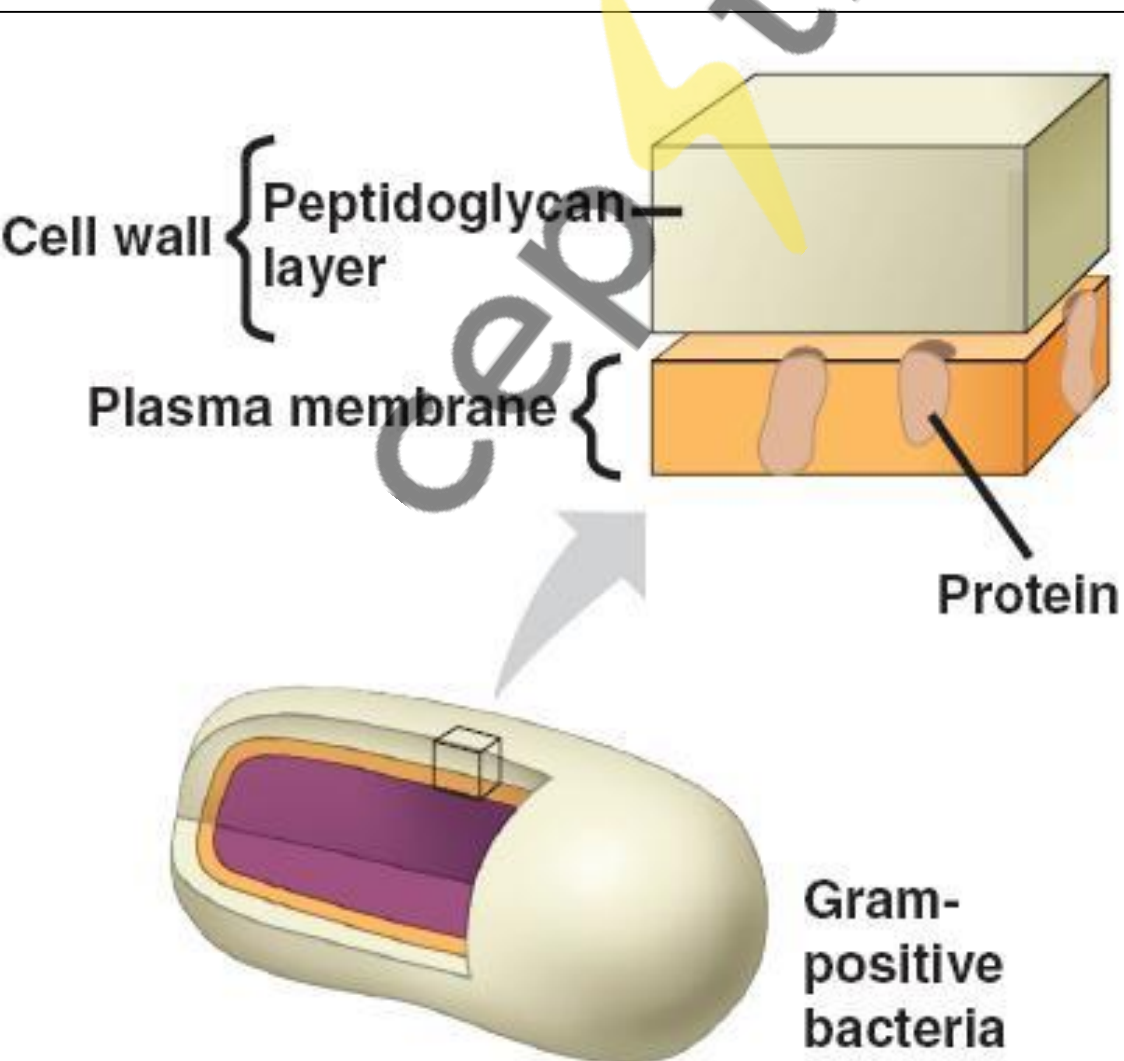
- ❑ **Overview of Cell Structure:**
- ❑ <https://www.youtube.com/watch?v=0xe1s65IH0w>
- ❑ <https://www.youtube.com/watch?v=RKmaq7jPnYM>

Cell wall

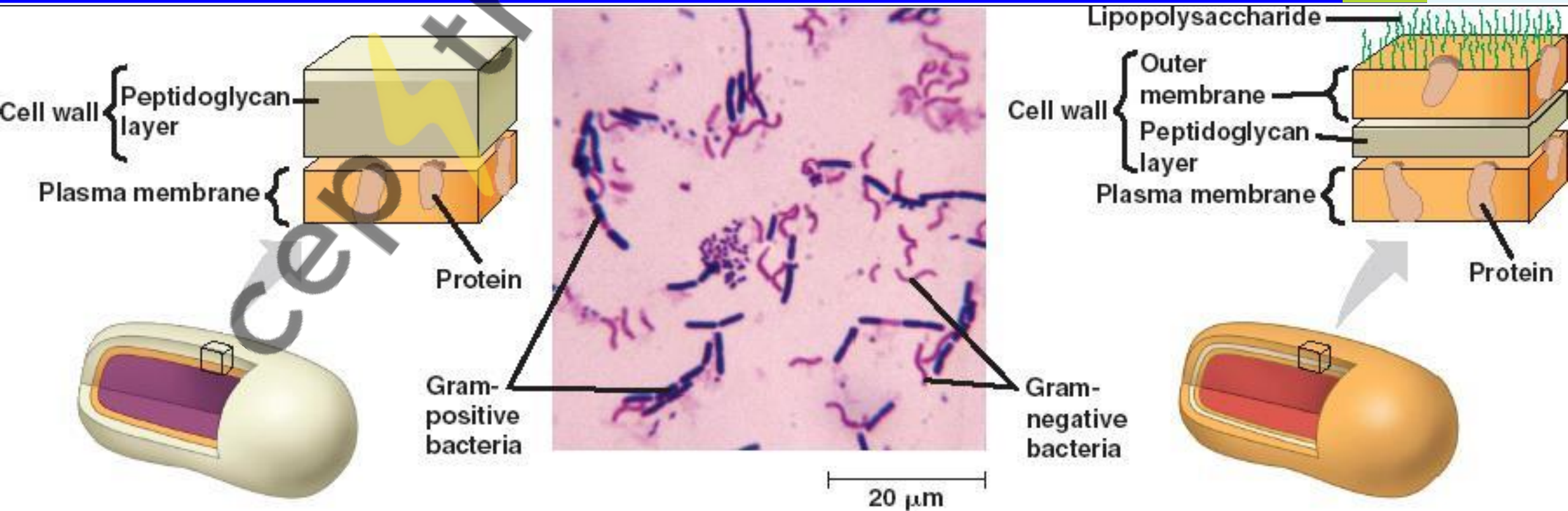


- ❑ The cell wall is a rigid and protective layer around the plasma membrane which provides mechanical support to the cell.
- ❑ Animal cells do not have a cell wall. They are present in most plant cells, fungi, bacteria, and some archaea.
- ❑ In plant cells, the cell wall is made up of cellulose (polysaccharide), hemicellulose (polysaccharide), pectin (polysaccharide), and protein.
- ❑ In many fungi, the cell wall is formed of chitin (polysaccharide).
- ❑ In bacteria, the cell wall contains protein-lipid-polysaccharide complexes. Bacterial cell walls are made of peptidoglycan (also called murein), which is made from polysaccharide chains cross-linked by peptides.
- ❑ The cell wall has many important functions in a cell including protection, structure, mechanical strength and support.
- ❑ The cell wall protect organisms against viruses and other pathogens.

Bacteria



Bacteria

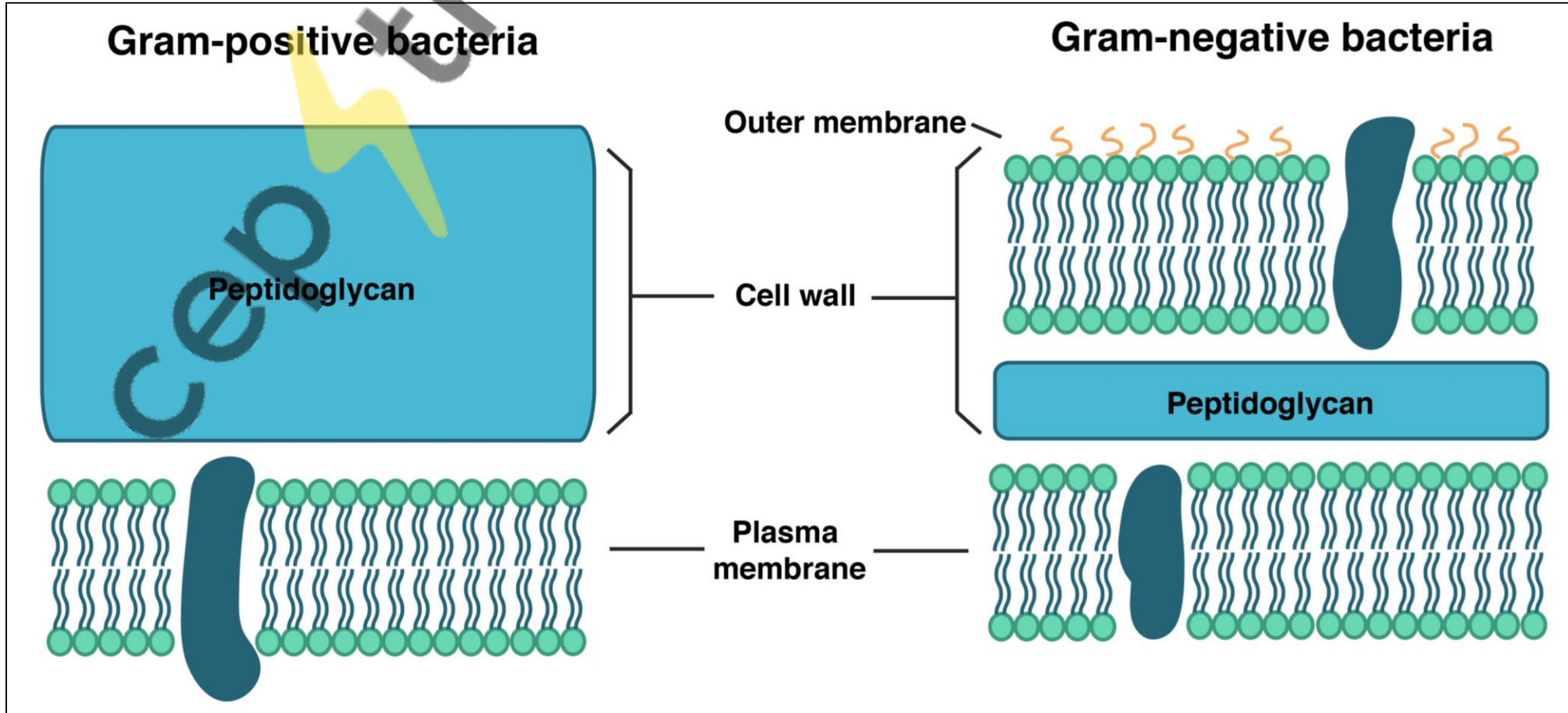


(a) Gram-positive

(b) Gram-negative

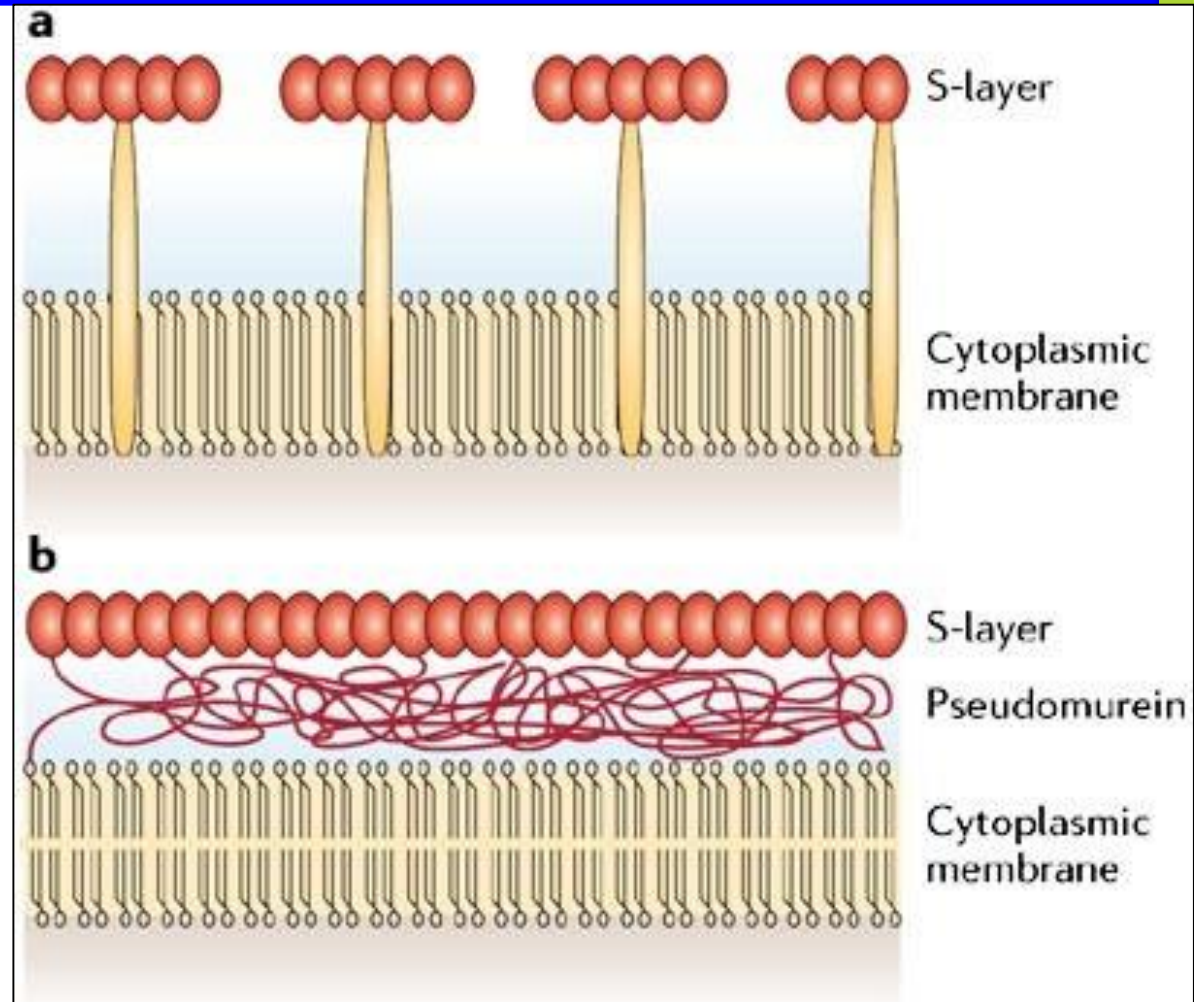
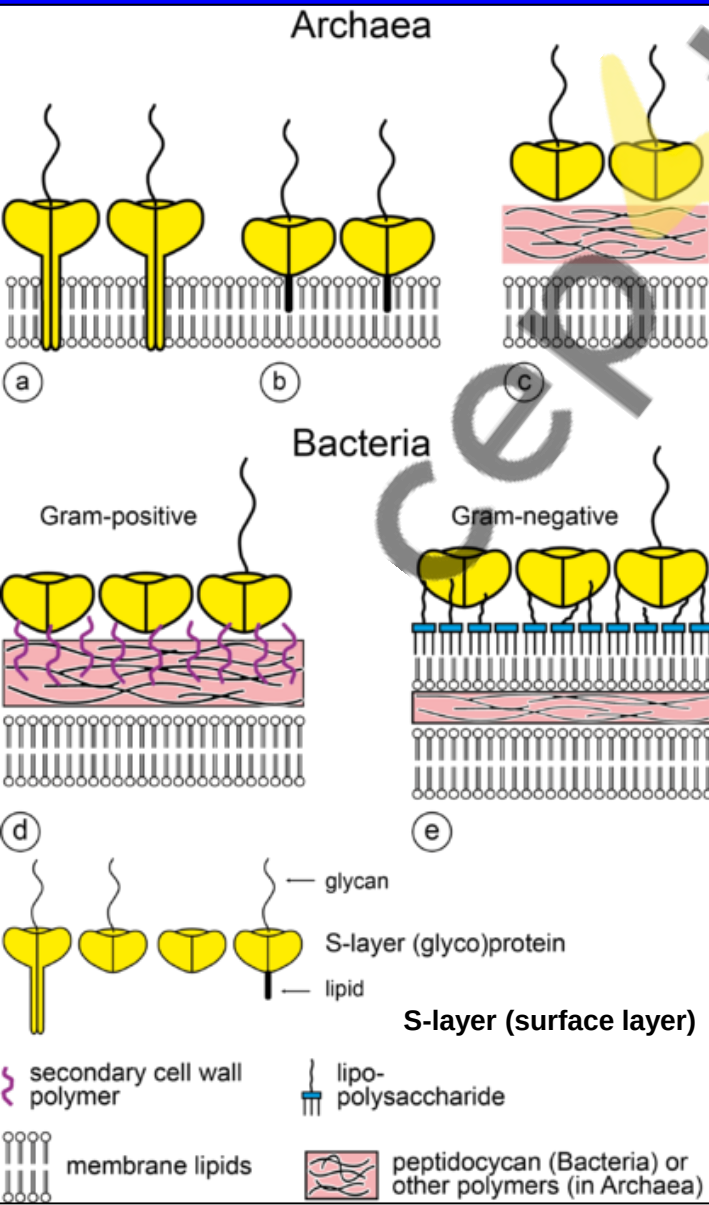
- ❑ Gram stain or Gram staining is a method of staining used to classify bacterial species into two large groups: Gram-positive bacteria and Gram-negative bacteria.
- ❑ Gram-positive bacteria have a thick mesh-like cell wall made of peptidoglycan (50–90% of cell envelope), and as a result are stained purple by crystal violet, whereas gram-negative bacteria have a thinner layer (10% of cell envelope), so do not retain the purple stain and are counter-stained pink by safranin.

Bacteria



- ❑ Gram stain or Gram staining is a method of staining used to classify bacterial species into two large groups: Gram-positive bacteria and Gram-negative bacteria.
- ❑ Gram-positive bacteria have a thick mesh-like cell wall made of peptidoglycan (50–90% of cell envelope), and as a result are stained purple by crystal violet, whereas gram-negative bacteria have a thinner layer (10% of cell envelope), so do not retain the purple stain and are counter-stained pink by safranin.

Archaea vs Eubacteria (cell wall)



Prokaryotic cells have a simpler internal organization than eukaryotic cells

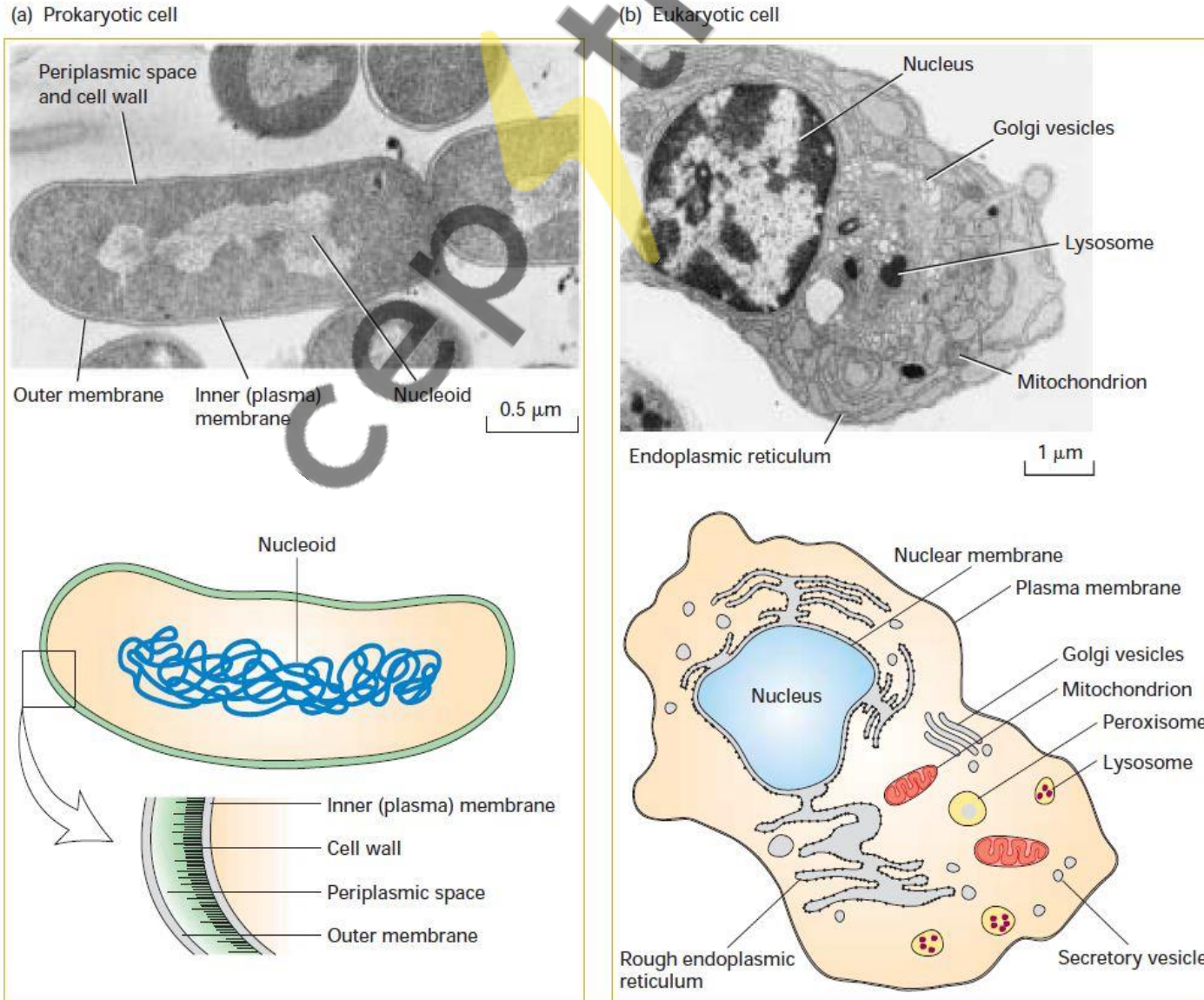


Figure 3.

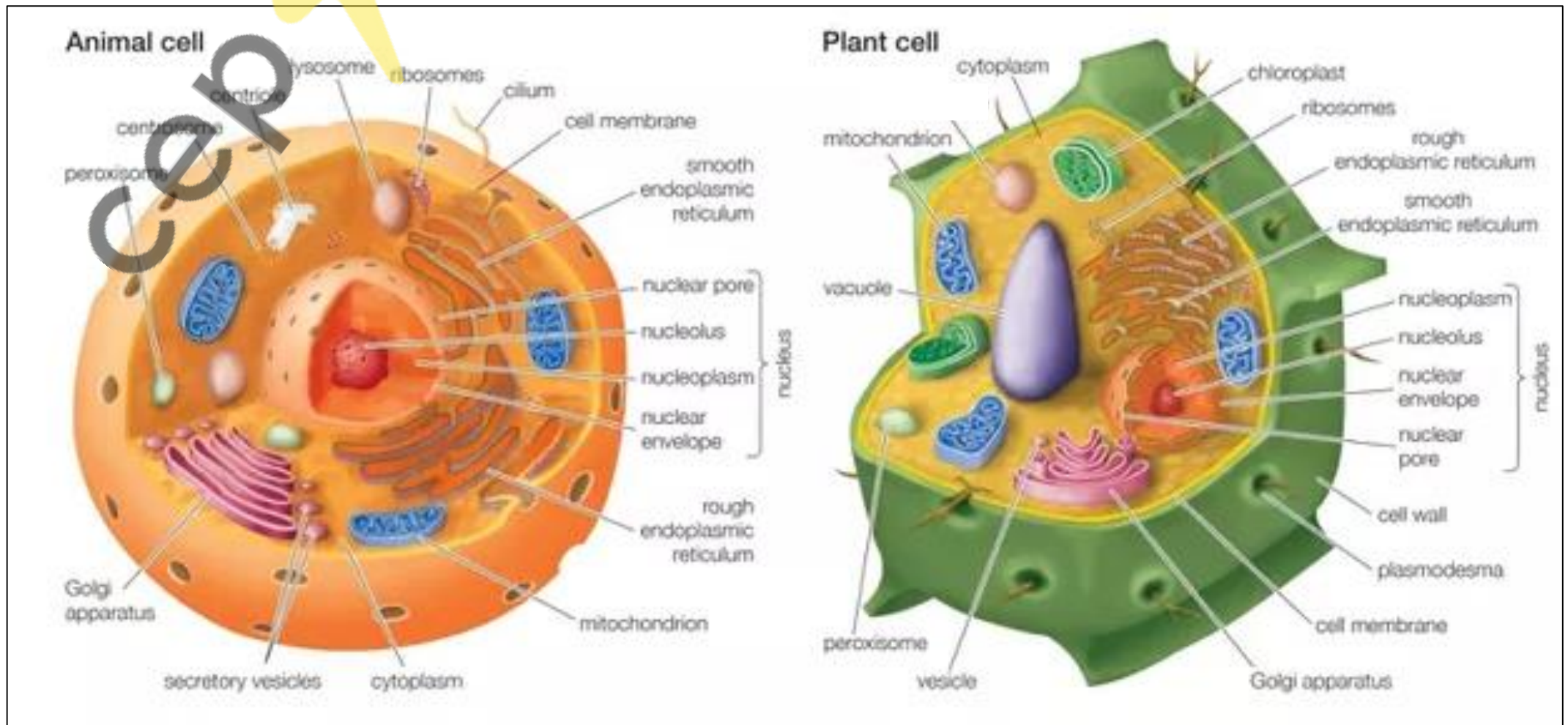
- a) Electron micrograph of a section of *Escherichia coli*, a common intestinal bacterium. **Nucleoid, consisting of DNA is not enclosed in a membrane.**
- b) Electron micrograph of White Blood Cell that secretes antibodies. **Only a single membrane (the plasma membrane) surrounds the cell, but the interior contains many membrane-bound organelles.**

An electron microscope is a microscope that uses a beam of accelerated electrons as a source of illumination. As the wavelength of an electron can be up to 100,000 times shorter than that of visible light photons, electron microscopes have a higher resolving power than light microscopes and can reveal the structure of smaller objects

*Credit: Molecular Biology of the Cell - Alberts

Eukaryotes

- ❑ It is any organism whose cells contain a nucleus and other organelles enclosed within membranes.



Animal cells vs Plant cells



- ❑ Both plant and animal cells are eukaryotic, so they contain membrane-bound organelles like the Nucleus and Mitochondria.
- ❑ Animal and Plant cells, both are Eukaryotic Cells, yet there are some differences among them, and they do not look exactly the same or have all of the same organelles, since each has different needs.

For example, plant cells contain Chloroplasts, since they need to perform photosynthesis, but animal cells do not.

Animal cells vs Plant cells



- ❑ **Both animal and plant cells have mitochondria, but only plant cells have Chloroplasts.**
- ❑ Plants don't get their sugar from eating food, so they need to make **sugar from sunlight**. This process (**Photosynthesis**) takes place in chloroplast. Once the sugar is made, it is broken down by the mitochondria to make energy for the cell. Because animals get sugar from the food they eat, so they do not need chloroplasts, but just **Mitochondria**.
- ❑ **Both plant and animal cells have Vacuoles.** A plant cell contains a large, single vacuole that is used for storage and maintaining the shape of the cell. In contrast, animal cells have many, smaller vacuoles.
- ❑ **Plant cells have a cell wall, as well as a cell membrane. In plants, the cell wall surrounds the cell membrane.** This gives the plant cell its unique shape. Animal cells simply have a **Cell membrane**, but no Cell wall.
- ❑ **Centriole** is present only in **Animal cells** and in some **lower plant cells** e.g. mosses and ferns, but **not present in higher plant cells**. Plants do not have centrioles but they possess microtubules which acts just like a centriole that is it helps in the spindle fibre (aggregates of microtubules that move chromosomes during cell division) formation during cell division. A **Centrosome** comprises two microtubule rings known as **Centrioles**. Its main function is to organize the microtubules and provide a **structure** to the **cell**.

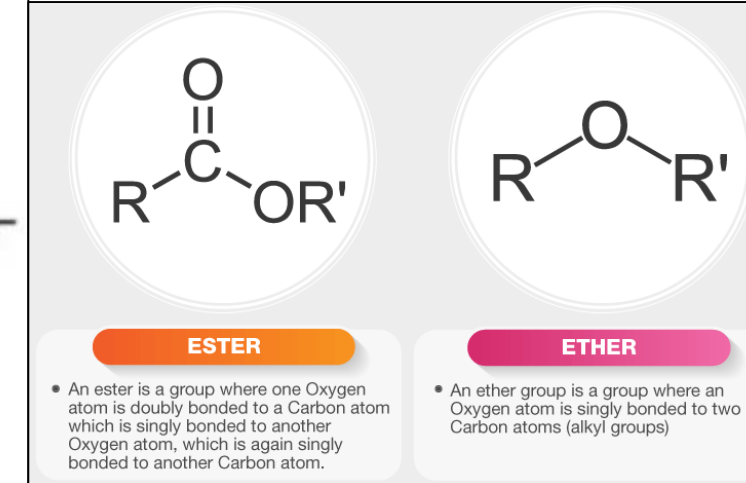
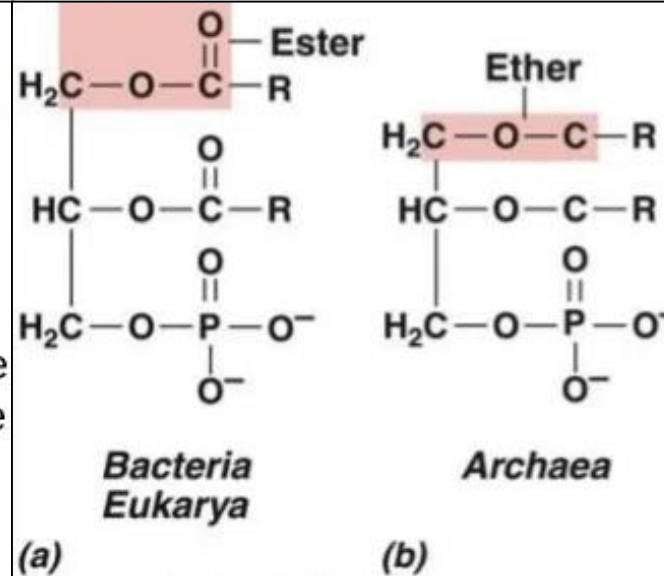
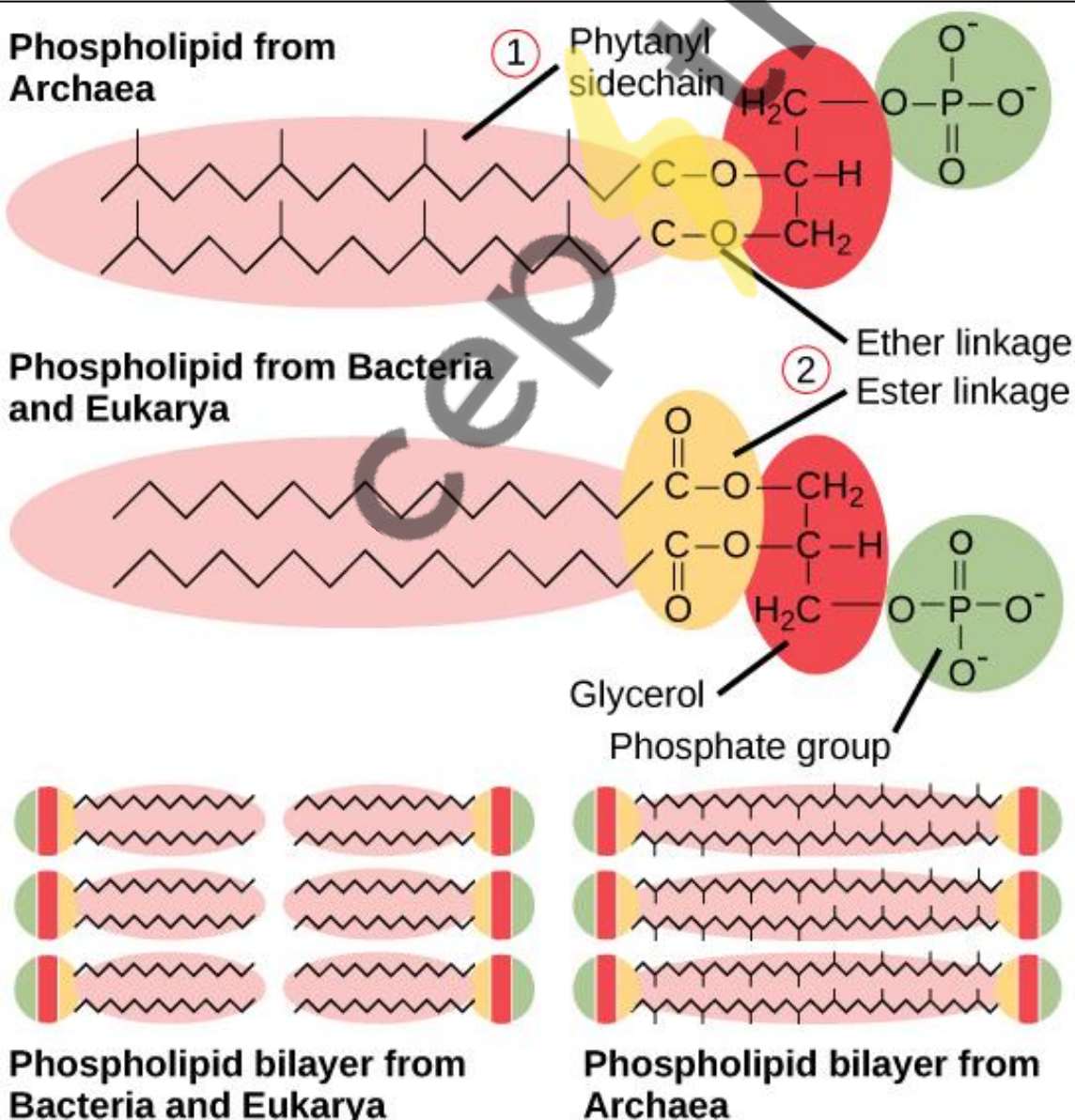
Biological Membranes



- ❑ **Membranes form boundaries both around cells and around subcellular organelles e.g. nucleus, mitochondria etc.**
- They act as selectively, permeable barriers.
- All membranes contain two basic components: lipids and proteins. Some membranes also contain carbohydrate.
- The composition of lipid, protein and carbohydrate varies from one membrane to another.

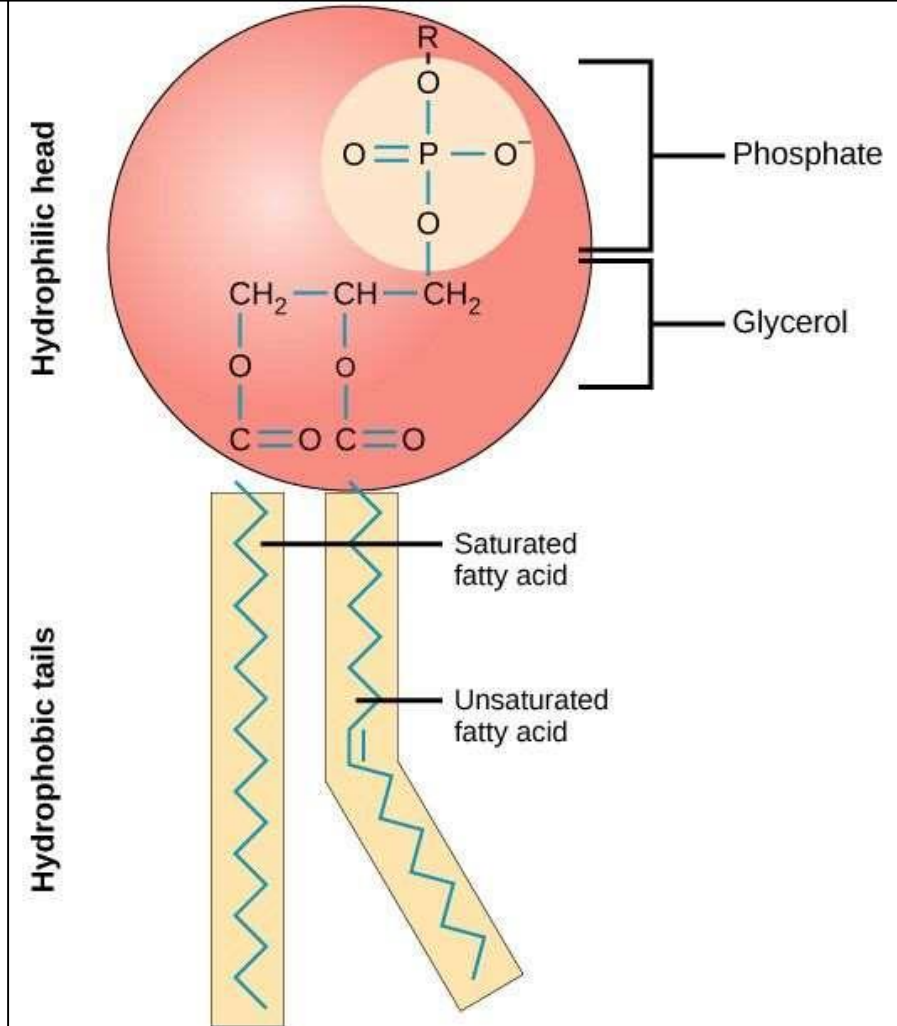
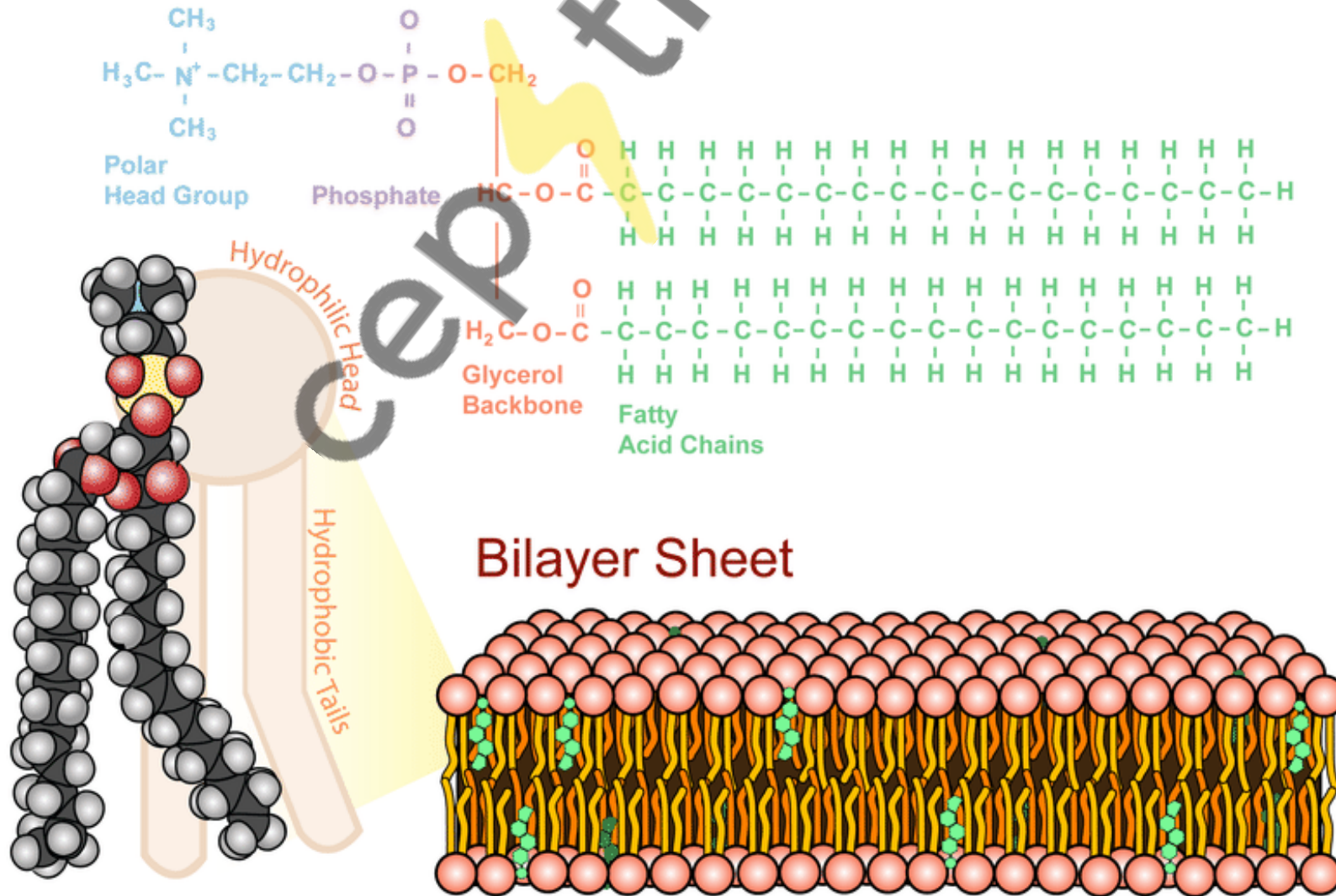
- ❑ **Membrane lipids**
- Lipids are water insoluble oily organic substances that are soluble in non-aqueous solvents such as chloroform, ether, etc.
- In addition to being structural components of membranes, lipids serve as fuel molecules.

Archaea vs Eubacteria vs Eukarya (cell membrane)



- ❑ Archaeol is one of the main core membrane lipids of archaea, one of the three domains of life.
- ❑ Standard archaeol is 2,3-di-O-phytanyl-sn-glycerol, with two phytanyl chains binding to the position of sn-2 and sn-3 of glycerol by ether bonds.
- ❑ Archaeol is generally composed by linking two phytanyl chains to the sn-2 and sn-3 positions of a glycerol molecule. The highly branched side chains are speculated to account for the very low permeability of archaeol-based membrane, which may be one of the key adaptations of archaea to extreme environment.
- ❑ One of the key features that distinguishes archaea from bacteria and eukarya is their membrane lipids, where archaeol plays an important role.
- ❑ Because of this, archaeol is also broadly used as a biomarker for ancient archaea.

Lipid bilayer



Lipid bilayer

Saturated Fat

meats, butter,
dairy products

solid at room
temperature

increase levels of
“**bad**” cholesterol
(low-density lipoprotein)

low-density lipoprotein
clogs arteries

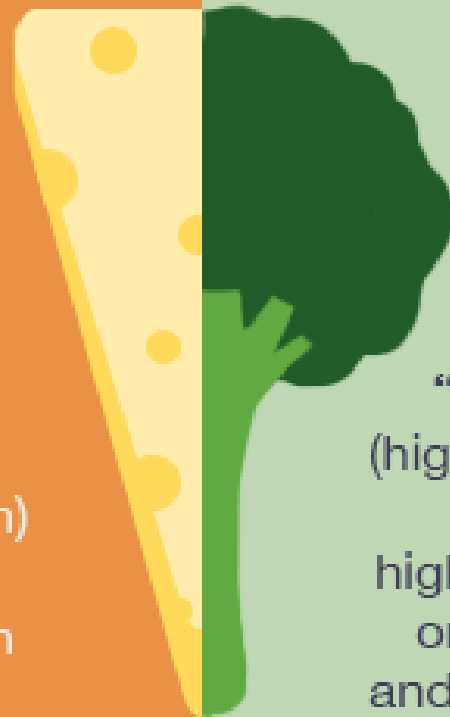
Unsaturated Fat

vegetable oils

liquid at room
temperature

increase levels of
“**good**” cholesterol
(high-density lipoprotein)

high-density lipoprotein,
or HDL, “grabs” LDL
and escorts it to the liver
where **LDL is broken down**
and eventually removed
from the body



Important Links to see Videos



- ❑ **Overview of Cell Membrane:**
- ❑ <https://www.youtube.com/watch?v=KGTcgShKJpw>
- ❑ <https://www.youtube.com/watch?v=fJfTDc3WzQ8>
- ❑ <https://www.youtube.com/watch?v=RT61MUjogRo>
- ❑ <https://www.youtube.com/watch?v=FYwB1EETakY>
- ❑ <https://www.youtube.com/watch?v=3k7Y6NcXe6s>

Organelles of the Eukaryotic Cells



- The Organelles are small, specialized structures in cells which perform specific functions.
- The cell organelles are of three types.

Those containing a, i) Double membrane, ii) Single membrane or iii) No membrane.

1. Double membrane bound cell organelles

1.1 Nucleus

1.2 Mitochondria

1.3 Chloroplast

2. Single membrane bound cell organelles

2.1 Endoplasmic reticulum

2.2 Golgi apparatus

2.3 Vacuole

2.4 Lysosomes and endosomes

2.5 Peroxisomes

3. Cell organelles with out the membrane

3.1 Ribosomes

3.2 Centrosomes (Centriole)

3.3 Cytoskeleton (Microtubules, actin filaments and intermediate filaments)

1.1 Nucleus

- The nucleus, the largest organelle in animal cells and is surrounded by **two membranes (outer and inner nuclear membrane)**, each one a phospholipid bilayer containing different types of proteins (**Figure 2**).
- The two nuclear membranes appear to fuse at **nuclear pores** (**Figure 2**), the ring-like complexes through which material moves between the nucleus and cytosol or cytoplasm.
- **Nuclear lamina** along the surface of the inner membrane is composed of proteins called **Lamins** which gives the shape to nucleus.
- In a growing or differentiating cell, the nucleus is metabolically active engaged in replicating DNA and synthesizing RNA (rRNA, tRNA and mRNA).

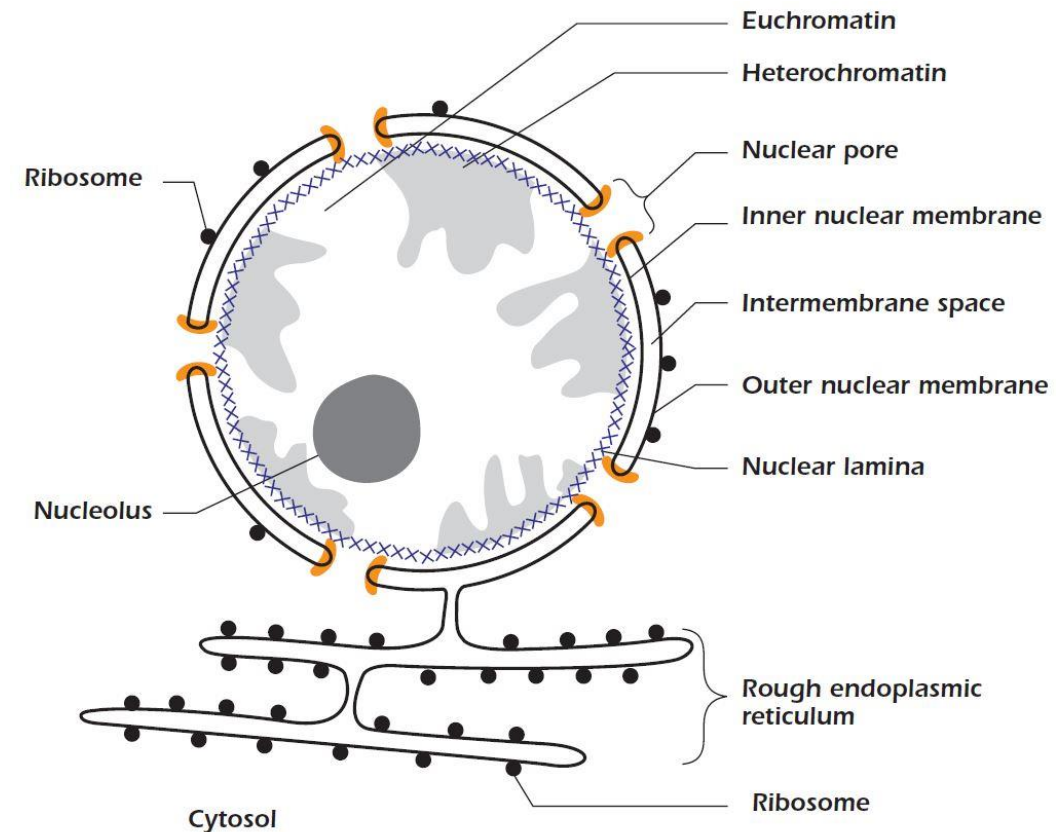


Figure 2. Nucleus and the relationship of its membranes with Endoplasmic Reticulum.

1.1 Nucleus

- In the electron microscope, the non-nucleolus region of the nucleus, called the **Nucleoplasm**, can be seen to have dark- and light-staining areas.
- The Dark areas, which are often associated with the nuclear membrane, contain condensed DNA (inactive for transcription), are called **Heterochromatin**.
- The Light areas containing the loose chromatin (DNA and protein) structure (active for transcription) are called **Euchromatin**.
- Most of the cell's **ribosomal RNA** is synthesized in the **Nucleolus**, present as a subcompartment in the Nucleus, that is not bound by any membrane.

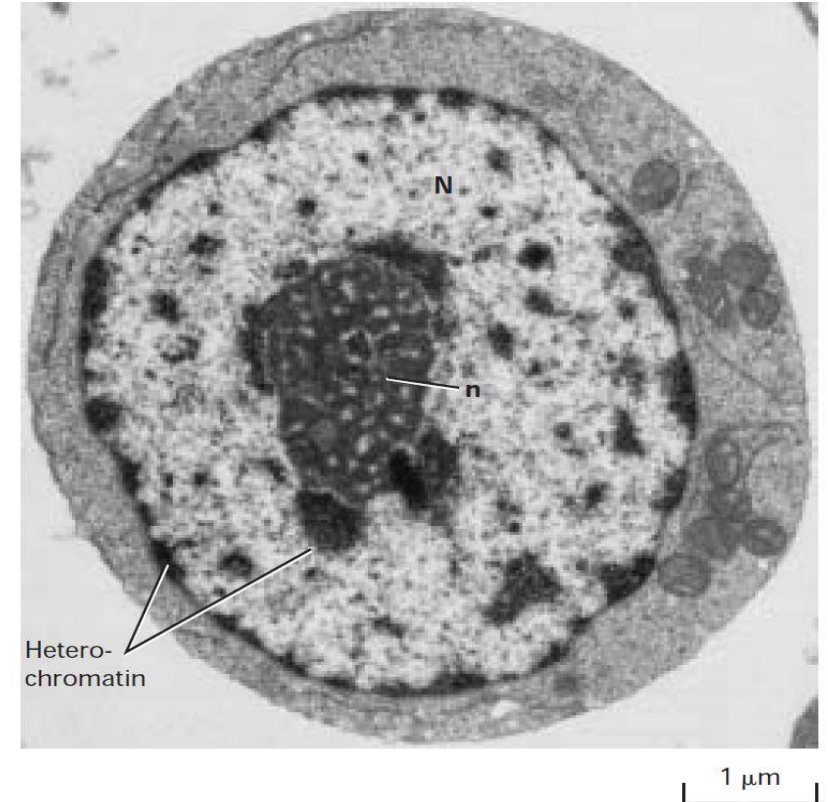
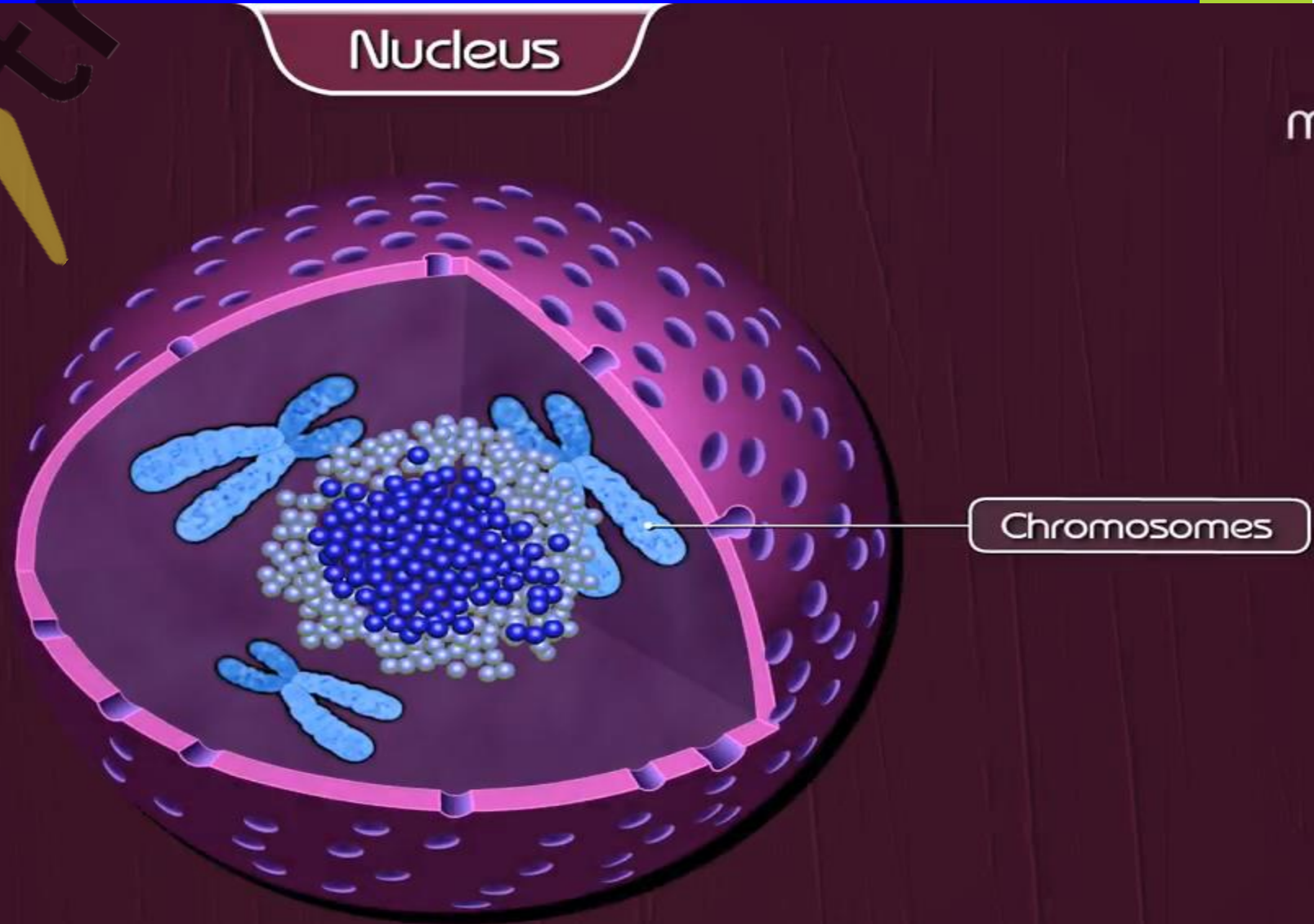


Figure 3. Electron micrograph of a thin section of a **Bone marrow** stem cell. **The nucleolus (n)** is a sub-compartment of nucleus (N) and is not surrounded by a membrane. **Dark stained areas in nucleus outside the nucleolus are heterochromatin.**

1.1 Nucleus



Important Links to see Videos



- ❑ **Nucleus:**
- ❑ <https://www.youtube.com/watch?v=UYj5aofYhpo>

1.2 Mitochondria: Structure

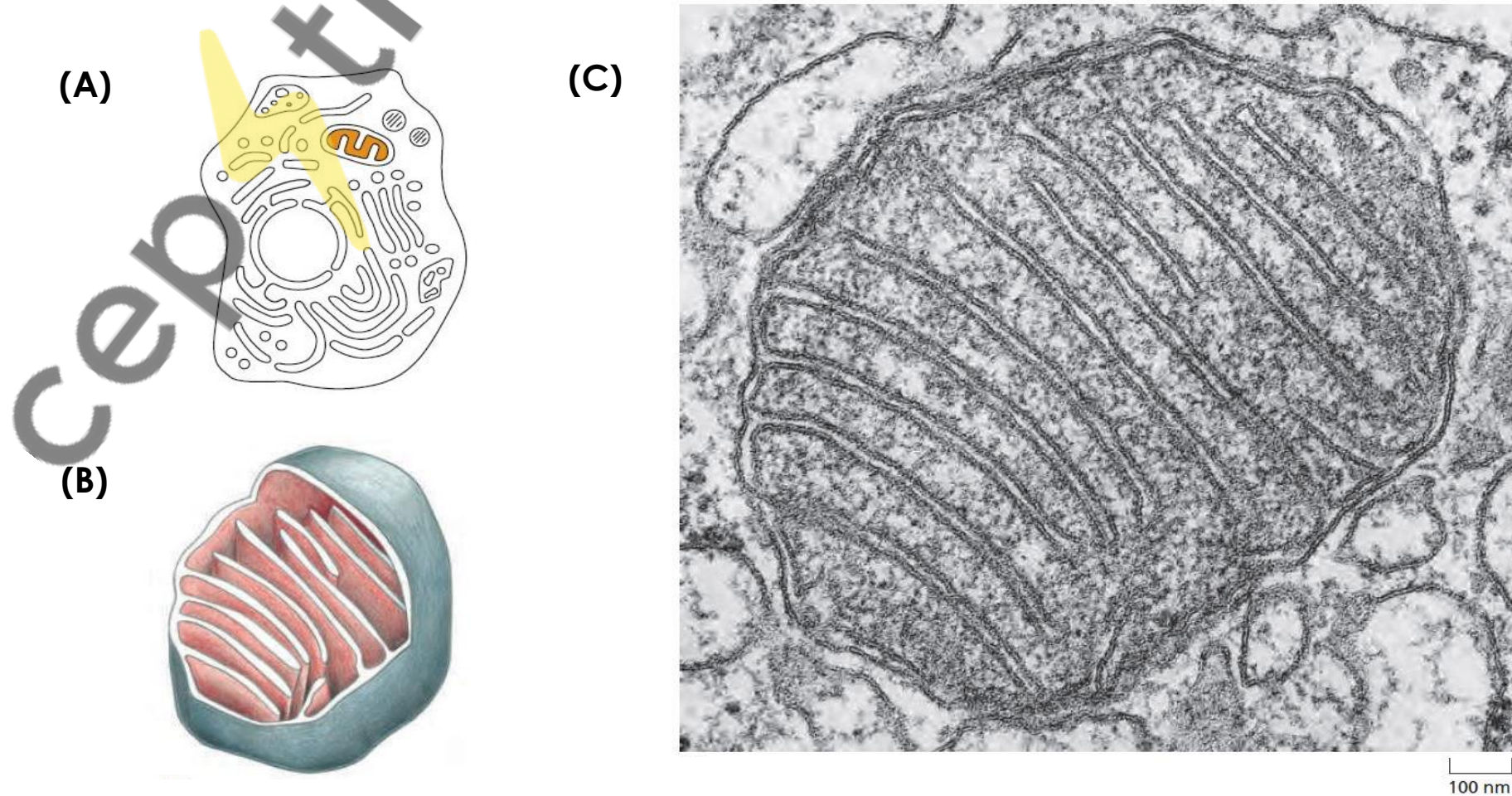


Figure 4. A) Schematic of Eukaryotic cell.

B) Drawing of a Mitochondrion with a part cut away, showing the 3-dimensional structure.

C) Cross-section of a Mitochondrion, as seen through image by Transmission Electron Microscope.

1.2 Mitochondria, The Power Plants of a cell

- Most eukaryotic cells contain many Mitochondria, which occupy up to 25% of the volume of the Cytoplasm.
- Mitochondrion (singular) is the main site for ATP synthesis during aerobic metabolism and is smaller in size than nucleus, vacuoles or chloroplasts.
- The two membranes (outer and inner) that bound a mitochondrion differ in composition and function.
- The surface area of the inner membrane is greatly increased by a large number of infoldings called **Cristae**, that protrude into the Central space or **Matrix** (Figure 5).

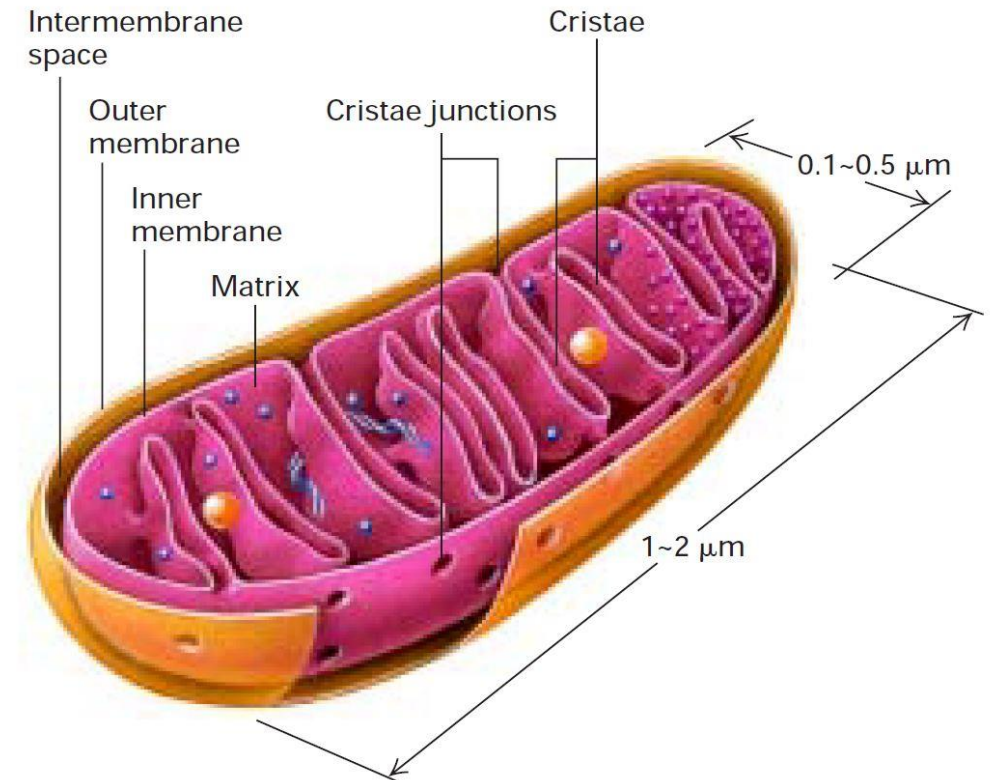
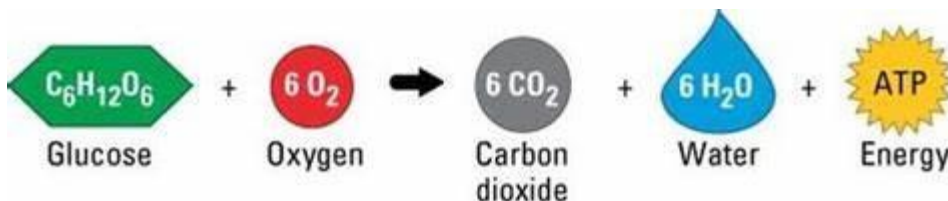


Figure 5. Schematic diagram of Mitochondrion showing membranes and compartments.

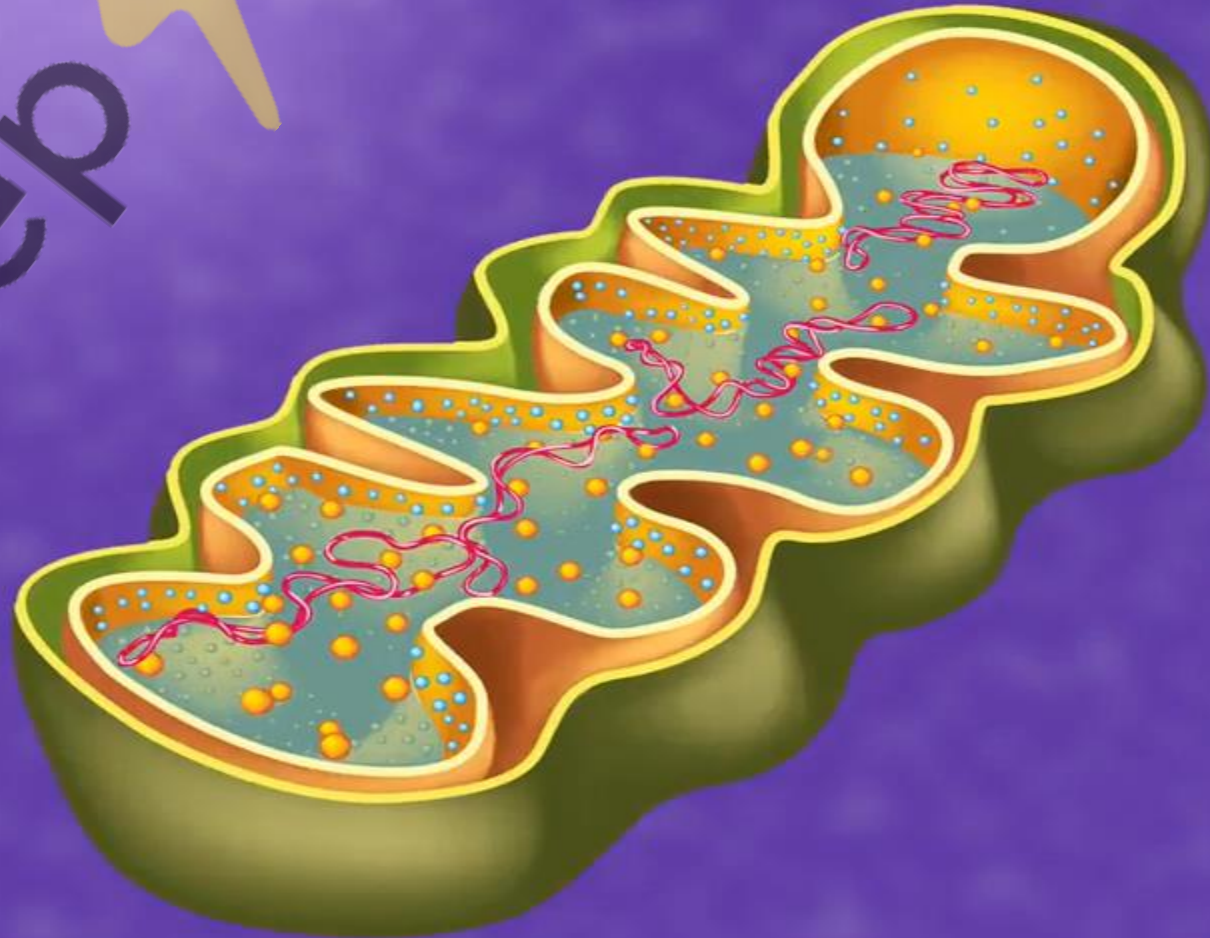
*Credit: Molecular Cell Biology-Lodish



Mitochondria

MITOCHONDRIA

STRUCTURE



MEDVIZZ

How Mitochondria Produce Energy?

Cepstrum



cell

CORTICAL STUDIOS

Important Links to see Videos



- ❑ **Mitochondria:**
- ❑ <https://www.youtube.com/watch?v=c4JsEBI9u6I>
- ❑ <https://www.youtube.com/watch?v=39HTpUG1MwQ>
- ❑ <https://www.youtube.com/watch?v=rdF3mnyS1p0>

1.3 Chloroplasts

Chloroplasts are the largest and most characteristic organelles in the cells of plants and green algae after the Vacuoles.

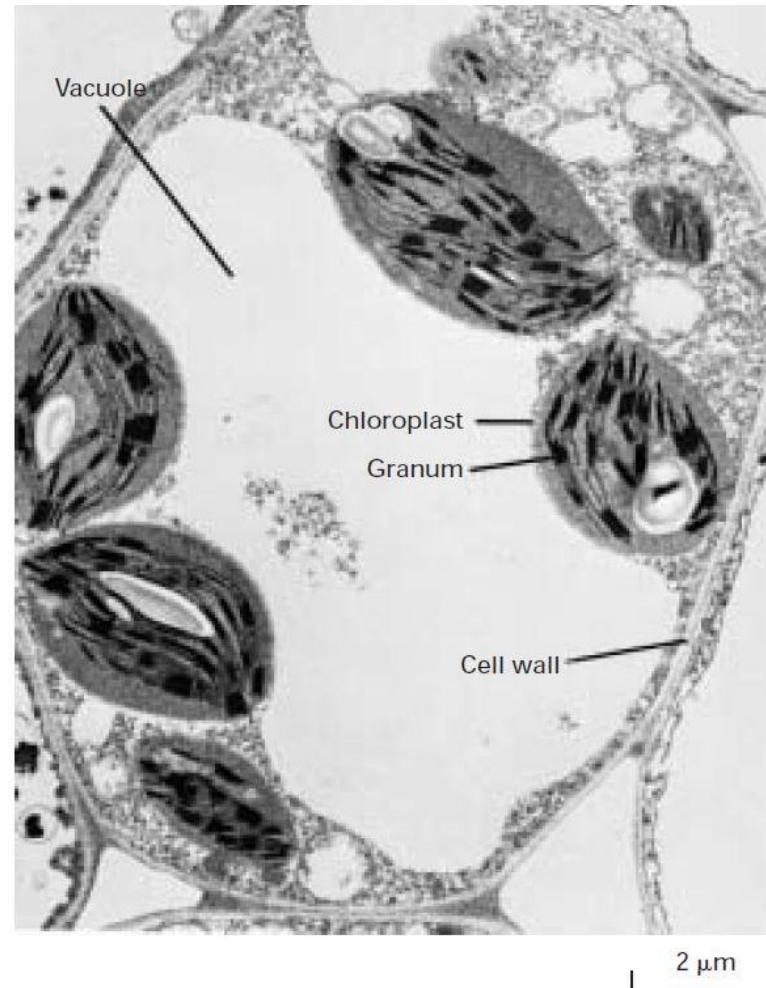


Figure 6. Chloroplast of Plant cell as seen by an image from Transmission Electron Microscope (TEM).

1.3 Chloroplasts

Chloroplasts are $\sim 10\text{ }\mu\text{m}$ long and $0.5 - 2.0\text{ }\mu\text{m}$ thick, but vary in size and shape in different cells, especially in algae.

Chloroplasts capture energy from sunlight in Plant cells and in single-celled Eukaryotes such as, Volvox (a Green alga).

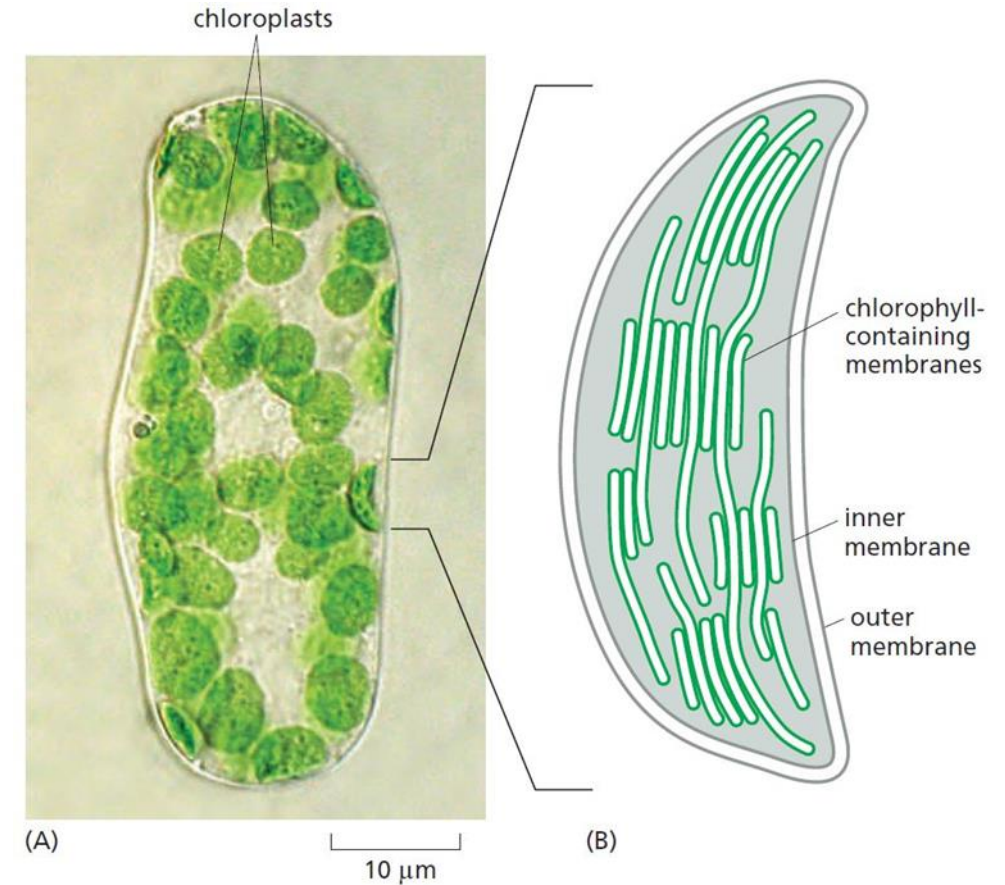


Figure 7. Chloroplasts surrounded by a double membrane (an outer and inner membrane).

1.3 Chloroplasts

- Chloroplast contains an extensive interconnected membrane bound sacs called **Thylakoids**.
- Thylakoids form stacks called **Grana** (singular Granum) and are embedded in a matrix called **Stroma**.
- Two Grana may be connected by a **Lamella**.
- The Thylakoid membranes contain green pigment (Chlorophyll) and other pigments that absorb light.

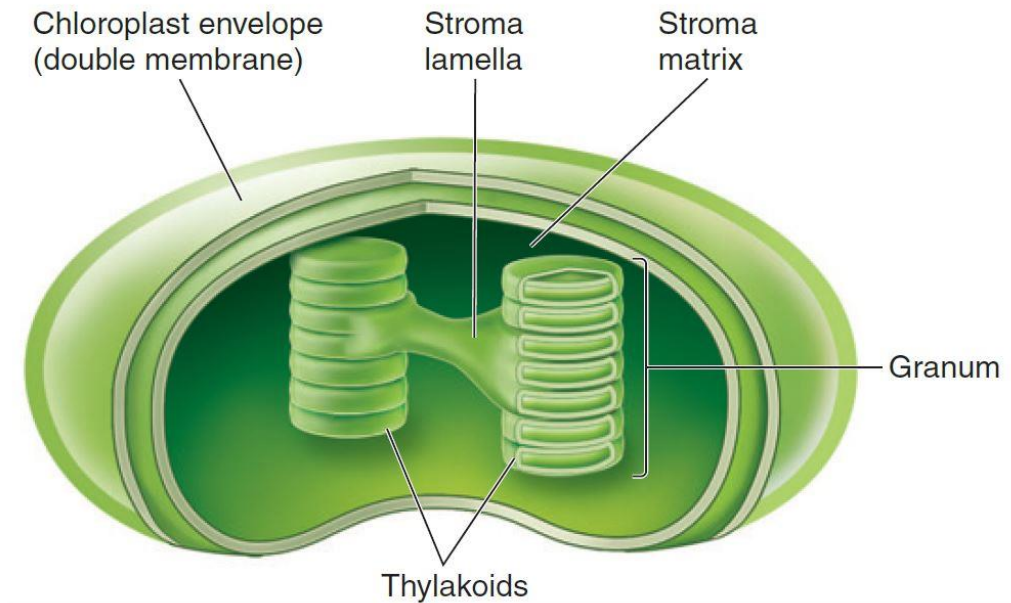


Figure 8. Chloroplast structure.

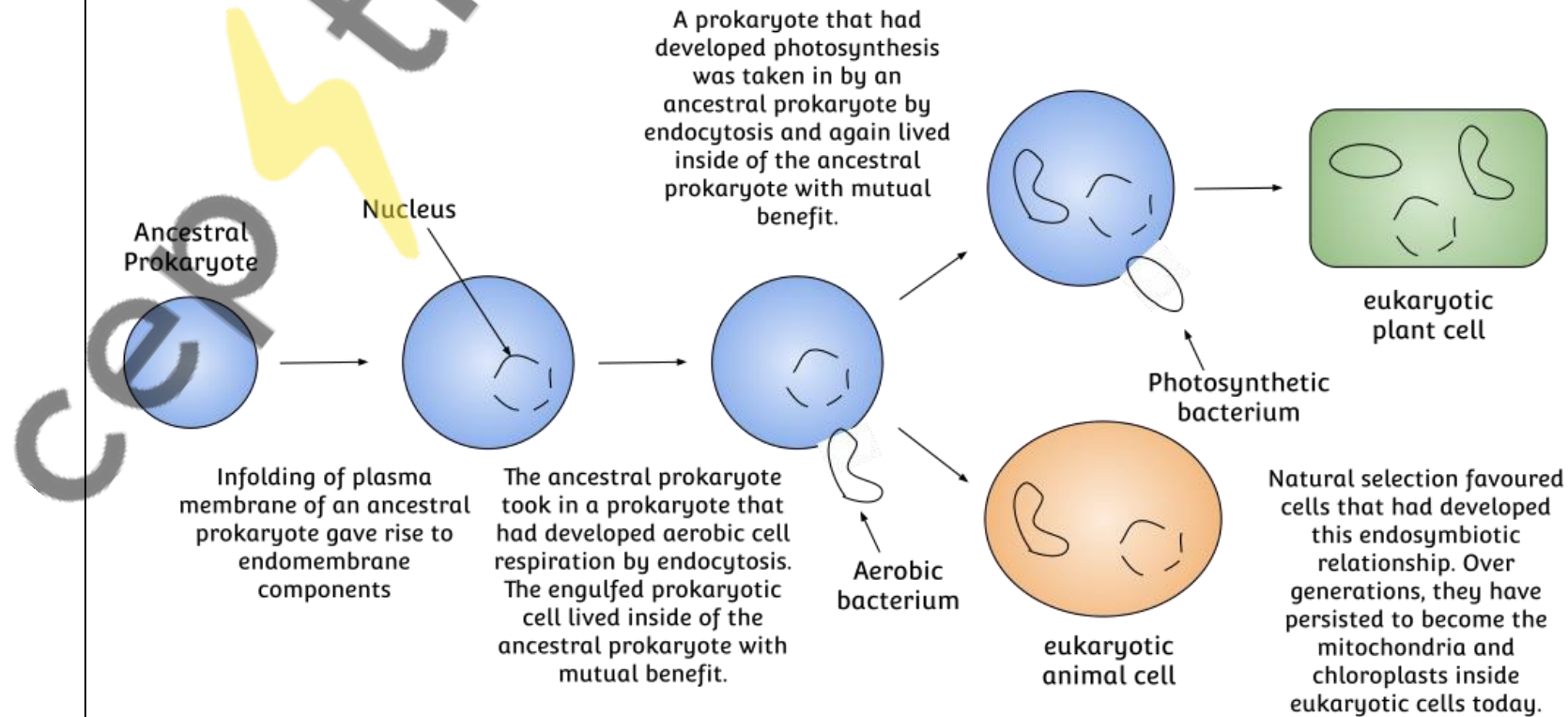
Cyanobacteria and Chloroplasts



- ❑ Cyanobacteria are globally widespread photosynthetic prokaryotes and are major contributors to global biogeochemical cycles. They are the only oxygenic photosynthetic prokaryotes, and prosper in diverse and extreme habitats. They are among the oldest organisms on Earth with fossil records dating back 3.5 billion years. Since then, cyanobacteria have been essential players in the Earth's ecosystems.
- ❑ Cyanobacteria are the first organisms known to have produced oxygen.
- ❑ By producing and releasing oxygen as a byproduct of photosynthesis, cyanobacteria are thought to have converted the early oxygen-poor, reducing atmosphere into an oxidizing one, causing the Great Oxidation Event and the "rusting of the Earth", which dramatically changed the composition of the Earth's life forms and led to the near-extinction of anaerobic organisms.
- ❑ Cyanobacteria use the energy of sunlight to drive photosynthesis, a process where the energy of light is used to synthesize organic compounds from carbon dioxide.
- ❑ Chloroplasts have many similarities with cyanobacteria, including a circular chromosome, prokaryotic-type ribosomes, and similar proteins in the photosynthetic reaction center.
- ❑ The endosymbiotic theory suggests that photosynthetic bacteria were acquired (by endocytosis) by early eukaryotic cells to form the first plant cells.
- ❑ Therefore, chloroplasts may be photosynthetic bacteria that adapted to life inside plant cells.
- ❑ Like mitochondria, chloroplasts still possess their own DNA, separate from the nuclear DNA of their plant host cells and the genes in this chloroplast DNA resemble those in cyanobacteria.



Endosymbiosis (Mitochondria and Chloroplasts)



Chloroplast and Mitochondrial Evidence

- They have double membranes
- They can only be produced by division of pre-existing mitochondria and chloroplasts
- They have their own DNA which is naked and circular
- They have ribosomes which are 70S in size

Organelles of the Eukaryotic Cells



- The Organelles are small, specialized structures in cells which perform specific functions.
- The cell organelles are of three types.

Those containing a, i) Double membrane, ii) Single membrane or iii) No membrane.

1. Double membrane bound cell organelles

1.1 Nucleus

1.2 Mitochondria

1.3 Chloroplast

2. Single membrane bound cell organelles

2.1 Endoplasmic reticulum

2.2 Golgi apparatus

2.3 Vacuole

2.4 Lysosomes and endosomes

2.5 Peroxisomes

3. Cell organelles with out the membrane

3.1 Ribosomes

3.2 Centrosomes (Centriole)

3.3 Cytoskeleton (Microtubules, actin filaments and intermediate filaments)

Single membrane Cell Organelles



2.1 Endoplasmic Reticulum (ER)

- Endoplasmic Reticulum (ER) is a network of membrane system constituting the channels and is separated from cytoplasm by a single membrane.
- The extensive network of closed, flattened membrane-bound sac like structures of ER are called **Cisternae**, whose interior is called **Cisternal space** or **Lumen**.
- ER is involved in Protein synthesis, Lipid metabolism and Calcium storage.

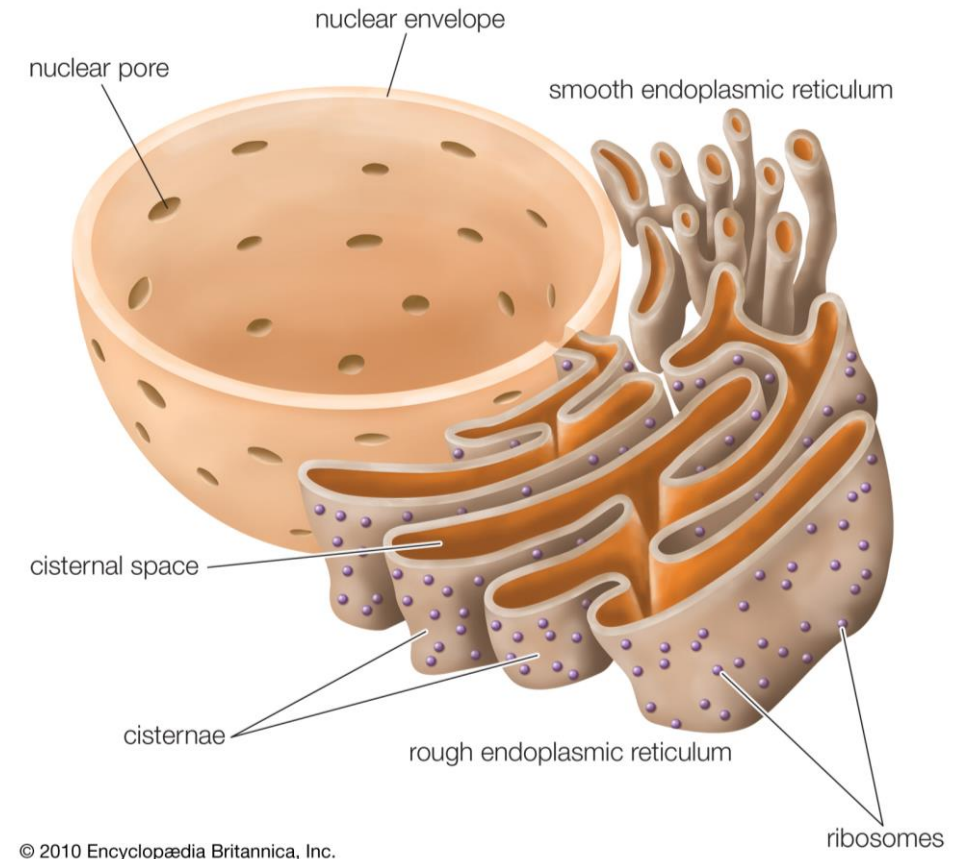
There are two types of Endoplasmic Reticulum

1. Smooth Endoplasmic Reticulum (SER).

The ER without the Ribosomes are called Smooth ER. The synthesis of fatty acids and phospholipids takes place in the smooth ER.

2. Rough Endoplasmic Reticulum (RER).

The ER attached with Ribosomes are called Rough ER. They synthesize membrane and organelle proteins and virtually all proteins required by the cell. They are involved in the synthesis, folding, modification and transport of proteins to rest of the cell.

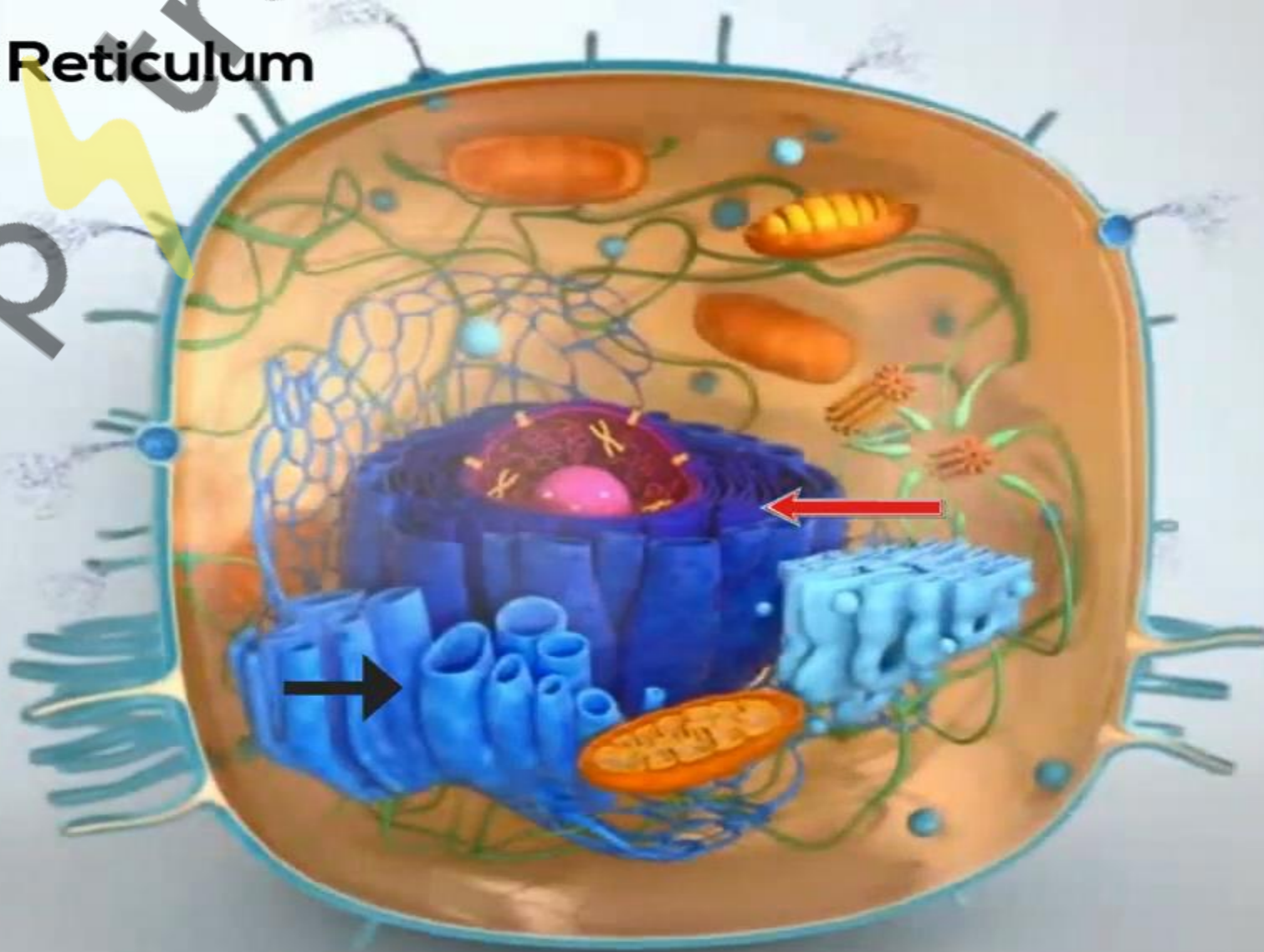


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**Fig. 1. Endoplasmic Reticulum (ER).
Smooth and Rough ER**

2.1 Endoplasmic Reticulum

Endoplasmic Reticulum



Important Links to see Videos



❑ **ER:**

❑ <https://www.youtube.com/watch?v=an7tpWR16mo>

2.2 Golgi Apparatus

Golgi apparatus, is named after its discoverer, Camillo Golgi (1906 Nobel prize winner).

- The Golgi apparatus (also called Golgi complex or Golgi body) is a membrane-bound organelle.
- It is made up of a series of flattened stacked pouches called **Cisternae**. It is located in the Cytoplasm next to the Endoplasmic Reticulum and near the Nucleus
- It helps in the processing and packaging of proteins and lipid molecules.
- Proteins after synthesis in Rough ER leaves the organelle within a small membrane bound **Vesicles** (which originate from Rough ER).
- These **Vesicles** act as transport vesicles which carry the proteins to **Golgi complex**.
- Stack of Golgi Cisternae has three defined regions
i) cis (face towards ER), ii) the medial, iii) the trans.

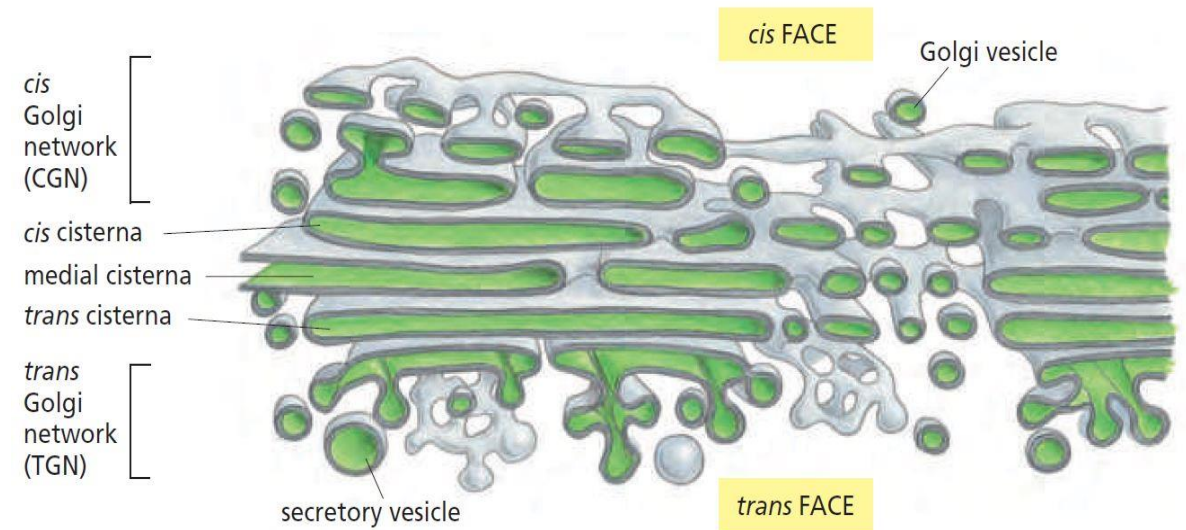
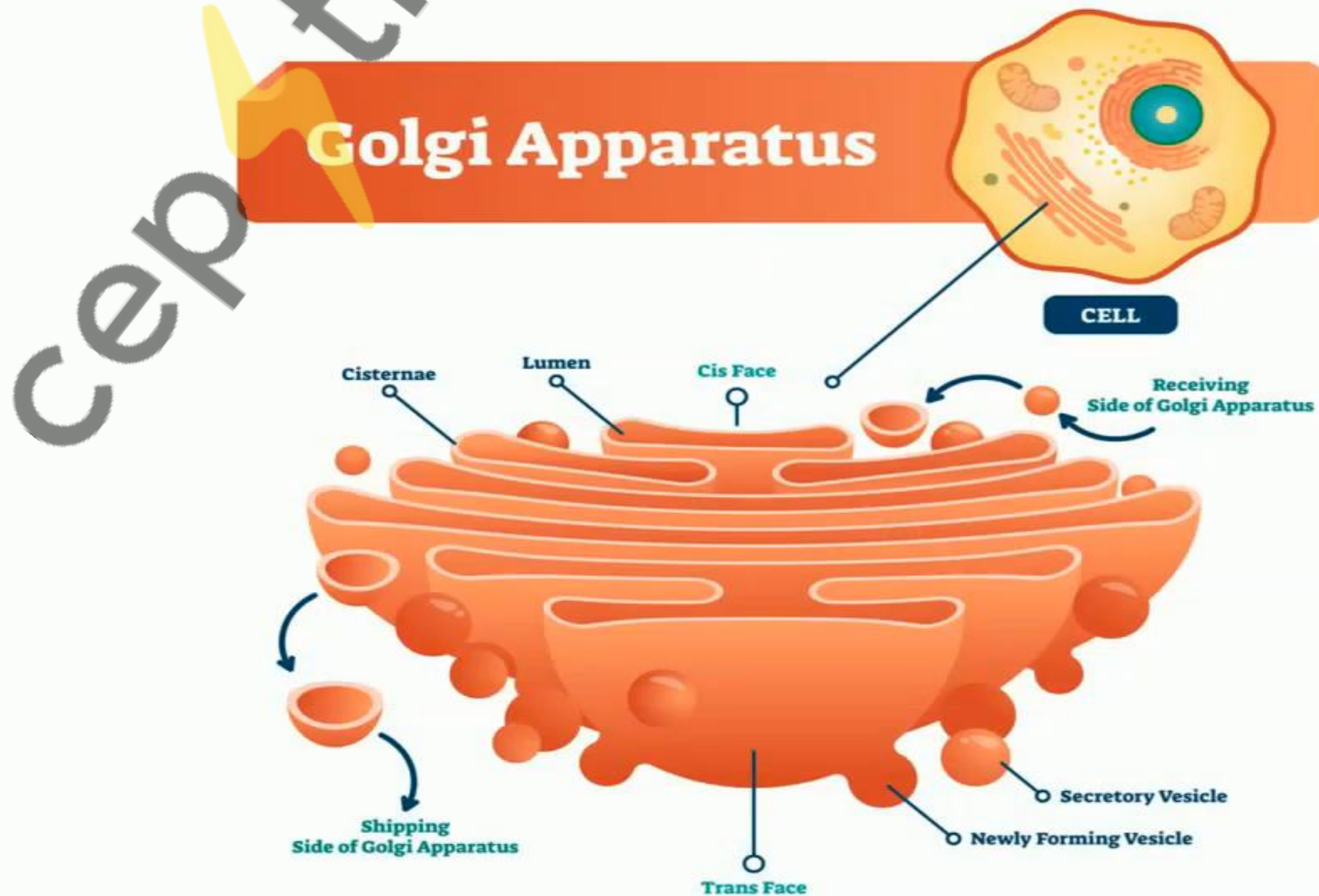


Fig. 3. The Golgi Apparatus.

2.2 Golgi Apparatus



Important Links to see Videos



- ❑ **Golgi Apparatus:**
- ❑ <https://www.youtube.com/watch?v=d0QljHf7REM>

2.3 Vacuole (Structure)

- **Vacuoles are membrane-bound organelles found in both animal and plant cells.**
- **The vacuoles in plant cells are larger than those in the animal cells. The plant vacuole can occupy more than 80% of the volume of the cell.**

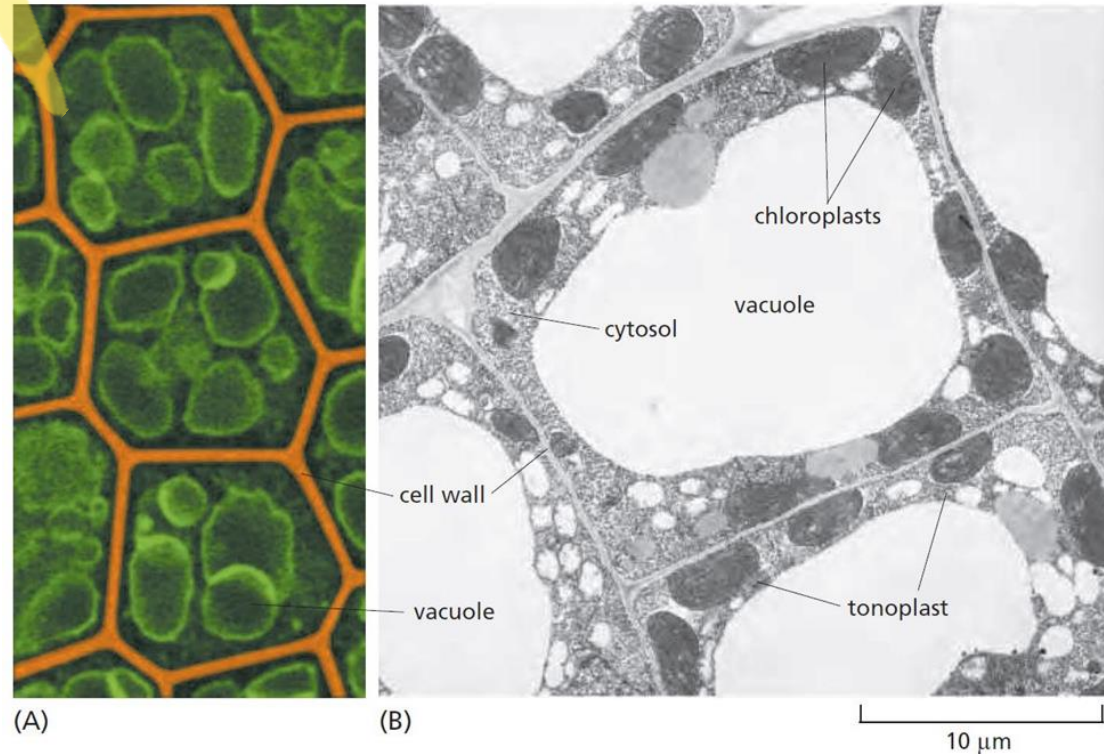


Fig. 4. Plant Cell Vacuole.

A) Confocal Microscopic image of Arabidopsis (Plant, mustard family) cells showing **Aquaporin (a yellow fluorescent protein)** in Vacuole membrane (**green**). The cell walls are shown in **orange** colour. Each cell contains several large vacuoles.

B) Electron micrograph of Tobacco Leaf cells showing an Enormous vacuole pressed against the cell wall.

2.3 Vacuole (Function)

The important functions of Vacuole are

1. **Storage:** A vacuole stores salts, minerals, pigments and proteins within the cell.
2. **Turgor Pressure:** The vacuoles are completely filled with water and exert force on the cell wall. This is known as **Turgor Pressure**. It provides shape to the cell and helps to withstand the extreme conditions.
3. **Endocytosis and Exocytosis:** The substances are taken in by a vacuole through **Endocytosis** and excreted through **Exocytosis**.
4. **Lumen** (the 3-D interior space) of a vacuole in plant cells contains degradative enzymes. Therefore, plant vacuoles also have a degradative function, similar to that of Lysosomes in animal cells.

2.4 Lysosomes and endosomes



- Lysosomes are membrane-bound cell organelles, which contain digestive enzymes due to which they are also called as **Cell Stomachs**.
- The major function of Lysosomes is digestion and removal of waste.
- Lysosomes are plentiful in cells (**such as macrophages**) involved in digesting and destroying other cells.
- Lysosomes contain enzymes such as
 - i) **Nucleases** which degrade RNA and DNA into their respective mononucleotides.
 - ii) **Proteases** which degrade a variety of proteins and peptides.
- Isolating such potentially destructive enzymes in the Lysosome, protects the cytosol.

2.4 Lysosomes and Endosomes

Primary Lysosomes – They are small vesicles produced by Golgi apparatus.

Secondary Lysosomes – They are formed when Primary Lysosomes fuse with **Endosomes**.

Lysosomes digest the material by 3 pathways: Endocytosis, Phagocytosis and Autophagy

- 1. Endocytosis** is the process of actively transporting molecules **into the cell** by engulfing it with its membrane.
- 2. Phagocytosis** is a specific form of endocytosis, by which the cells internalise external solid matter, including microbial pathogens.
- 3. Autophagy** (Self eating) is a self-digesting mechanism responsible for removal of damaged organelles or other cellular components by lysosome.

Endosomes are membrane-bound vesicles involved in

- Sorting and delivering of the internalized material (such as nutrient uptake) from the cell surface and
- the transport of material from Golgi complex to Lysosome or Vacuole.

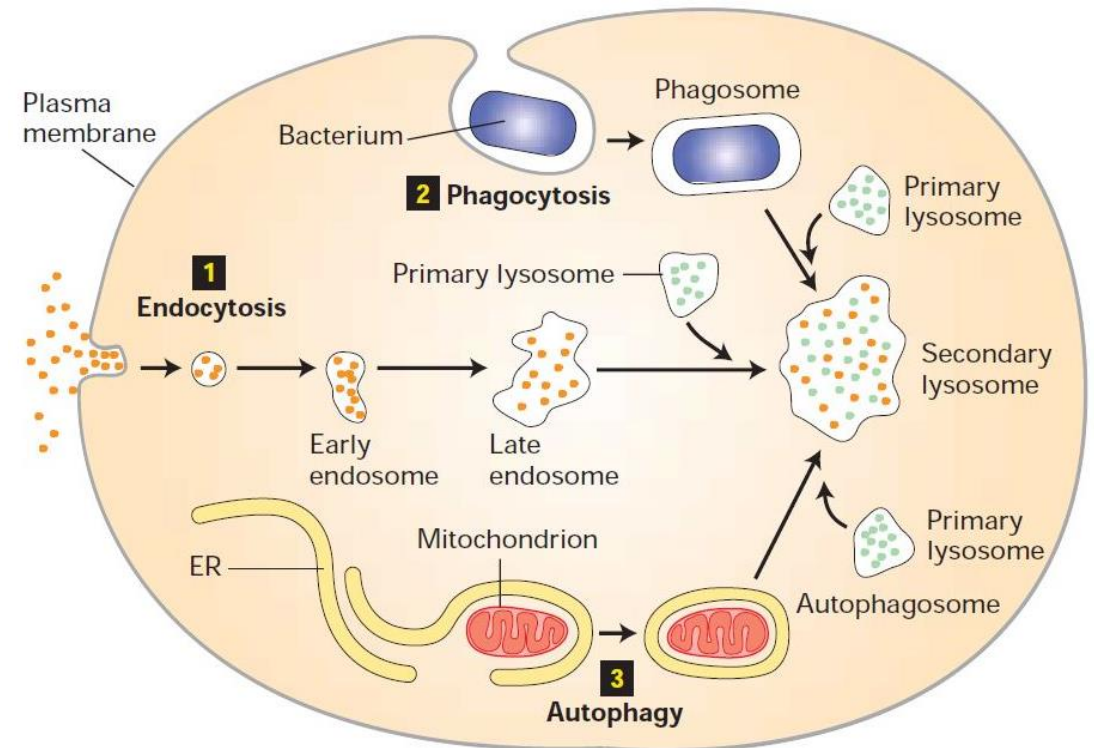
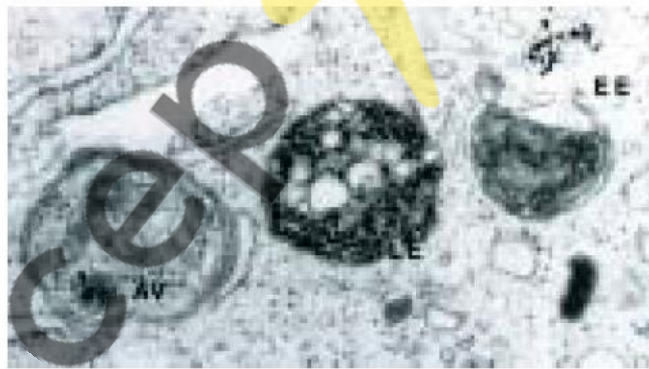


Fig. 5. Schematic view of Three pathways by which materials are moved to Lysosomes.

**Credit: Molecular Cell Biology-Lodish*

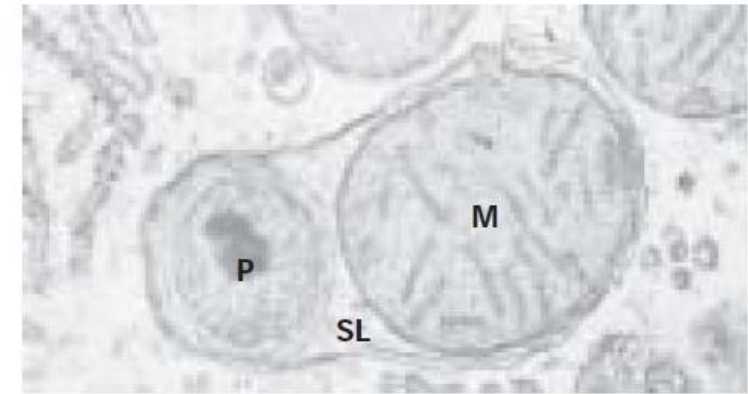
2.4 Lysosomes and Endosomes



0.1 μm

Fig. 6A. Electron micrograph of a section of a **mammalian cell** with internalized gold particles coated with the egg protein, Ovalbumin.

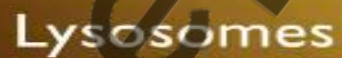
Gold-labeled Ovalbumin (black spots) is found in **Early Endosomes (EE)** and **Late Endosomes (LE)**, but very little is present in **Autophagosomes (AV)**.



1 μm

Fig. 6B. Electron micrograph of a section of **Rat Liver cell** showing a **Secondary Lysosome** containing a **Mitochondrion (M)** and a **Peroxisome (P)**.

**Credit: Molecular Cell Biology-Lodish*



Hydrolytic enzymes

Important Links to see Videos



- ❑ **Lysosomes:**
- ❑ https://www.youtube.com/watch?v=cA-Ou_t2sag

2.5 Peroxisomes (Structure)



- Peroxisomes are small, **single membrane-enclosed organelles** which contain enzymes involved in a variety of metabolic reactions.
- All animal cells (except Erythrocytes, the Red Blood cells) and many plant cells contain peroxisomes, a class of roughly spherical organelles, 0.2–1.0 μm in diameter.

2.5 Peroxisomes (Structure)

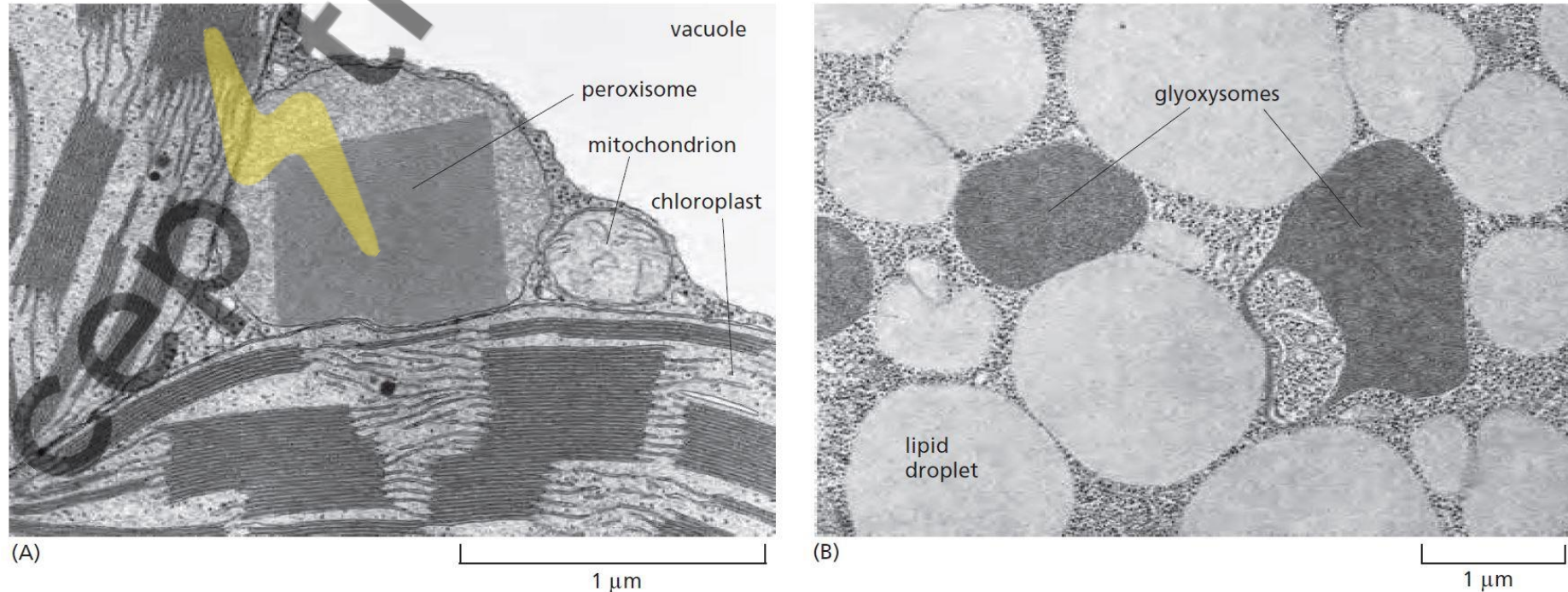


Fig. 1. Electron micrographs of two types of peroxisomes found in plant cells.

A. A peroxisome with a paracrystalline core in a Tobacco Leaf Mesophyll cell.

B. Peroxisomes present in a fat-storing cotyledon cell of a Tomato seed 4 days after germination. Where Peroxisomes convert the fatty acids stored in seed lipids into sugars (called **Glyoxysomes) needed for the growth of young plant.**

**Credit: Molecular Biology of the Cell - Alberts*

2.5 Peroxisomes (Functions)

- Peroxisomes contain several **Oxidase** enzymes, which use molecular oxygen (O_2) to oxidize organic substances (designated as R here), leading to the formation of **Hydrogen peroxide** (H_2O_2).



- When excess H_2O_2 accumulates in the cell, another enzyme, **Catalase** converts it into H_2O by the reaction.



- In humans, peroxisomes are primarily associated with **Lipid metabolism**, While in photosynthetic plant tissues, their most prominent role is in **Photorespiration**.
- Peroxisomes have the ability to adapt to the changing conditions.
 - e.g. Yeasts grown on sugar, have few small peroxisomes, but when they are grown on methanol, numerous large peroxisomes are formed that oxidize methanol.

Cell organelles without membrane



3.1 Ribosomes

- A ribosome is a complex molecular machine found inside the living cells that produces proteins from amino acids during the process called **Translation or Protein Synthesis**.
- Ribosomes are found in both Prokaryotic and Eukaryotic cells.
- Ribosomes typically contain about 80% of the total cellular RNA. They are complex structures, each consisting of more than 50 different proteins and RNA molecules (Table 1).
- A functional ribosome consists of two subunits, a Large Ribosomal Subunit and a Small Ribosomal Subunit.
- The size of Ribosomes and their RNA components are given in **Svedberg (S)** Units (a measure of how rapidly it sediments in a centrifugal field).

Table 1. Composition of ribosomes in Bacterial and Eukaryotic cells.

Cell Type	Ribosome Size	Subunit	rRNA Component	Proteins
Bacterial	70S	Large (50S)	23S (2900 nucleotides), 5S (120 nucleotides)	31
		Small (30S)	16S (1500 nucleotides)	21
Eukaryotic	80S	Large (60S)	28S (4700 nucleotides), 5.8S (160 nucleotides), 5S (120 nucleotides)	49
		Small (40S)	18S (1900 nucleotides)	33

It is important to note that Svedberg units are not additive.

3.1 Structure of Ribosome

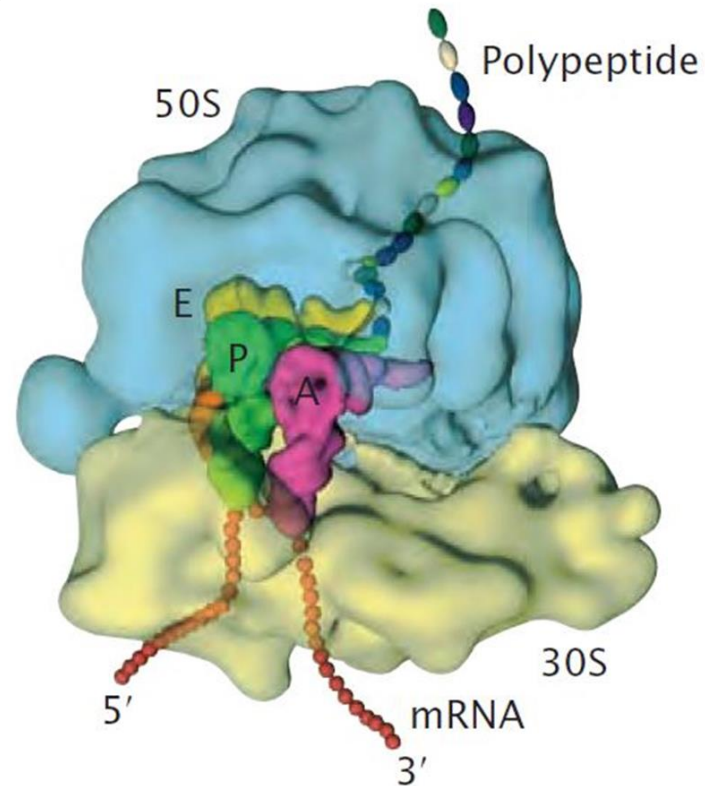


Fig. 2. Structure of the Bacterial Ribosome. Low-resolution model of Ribosome, showing the A (Amino acid binding), P (peptide and also tRNA binding site) and E (the exit site) where the growing polypeptide chain reside.

3.2 The Centrosome (Structure)

Centrosomes are made up of **two, Barrel-shaped clusters of Microtubules** called “**Centrioles**” and a complex of proteins which help in the formation of additional microtubules.

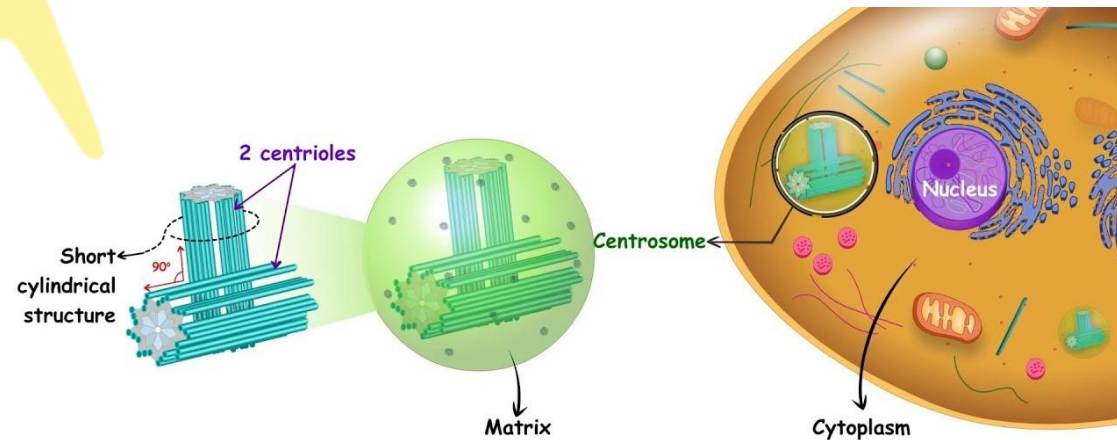


Fig. 3. Centrosome

Centrosomes are not found in higher plants, but it can be found in some lower plants.
e.g. Mosses and Ferns.



Fig. 4. a) Moss



b) Fern

3.2 The Centrosome (Function)



- Centrosomes play an important role during **cell division** as the organizers of the microtubules that **pull** sister chromatids apart, ensuring that each daughter cell gets a full compliment of the parent cell's DNA.
- Apart from cell division, they are also involved in the formation of **Cilia and Flagella** and thus contribute to the cell movement.
- Centrosomes are organelles that serve as **Microtubule-Organizing Centers (MTOC)**.

3.3 The Cytoskeleton: Components, Structure & Functions

- The cytoskeleton is a network of filaments and tubules that extends throughout the cell, through the Cytoplasm.
- It is found in all cells, though the proteins that it is made up of, vary among the organisms.
- Three types of filaments compose the Cytoskeleton:
 1. **Actin filaments**, also called microfilaments, provides mechanical support, determines cell shape and allows the movement of the cell surface.
 2. **Intermediate filaments** (IFs), their function is purely structural. They bear tension, thus maintaining the shape of the cell, and anchor the nucleus and other organelles in place.
 3. **Microtubules**, play important roles in cell movement, cell division and cellular transport.

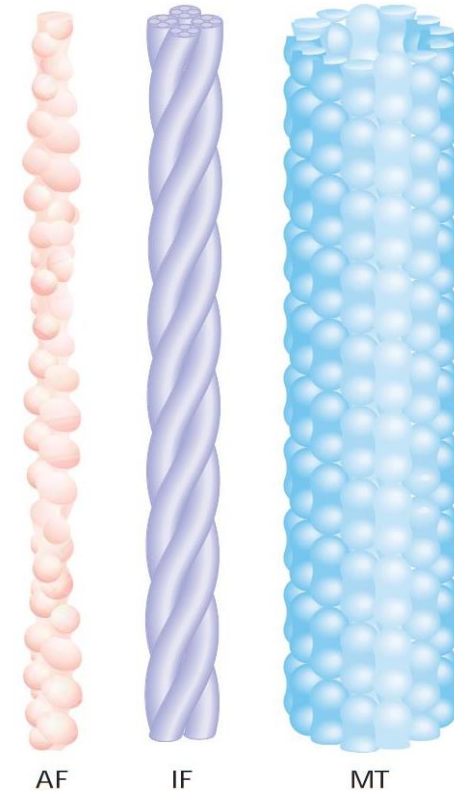


Fig. 5. Three types of filaments forming Cytoskeleton.

Actin filament (AF),
Intermediate filament (IF),
Microtubule (MT).

3.3 The Cytoskeleton: Components and Structure

- 1. Microfilaments:** They are 8-9 nm in diameter and have a twisted two-stranded structure. They are composed of the Actin protein.
- 2. Intermediate filaments:** are about 8-12 nm wide. They are called intermediate, because their size is in-between the sizes of microfilaments and microtubules. They are made up of different proteins, such as Keratin Vimentin, Desmin and Lamin.
- 3. Microtubules:** are **Hollow tube-like structures**, 24 nm in diameter. They are made up of **Tubulin** protein.

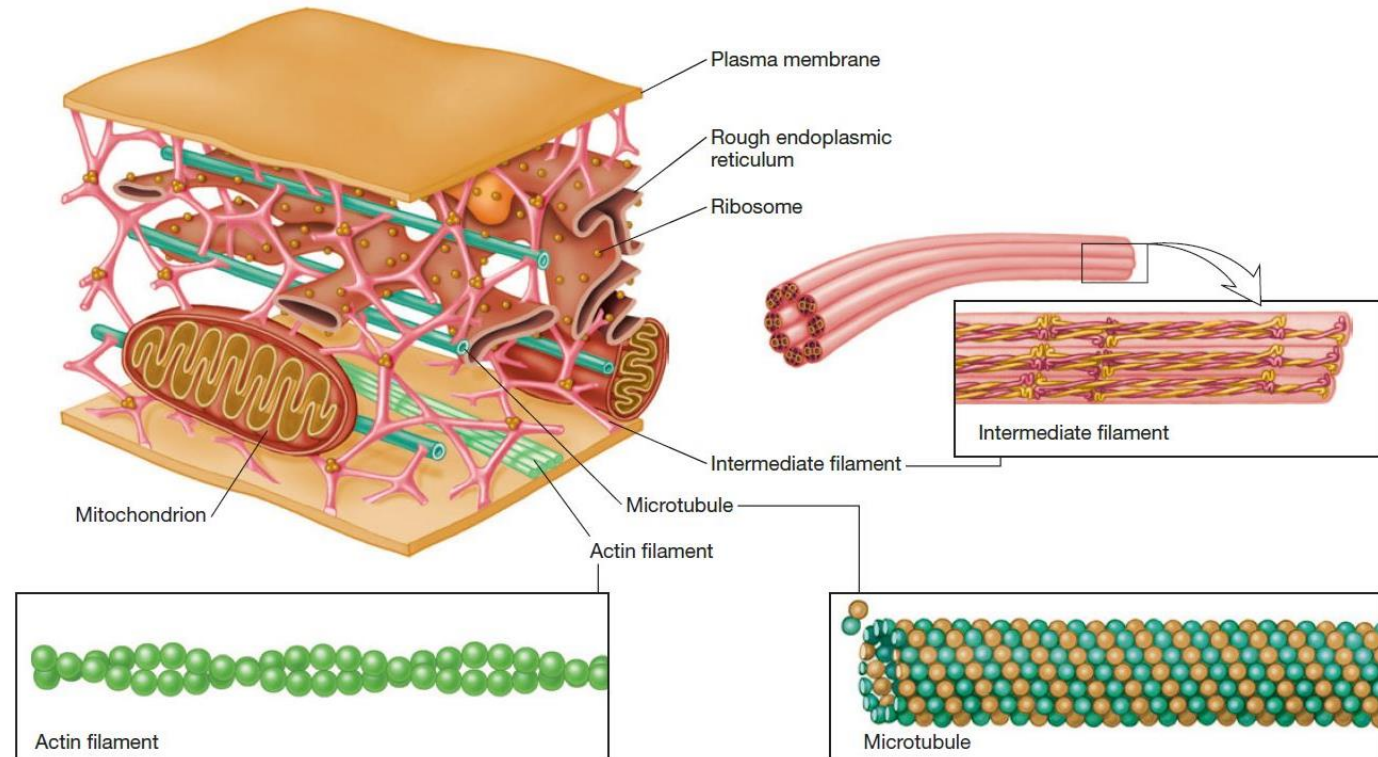
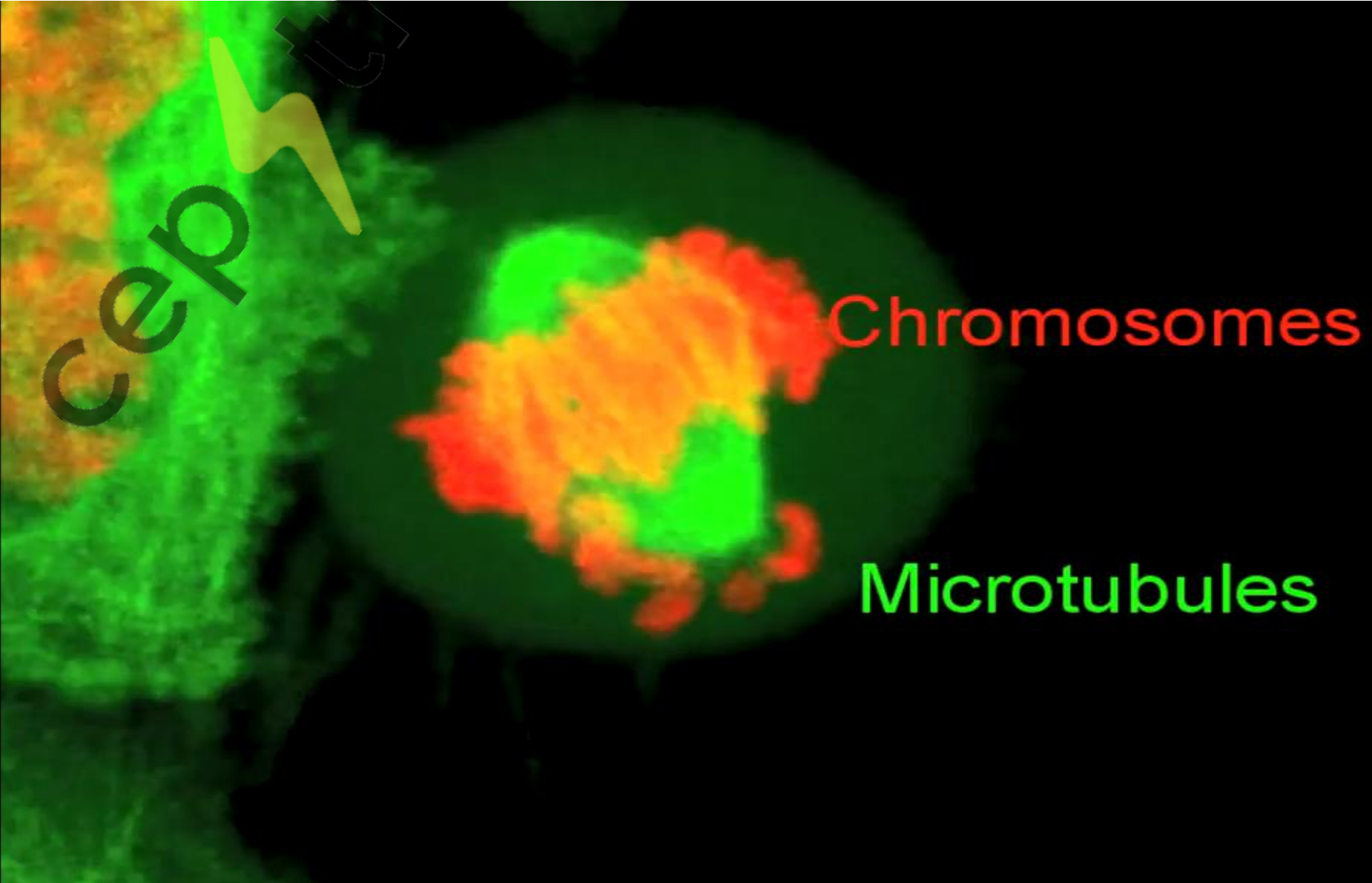


Fig. 6. The Eukaryotic Cytoskeleton.

Cytoskeleton



Important Links to see Videos



- ❑ **Cytoskeleton:**
- ❑ <https://www.youtube.com/watch?v=YTv9ItGd050>

Various organelles in a human cell and its major function(s)



- 1. Cell membrane** – (is a membrane that separates the interior of all cells from the outside environment (the extracellular space) and protects the cell from its environment; acts as a barrier for cell contents. It consists of double phospholipid layer and monolayer of protein scattered around phospholipid layer. The other components in plasma membrane are cholesterol and glycoproteins).
- 2. Cytosol** – (fluid that contains organelles; represents the material outside the nucleus and inside the cell membrane)
- 3. Cytoskeleton** – (a network of protein structures that extend throughout the cytoplasm; provides the cell with an internal framework (e.g. microfilaments and microtubules))
- 4. Nucleus** – (control center of the cell; contains genetic material; It is the center of the cell because it contains genetic material (DNA). It consists of three main regions: the nuclear membrane, the nucleolus and chromatin.
Nuclear membrane: Nuclear membrane serves as a barrier of nucleus. It consists of a double phospholipid membrane and contains nuclear pores that allow for the exchange of material with the rest of the cell.
Chromatin: It is composed of DNA and protein scattered throughout the nucleus. Chromatin condenses to form chromosomes when the cell divides)
- 5. Nucleolus** – (functions as site of ribosome production; ribosomes then migrate to the cytoplasm through nuclear pores)
- 6. Ribosome** (little dots) – sites of protein synthesis in the cell.

Various organelles in a human cell and its major function(s)



7. **Rough endoplasmic reticulum** – (carry ribosomes that represent sites of protein synthesis)
8. **Smooth endoplasmic reticulum** – (function in cholesterol synthesis and breakdown, fat metabolism, and detoxification of drugs)
9. **Golgi apparatus** (or "Golgi body") – (modifies and packages proteins, secrete vesicles, plasma membrane components and lysosomes)
10. **Mitochondrion** – (powerhouse of the cell; generate ATP for cellular activities)
11. **Lysosomes** – (suicidal bag of the cell; contain enzymes that digest non-usable materials within the cell)
12. **Peroxisomes** – (detoxify harmful substances and break down free radicals; involved in catabolism of very long chain fatty acids,)
13. **Centrioles** – (involved in the organization of mitotic spindle fibers during cell division and in the completion of cytokinesis)
14. **Centrosomes** – (serves as the main microtubule organizing center (MTOC) of the animal cell, as well as a regulator of cell-cycle progression; are composed of two centrioles arranged at right-angles to each other)
15. **Vesicles** – (involved in metabolism, transport, buoyancy control, and temporary storage of food and enzymes)

Important Links to see Videos



☐ Cells:

- ☐ <https://www.youtube.com/watch?v=URUJD5NEXC8>
- ☐ <https://www.youtube.com/watch?v=FYwB1EETakY>
- ☐ <https://www.youtube.com/watch?v=zZtcMBTQaS4>
- ☐ <https://www.youtube.com/watch?v=Pxujitlv8wc>

☐ DNA to protein:

- ☐ <https://www.youtube.com/watch?v=gG7uCskUOrA>

BT101

Introductory Biology

(Lecture 8)

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Assistant Professor
Department of Biosciences and Bioengineering
IIT Guwahati
Guwahati

Introduction



Evolution of life:

- ☐ Origin of Life
- ☐ Darwin's concepts of evolution
- ☐ Biodiversity

Cell, the structural and functional unit of life:

- ☐ Three domains of life
- ☐ Cell types, cell organelles and structure
- ☐ Basic biomolecules of cell

Biomolecules or Biological Molecules



- ❑ **Biomolecules (also called biological molecules), are substances that are produced by the cells and living organisms.**
- ❑ **Biomolecules are of wide range of size and structures and perform a vast array of functions.**
- ❑ **The four major types of biomolecules are:**
 - 1) Nucleic acids**
 - 2) Proteins**
 - 3) Lipids**
 - 4) Carbohydrates**

Biomolecules or Biological Molecules



- ❑ The four major types of biomolecules are carbohydrates, lipids, nucleic acids, and proteins.
- ❑ Among biomolecules, nucleic acids, namely DNA and RNA, have the unique function of storing an organism's genetic code—the sequence of nucleotides that determines the amino acid sequence of proteins, which are of critical importance to life on Earth.
- ❑ There are 20 different amino acids that can occur within a protein; the order in which they occur plays a fundamental role in determining protein structure and function. Proteins themselves are major structural elements of cells. They also serve as transporters, moving nutrients and other molecules in and out of cells, and as enzymes and catalysts for the vast majority of chemical reactions that take place in living organisms. Proteins also form antibodies and hormones, and they influence gene activity.
- ❑ Likewise, carbohydrates, which are made up primarily of molecules containing atoms of carbon, hydrogen, and oxygen, are essential energy sources and structural components of all life, and they are among the most abundant biomolecules on Earth. They are built from four types of sugar units—monosaccharides, disaccharides, oligosaccharides, and polysaccharides.
- ❑ Lipids, another key biomolecule of living organisms, fulfill a variety of roles, including serving as a source of stored energy and acting as chemical messengers. They also form membranes, which separate cells from their environments and compartmentalize the cell interior, giving rise to organelles, such as the nucleus and the mitochondrion, in higher (more complex) organisms.
- ❑ All biomolecules share in common a fundamental relationship between structure and function, which is influenced by factors such as the environment in which a given biomolecule occurs.
- ❑ Lipids, for example, are hydrophobic (“water-fearing”); in water, many spontaneously arrange themselves in such a way that the hydrophobic ends of the molecules are protected from the water, while the hydrophilic ends are exposed to the water. This arrangement gives rise to lipid bilayers, or two layers of phospholipid molecules, which form the membranes of cells and organelles.
- ❑ In another example, DNA, which is a very long molecule—in humans, the combined length of all the DNA molecules in a single cell stretched end to end would be about 1.8 metres (6 feet), whereas the cell nucleus is about 6 μm (6×10^{-6} metre) in diameter—has a highly flexible helical structure that allows the molecule to become tightly coiled and looped. This structural feature plays a key role in enabling DNA to fit in the cell nucleus, where it carries out its function in coding genetic traits.

Nucleic Acids

- A nucleic acid is a polymeric macromolecule made up of repeated units of monomers called '**Nucleotides**'.
- Nucleic acids serve as repositories and transmitters of genetic information.
- There are two types of Nucleic acids
 - I. Deoxyribonucleic acid (DNA)
 - II. Ribonucleic acid (RNA).
- A **nucleotide** has three components:
 1. A Nitrogenous base (Purine and Pyrimidine)
 2. A Sugar (ribose and deoxyribose)
 3. A Phosphate group

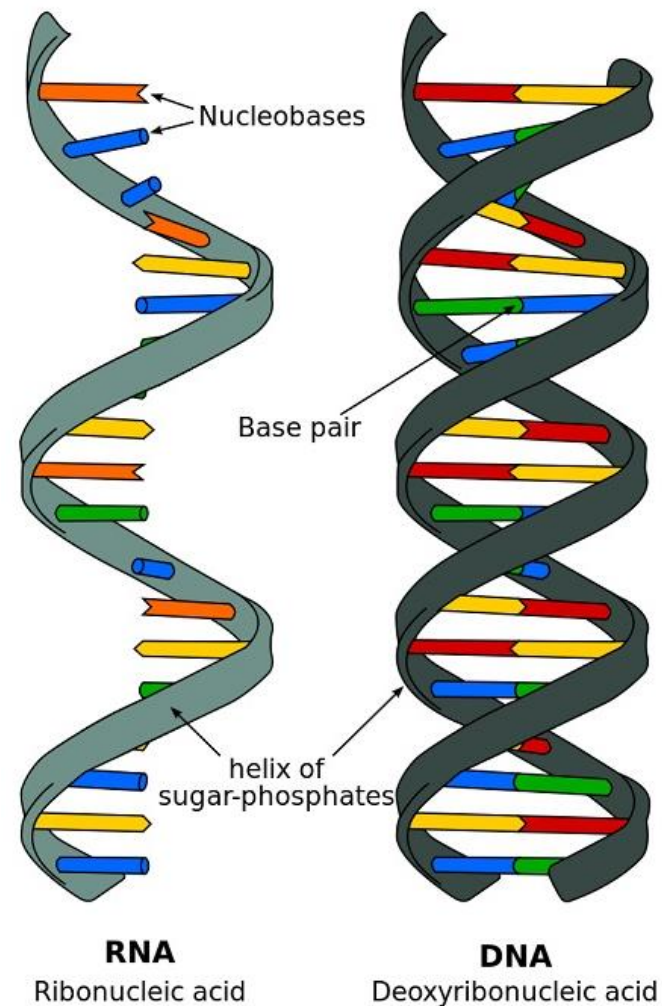
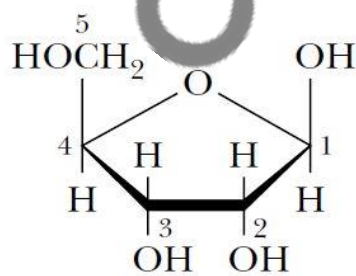


Fig. 9. General structure of RNA and DNA

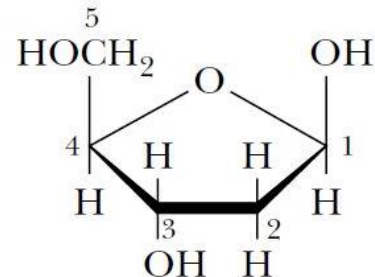
Nitrogenous bases and Sugars

- Nitrogenous bases found in nucleotides are aromatic heterocyclic compounds. The bases are of two types- **Purines** (Adenine and Guanine) and **Pyrimidines** (Cytosine, Thymine and Uracil).
- Five carbon containing monosaccharides (pentoses) are found in the nucleic acid structure. RNA contains **Ribose** while DNA contains **Deoxy-ribose**.



Furanose form of
D-Ribose

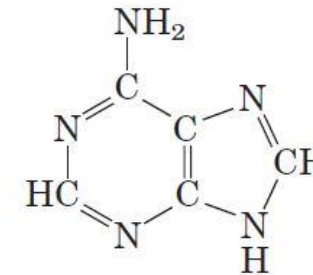
β -D-Ribofuranose



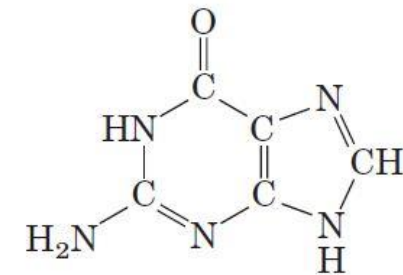
Furanose form of
D-2-Deoxyribose

β -D-2-Deoxyribofuranose

Fig. 10. Structures of Ribose and Deoxy-ribose 5 carbon sugars present in nucleic acids

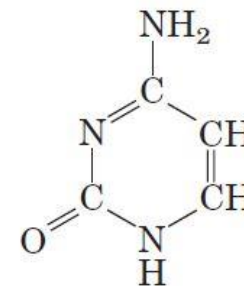


Adenine

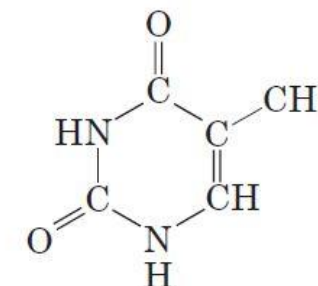


Guanine

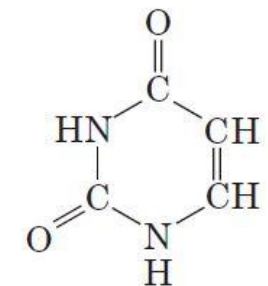
Purines



Cytosine



Thymine
(DNA)



Uracil
(RNA)

Pyrimidines

Fig. 11. Purines and Pyrimidines

Nucleotide

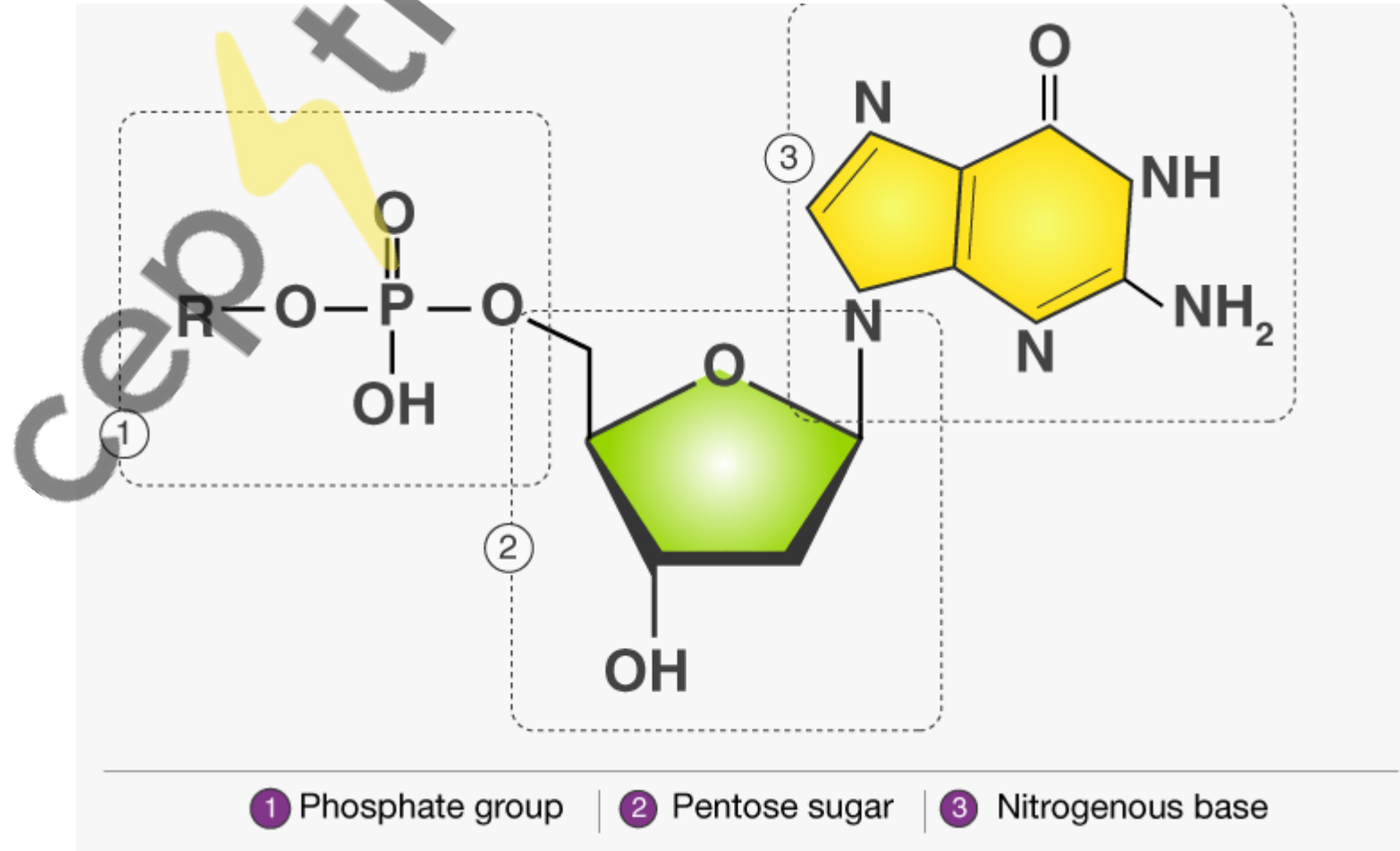
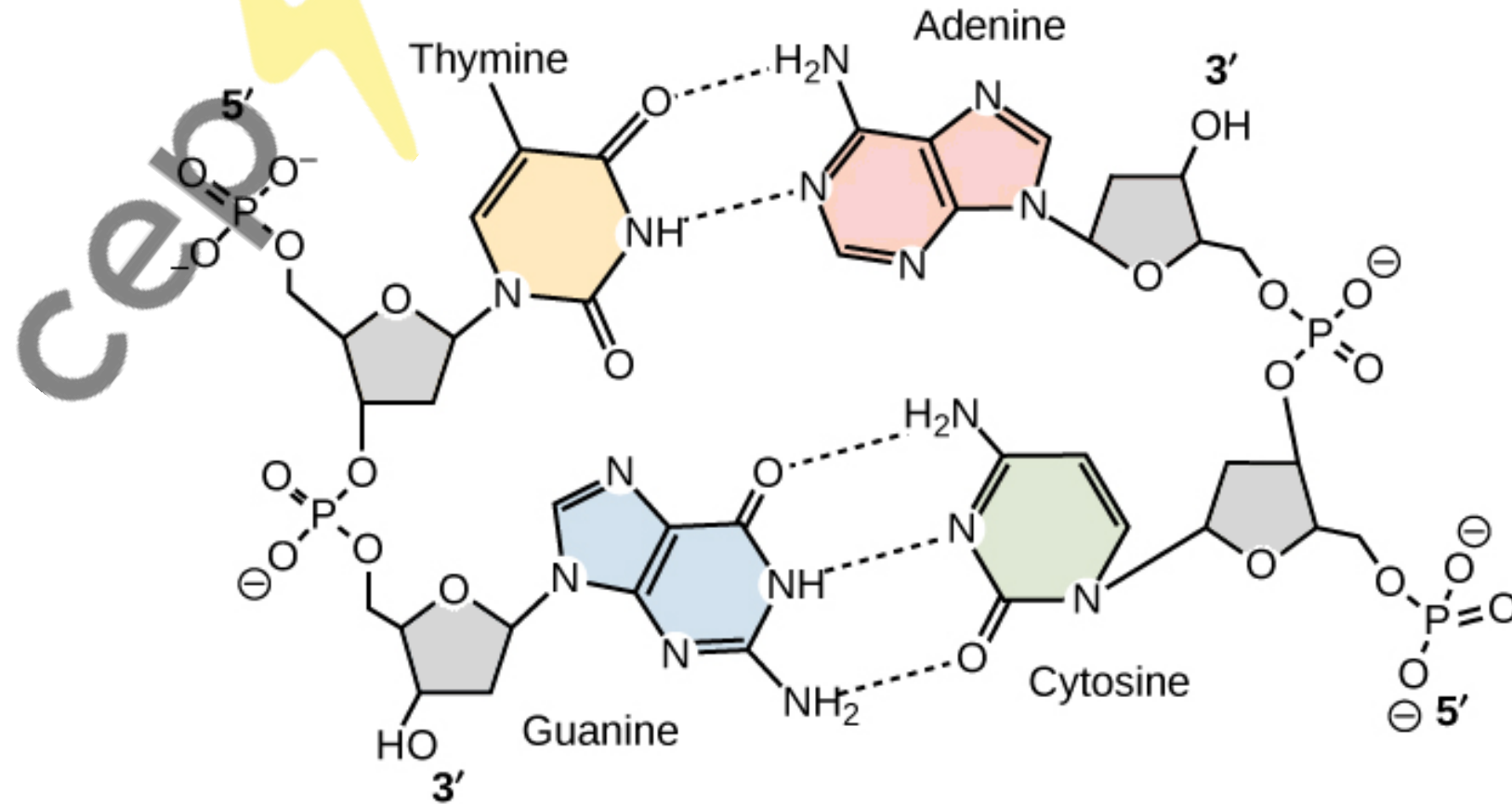
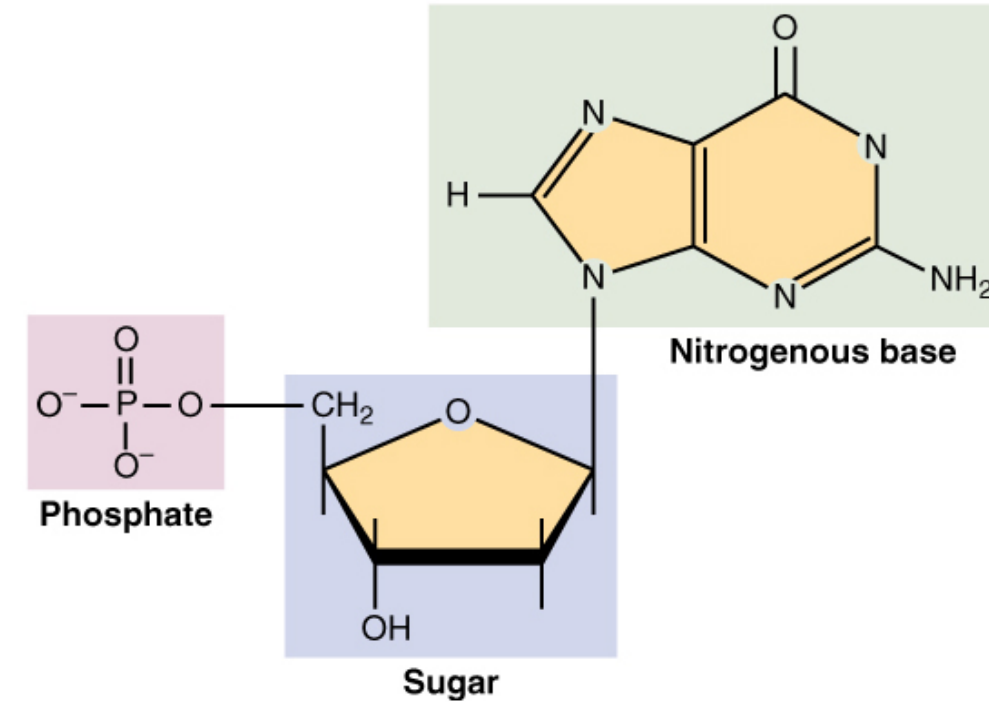
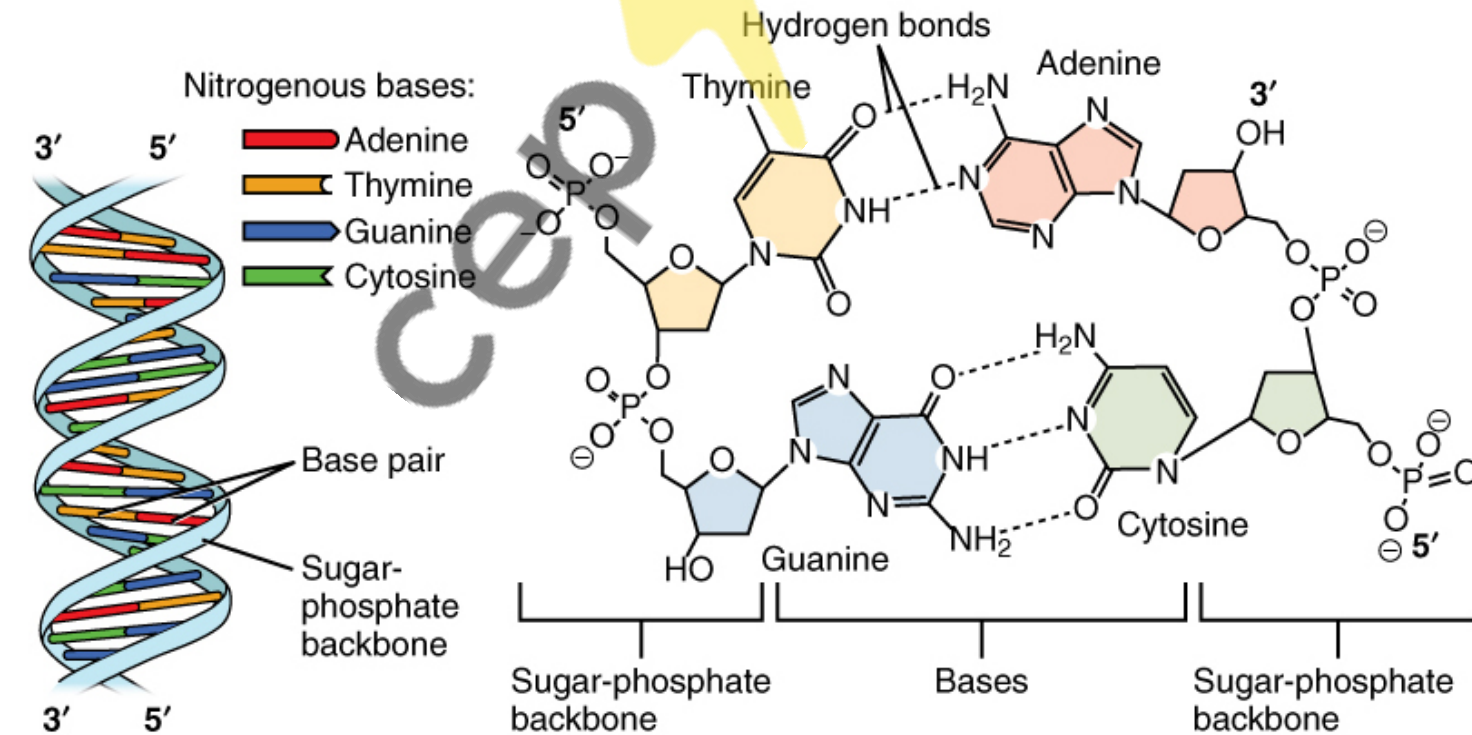


Fig. 12. A nucleotide (Adenosine mono phosphate, AMP) containing Adenine, Deoxyribose sugar and a Phosphohate group.



Nucleotide



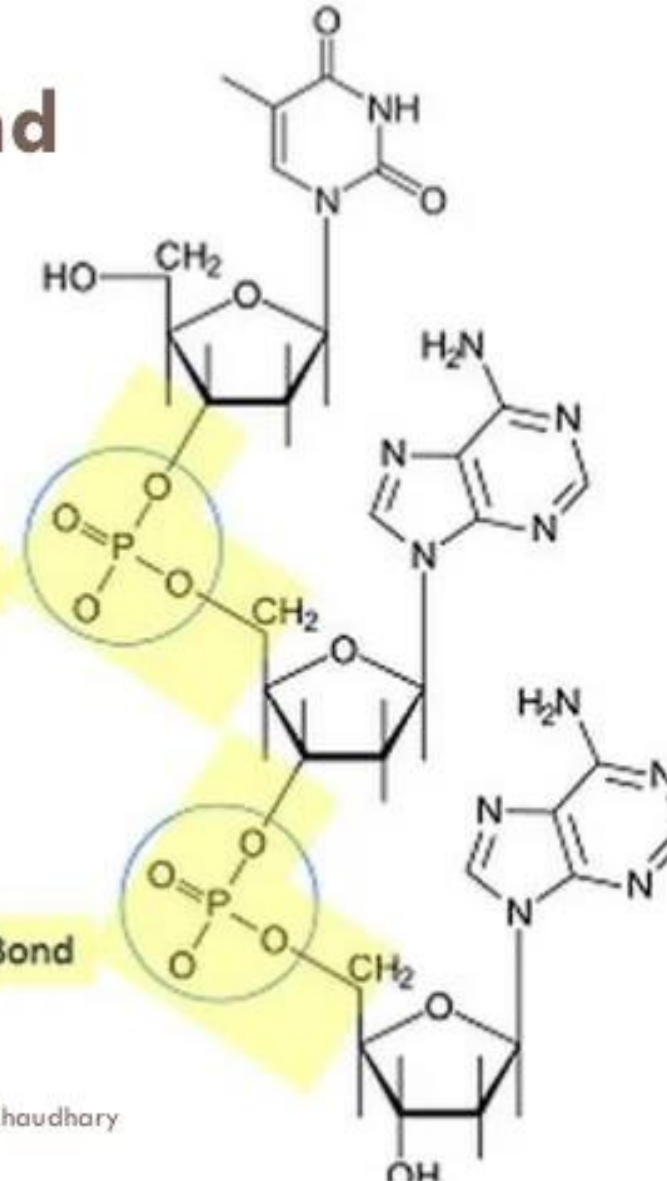
Phosphodiester bond

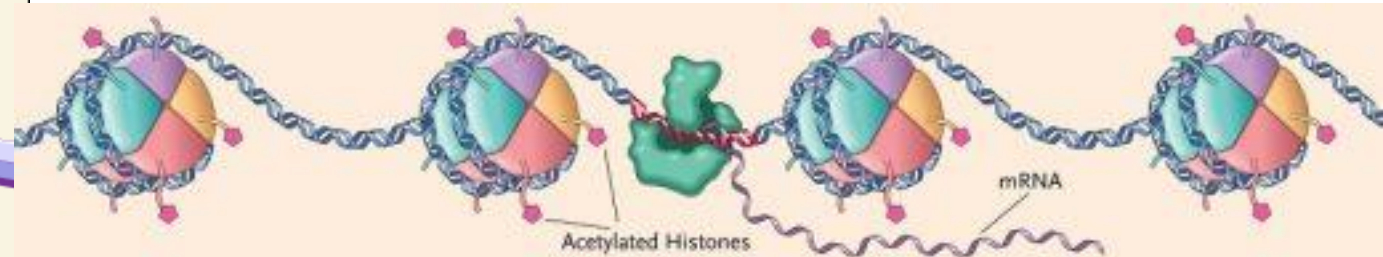
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Phosphodiester Bond

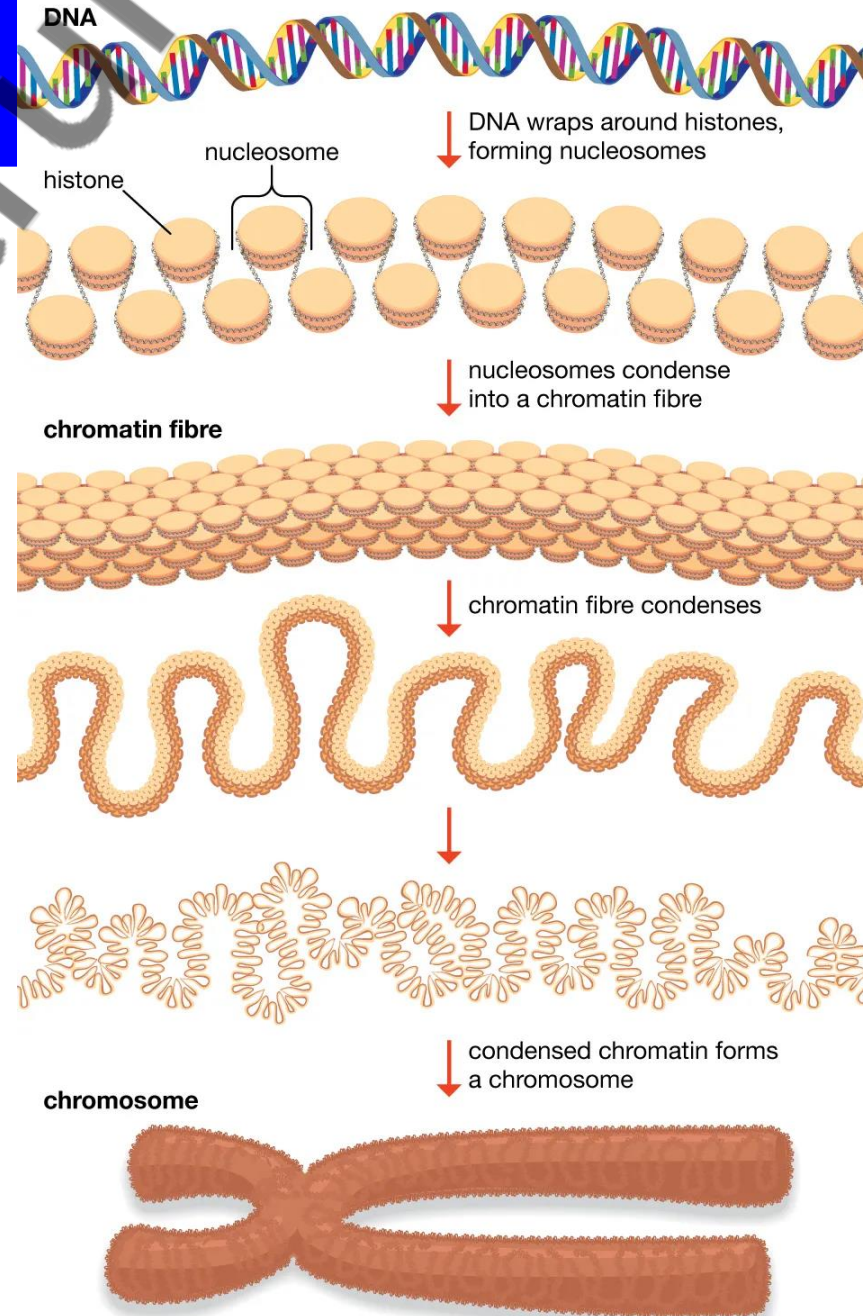
Phosphodiester bonds in DNA or RNA are cleaved hydrolytically by chemicals or enzymes such as **nucleases: deoxyribonucleases** for DNA and **ribonucleases** for RNA.

Phosphodiester Bond





DNA packaging into chromatin and chromosome



Proteins

- Proteins are a diverse and abundant class of biomolecules, constituting more than 50% of the dry weight of cells. This diversity and abundance reflect the central role of proteins in virtually all aspects of cell structure and function.
- Proteins are Linear Polymers of Amino Acids.
- Amino acid, as the name signifies a molecule that contains an amino group and an acid group. It is an organic molecule that consists of a basic amino group, an acidic carboxyl group and a unique organic side chain.
- Proteins are unbranched polymers of amino acids linked from carboxyl group to amino group, through a type of amide linkage called **peptide bond**.

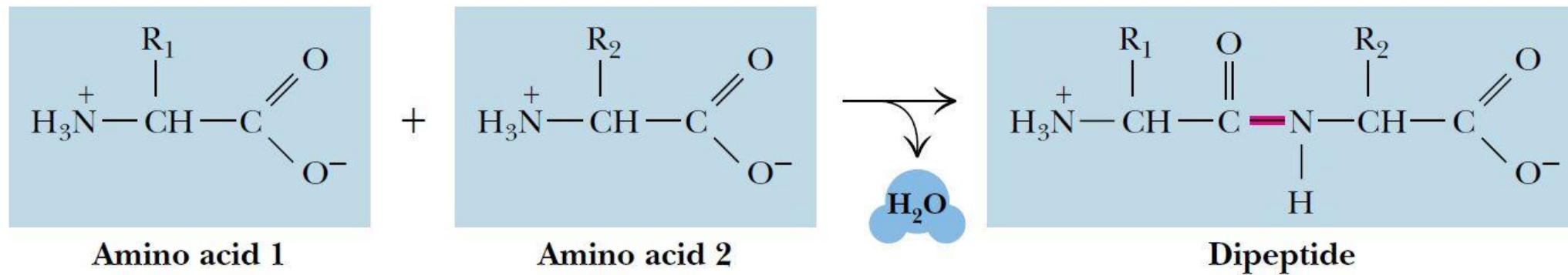


Fig. 8. Peptide bond formation is the creation of an amide bond between carboxyl group of one amino acid and the amino group of another amino acid. R_1 and R_2 represent, Side chain R groups of two different amino acids.

Amino acids, building blocks of Proteins

Amino acids are classified into two categories:

- i) **Essential amino acids**, which are not synthesized by the body and
- ii) **Non-essential amino acids**, which are synthesized by the body.

Table 1. Essential and Non-essential amino acids.

S. No.	Essential amino acids	S. No.	Non-essential amino acids
1	Histidine	01	Alanine
2	Isoleucine	02	Asparagine
3	Leucine	03	Aspartate
4	Lysine	04	Arginine
5	Methionine	05	Cysteine
6	Phenylalanine	06	Glutamate
7	Threonine	07	Glutamine
8	Tryptophan	08	Glycine
9	Valine	09	Proline
		10	Serine
		11	Tyrosine

Functional Classification of Proteins

Proteins are classified into various groups based on their function

- | | |
|----------------------------------|---|
| 1. Structural proteins | : Keratin of hair and nails, collagen of bone. |
| 2. Enzymes or catalytic proteins | : Hexokinase, pepsin (metabolism, digestion) |
| 3. Transport proteins | : Haemoglobin, serum albumin |
| 4. Hormonal proteins | : Insulin, growth hormone |
| 5. Contractile proteins | : Actin, myosin (muscle) |
| 6. Storage proteins | : Ovalbumin, glutelin (seeds) |
| 7. Genetic proteins | : Nucleoproteins (e.g. Histones) |
| 8. Defense proteins | : Immunoglobulins (antibodies), Snake venoms |
| 9. Receptor proteins | : Interferon receptor (e.g. cytokines) are signaling proteins released by host cells against virus. A virus-infected cell will release interferons causing nearby cells to heighten their anti-viral defenses. |

Lipids



- Lipids are a chemically diverse and heterogeneous group of compounds and it is difficult to define them precisely. The common and defining features of lipids are their low solubility in water and solubility in organic solvents. e.g. **fats, oils, waxes, phospholipids, sphingo-phospholipids, etc.,**
- **The biological functions of the lipids are as diverse as their chemistry.**
- Fats and oils are the principal stored forms of energy in many organisms.
- **Phospholipids and sterols are major structural elements of biological membranes.**
- Lipids play crucial roles as
 1. **Enzyme cofactors** : **Vitamin K (Phylloquinone)**
 2. Electron carriers : **Coenzyme Q (Mitochondria)**
 3. **Light-absorbing pigments** : **Chlorophyll**
 4. Hydrophobic anchors for proteins : **Glycosylphosphatidylinositol**
 5. **“Chaperones” to help membrane proteins fold** : **Phosphatidyl-ethanolamine**
 6. Emulsifying agents in the digestive tract : **Diglycerides**
 7. **Hormones** : **Testosterone**
 8. Intracellular messengers : **Diacylglycerol (DAG)**

Carbohydrates



- Carbohydrate is the most abundant biomolecule on earth. Each year, photosynthesis converts more than 100 billion metric tons of CO_2 and H_2O into cellulose and other plant products.
- Carbohydrate polymers (also called glycans) serve as structural and protective elements in the cell walls of bacteria, plants and in the connective tissues of animals.
- Complex carbohydrate polymers, that are covalently attached to the proteins or lipids act as signal molecules and are called **Glycoconjugates**.
- Carbohydrates are polyhydroxy aldehydes or ketones.
- There are three major size classes of carbohydrates: monosaccharides, oligosaccharides, and polysaccharides (the word “Saccharide” is derived from Greek *sakcharon*, meaning “sugar”).

Alteration of saccharon, from Ancient Greek, via Pali from Sanskrit शर्करा (śárkarā).

When the Greeks introduced from Asia Minor, it became in ancient Greece as Sakchar or Sakcharon.

Romans who subsequently took the word from the Greeks wrote it as Saccharum or Saccharon.

Carbohydrates

Carbohydrates

Monosaccharide

Glucose
Fructose
Galactose



Single sugar
molecule

Disaccharide

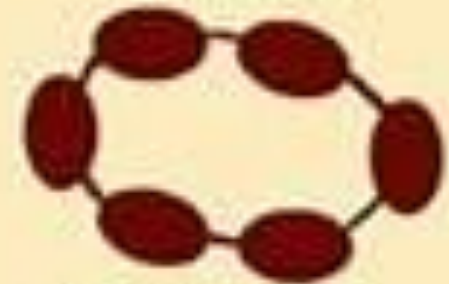
Maltose
Sucrose
Lactose



Two sugar
molecules linked

Polysaccharide

Starch
Glycogen
Cellulose



Many sugar
molecules linked

Mono-, 1.2 Di- and Oligo-saccharides

- **Monosaccharides**, or simple sugars, consist of a single polyhydroxy aldehyde or ketone unit. The two families of monosaccharides are aldoses and ketoses (Fig. 1a).
- The Most abundant monosaccharide in nature is the six-carbon reducing sugar, D-Glucose, also referred to as Dextrose.
- **Disaccharides** are made of two covalently linked monosaccharide units. The most abundant disaccharide is sucrose (from sugarcane), consisting of 6-carbon atom containing, D-glucose and D-fructose (Fig. 1b).
- **Oligosaccharides** are short chains of monosaccharide units, linked by glycosidic bonds.

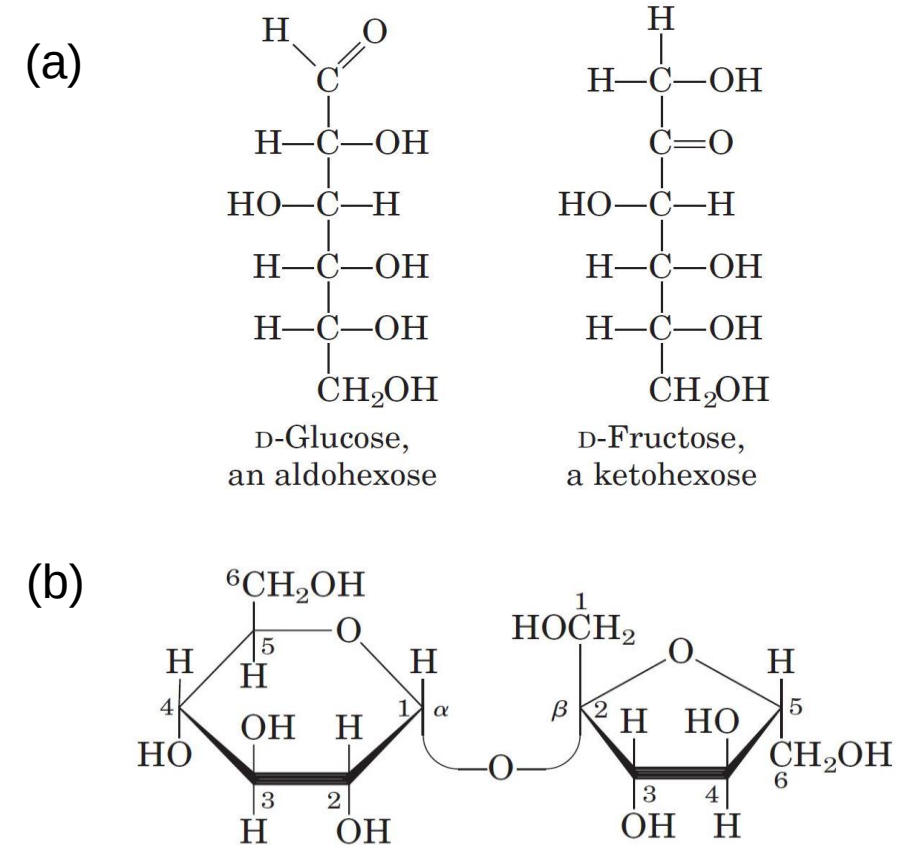


Fig. 1.

a) Representative monosaccharides.

D-Glucose an aldose and D-Fructose, a ketose.

b) Sucrose, a Disaccharide.

Polysaccharides (Cellulose, Starch and Glycogen)

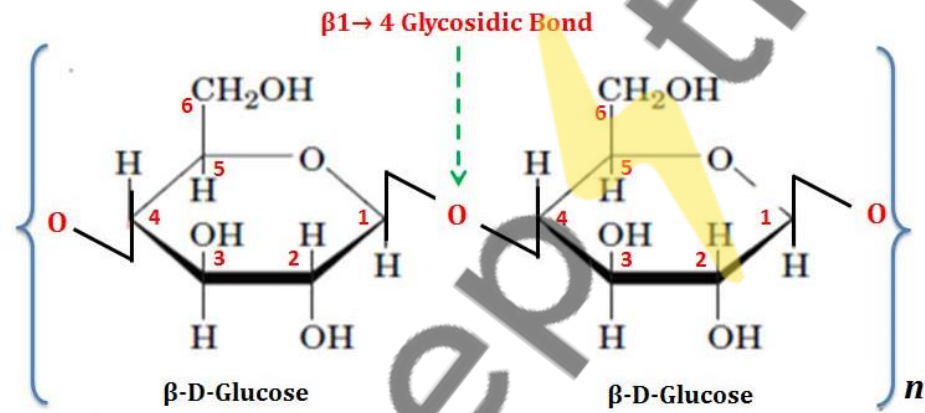


Fig. 3. Cellulose. Linear chain of D-glucose units.
Two D-glucose units joined by ($\beta(1 \rightarrow 4)$) linkage.

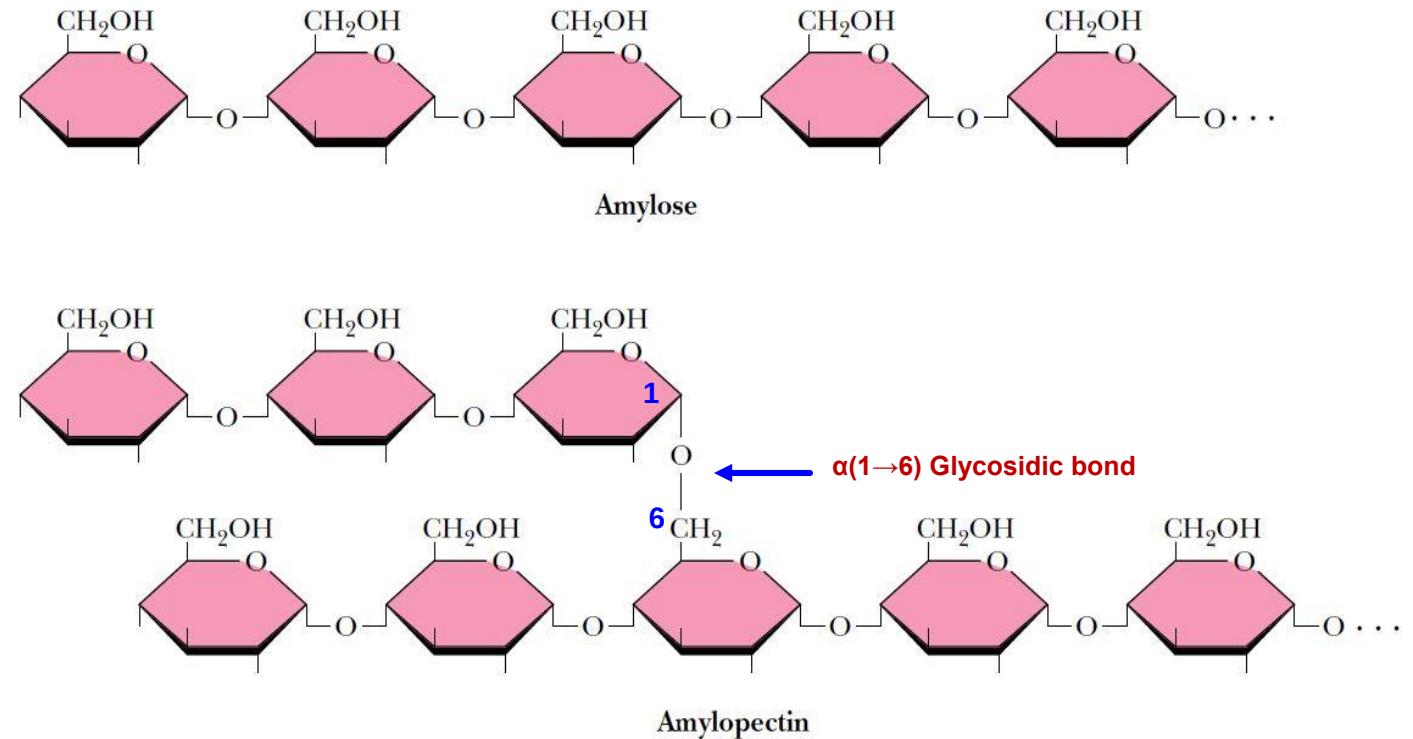


Fig. 4. Starch (Amylose and Amylopectin) and Glycogen (highly branched amylopectin).

Starch contains two types of glucose polymers, **Amylose** (Linear $\alpha(1 \rightarrow 4)$ linkages) and **Amylopectin** (Branched by $\alpha(1 \rightarrow 6)$ linkages), (less soluble in water and present in Plants as storage polysaccharide).

Glycogen has the same basic structure, but has more branching than **Amylopectin**. (water soluble and present in fungi and Animals as storage polysaccharide)

Glycoconjugates



- Glycoconjugates are conjugate (joined) molecules of Carbohydrates with other macromolecules such as Proteins or Lipids.
- They are ubiquitously (everywhere) distributed in microorganisms, cells and body fluids of animals and plants.
- These Glycoconjugates are central players in,
 1. Cell-cell recognition and adhesion
 2. Cell migration during tissue development
 3. Blood clotting and the immune response
 4. Wound healing and other cellular processes.

Glycoconjugates

There are 3 major types

1) **Proteoglycans** 2) **Glycoproteins** 3) **Glycolipids**

1. **Proteoglycans** are **macromolecules** composed of **Mucopolysaccharides** and **Proteins** linked by a **Tetra-saccharide** linkage.

Mucopolysaccharides are heteroglycans (made up of repeating units of **Amino sugars** and **Uronic acids**) commonly known as **Glycos-aminoglycans** (GAG).

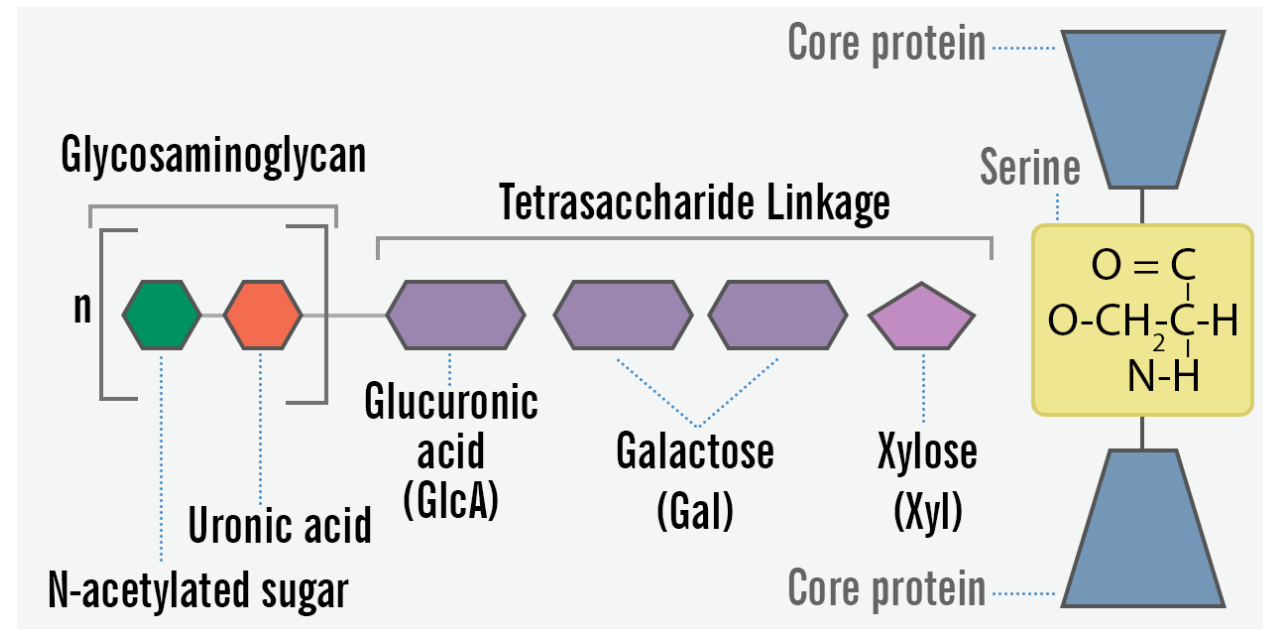


Fig. 5. Basic structure of Proteoglycan (**Peptidoglycan**).

Glycoconjugates

2. Glycoproteins have **one or more oligosaccharides** linked covalently **to a protein**. They are found on the outer face of Plasma membrane, in extracellular matrix.
e.g. **Mucins** are found at the mucosal surfaces in intestine, as they build the protective mucus and are also involved in Lubrication.
They are also found inside the cells, in Golgi complexes, Secretory vesicles and Lysosomes.

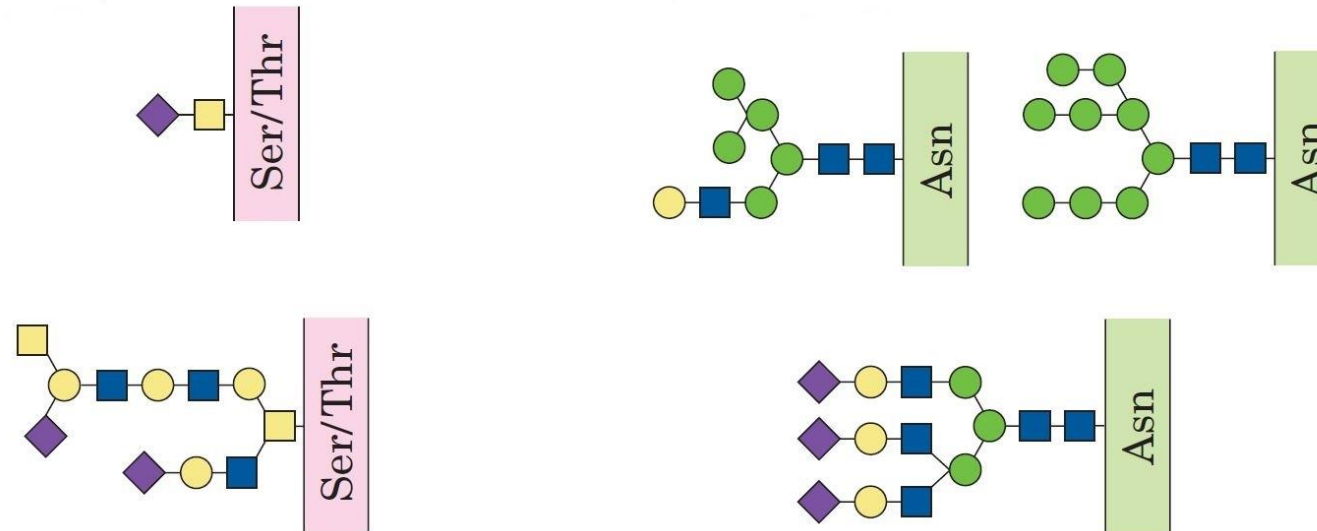


Fig. 6. Schematic structure of Glycoproteins.

Glycoconjugates

3. Glycolipids are **lipids with a carbohydrate moiety** attached by a glycosidic (covalent) bond. e.g. Membrane **Glyco-sphingolipids** in which, hydrophilic head groups are oligosaccharides. The **Neurons** in brain are rich in glycolipids, which help in nerve conduction and myelin formation. e.g. **Galactocerebroside**.

Glycolipid structure — cerebroside

- The carbohydrate component is linked by an O-glycosidic bond to ceramide
- Cerebroside contains a single sugar (Glu or Gal) or few sugars; they are abundant in brain and myeline sheath

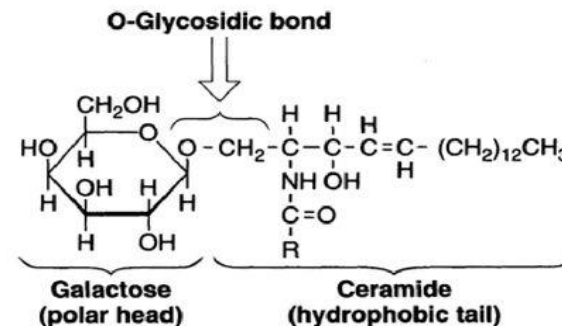


Fig. 7. Structure of **Galactocerebroside** (Glycosphingolipid).

Important Links



Macromolecules

- ☐ <https://www.youtube.com/watch?v=FgfkNBZaVTI>
- ☐ <https://www.youtube.com/watch?v=YO244P1e9QM>
- ☐ <https://www.youtube.com/watch?v=GhABWQC3YDs> (Nucleic Acids)
- ☐ <https://www.youtube.com/watch?v=SeOrvA9ikW8> (Nucleic Acids)
- ☐ <https://www.youtube.com/watch?v=kMg517MHDJs> (Proteins)
- ☐ <https://www.youtube.com/watch?v=ebScOnAJdu0> (Lipids)
- ☐ <https://www.youtube.com/watch?v=rQyWJIn1HYE> (Carbohydrates)
- ☐ <https://www.youtube.com/watch?v=Mb2bJgEZtwE> (Replication)
- ☐ <https://www.youtube.com/watch?v=YlOql3PQwjo> (Transcription)

Ghee vs Butter

- ☐ <https://www.youtube.com/watch?v=DXPUWbJgmIE>

BT101

Introductory Biology

(Lecture 9)

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Guwahati

Introduction



Evolution of life:

- ☐ Origin of Life
- ☐ Darwin's concepts of evolution
- ☐ Biodiversity

Cell, the structural and functional unit of life:

- ☐ Three domains of life
- ☐ Cell types, cell organelles and structure
- ☐ Basic biomolecules of cell

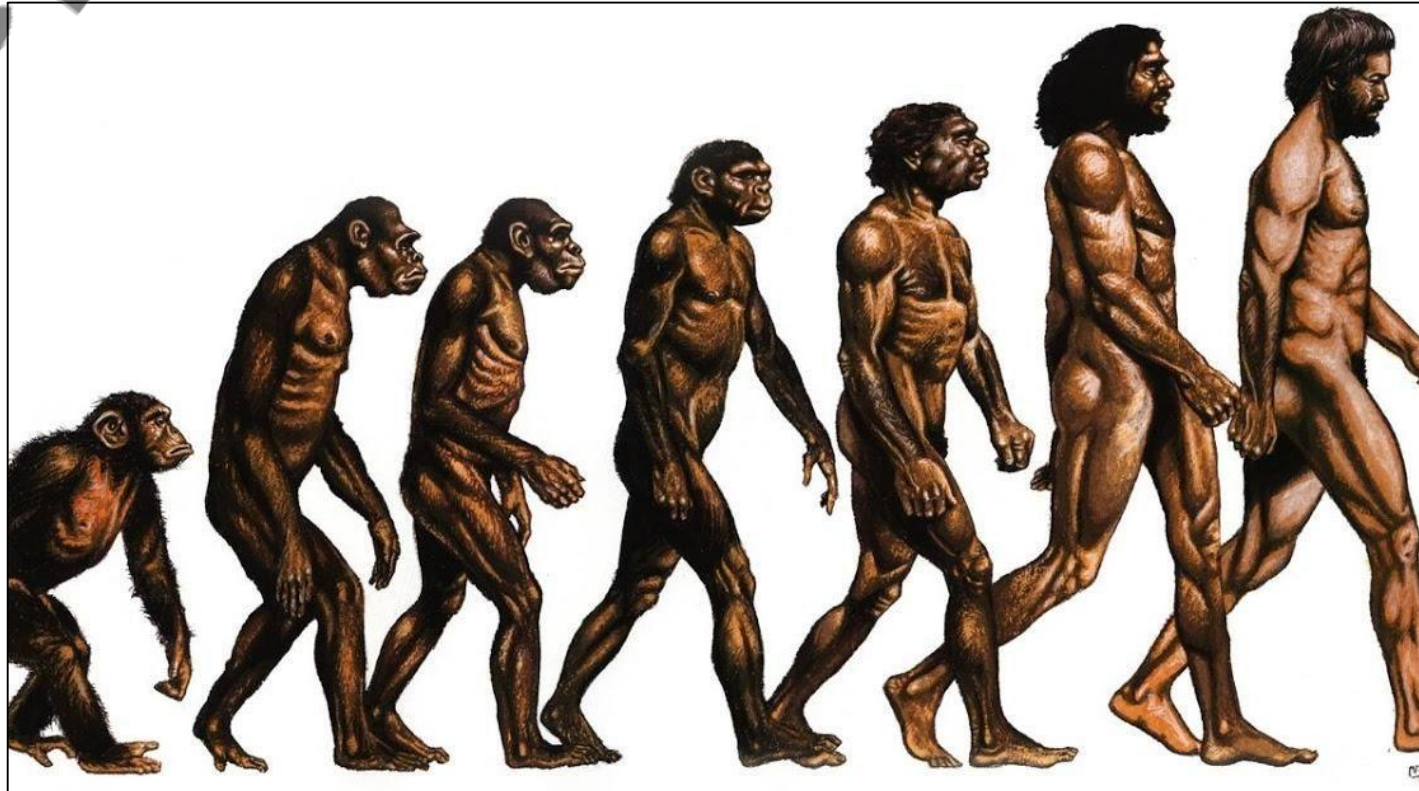
Introduction



- ❑ **Evolution** is the change in the inherited characteristics of biological populations over successive generations.
OR Simply
- ❑ **Evolution** is the change in the genetic composition of a population from generation to generation.
OR Simply
- ❑ **Evolution** is a process of gradual change that takes place over many generations, during which species of animals, plants, or insects slowly change some of their physical characteristics.
- ❑ **Evolutionary process** gives rise to diversity at every level of biological organization, including species, individual organism and molecules such as DNA and proteins.

Darwinism

- ❑ Darwinism is a **theory** of biological evolution.
- ❑ Darwinism is a **theory** of biological evolution by natural selection.
- ❑ Darwin defined **evolution** as "descent with modification," the idea that species change over time, give rise to new species, and share a common ancestor.



Darwinism

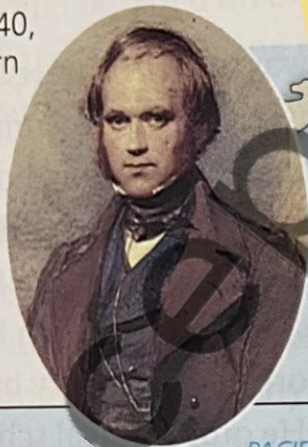


- ❑ It states that all species of organisms arise and develop through the process of natural selection of small, inherited variations that increase the individual's ability to compete, survive and reproduce. Also called Darwinian theory.
- ❑ The mechanism that Darwin proposed for evolution is natural selection. Because resources are limited in nature, organisms with heritable traits that favor survival and reproduction will tend to leave more offspring than their peers, causing the traits to increase in frequency over generations.
- ❑ Natural selection causes populations to become adapted, or increasingly well-suited, to their environments over time. Natural selection depends on the environment and requires existing heritable variation in a group.

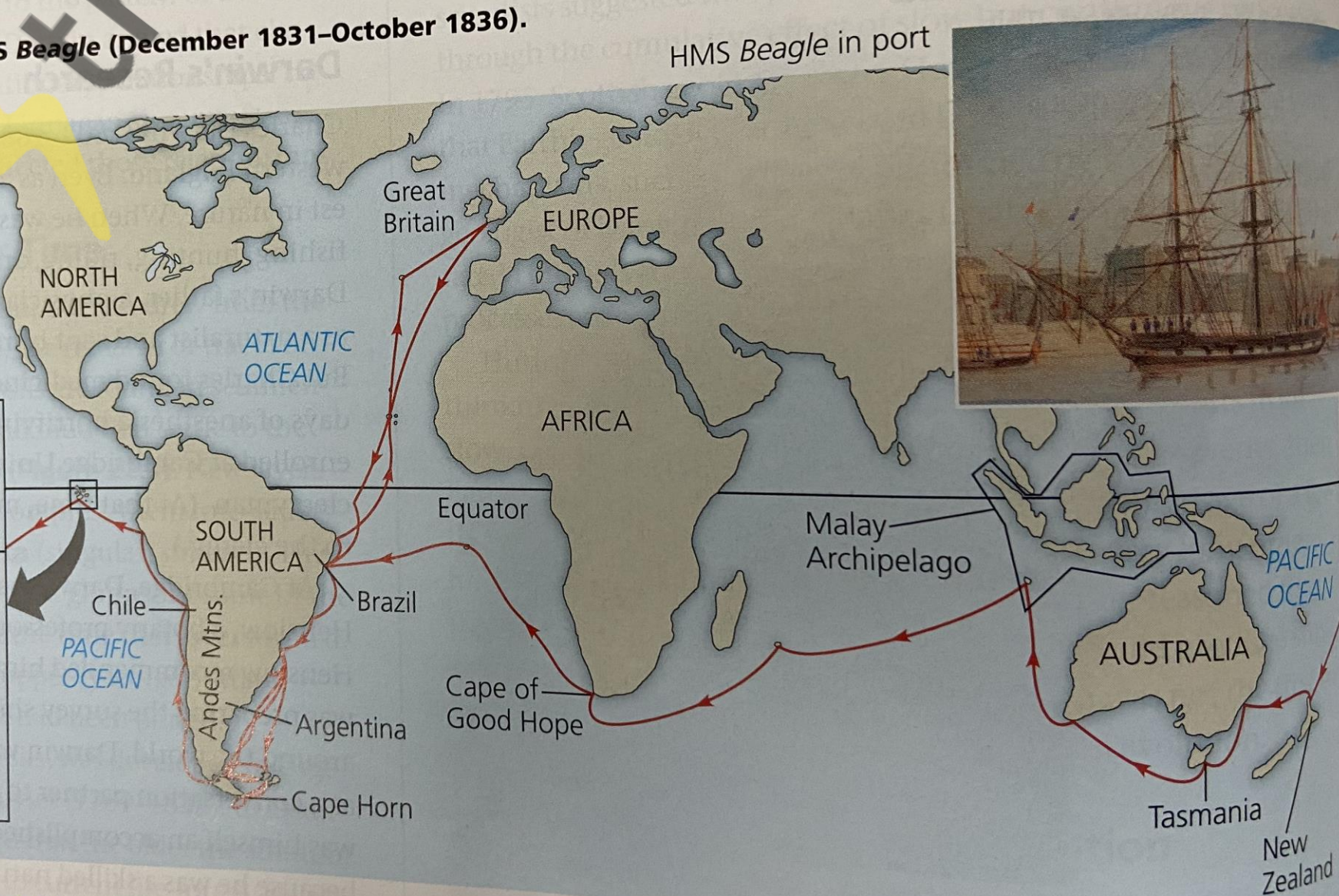
Introduction

▼ **Figure 22.5** The voyage of **HMS Beagle** (December 1831–October 1836).

Darwin in 1840,
after his return
from the
voyage



HMS Beagle in port



Lamarckism



- ❑ In the early nineteenth century, the French naturalist **Jean-Baptiste Lamarck** published a book proposing that evolution took place and suggesting a mechanism by which it might occur.
- ❑ **Lamarck proposed that** modifications in an individual caused by its environment, or the use or disuse of a structure during its lifetime, could be inherited by its offspring and lead to a change in a species.
- ❑ **For instance, Lamarck suggested that giraffes have long necks because some of them stretched their necks during their lifetimes, then passed the stretched neck on to their children.**
- ❑ This mechanism for evolutionary change did not turn out to be correct, and today, we often remember Lamarck as the one who "got the wrong answer."
- ❑ However, before we judge, we should remember that Lamarck was a pioneer despite his incorrect idea.
- ❑ He was one of the very first to seriously engage with the idea of evolution and to actually suggest how it might take place.

Darwin's theory

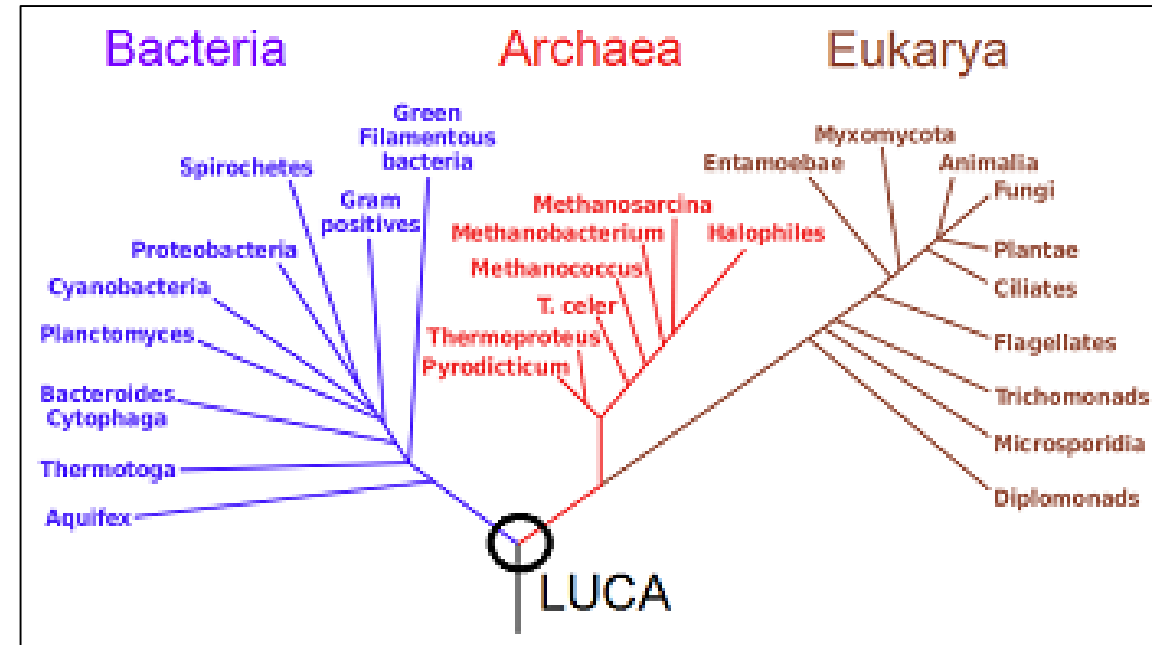


Four points of Theory

1. Common Descent
2. Gradualism
3. Multiplication
4. Natural Selection

1. Common Descent

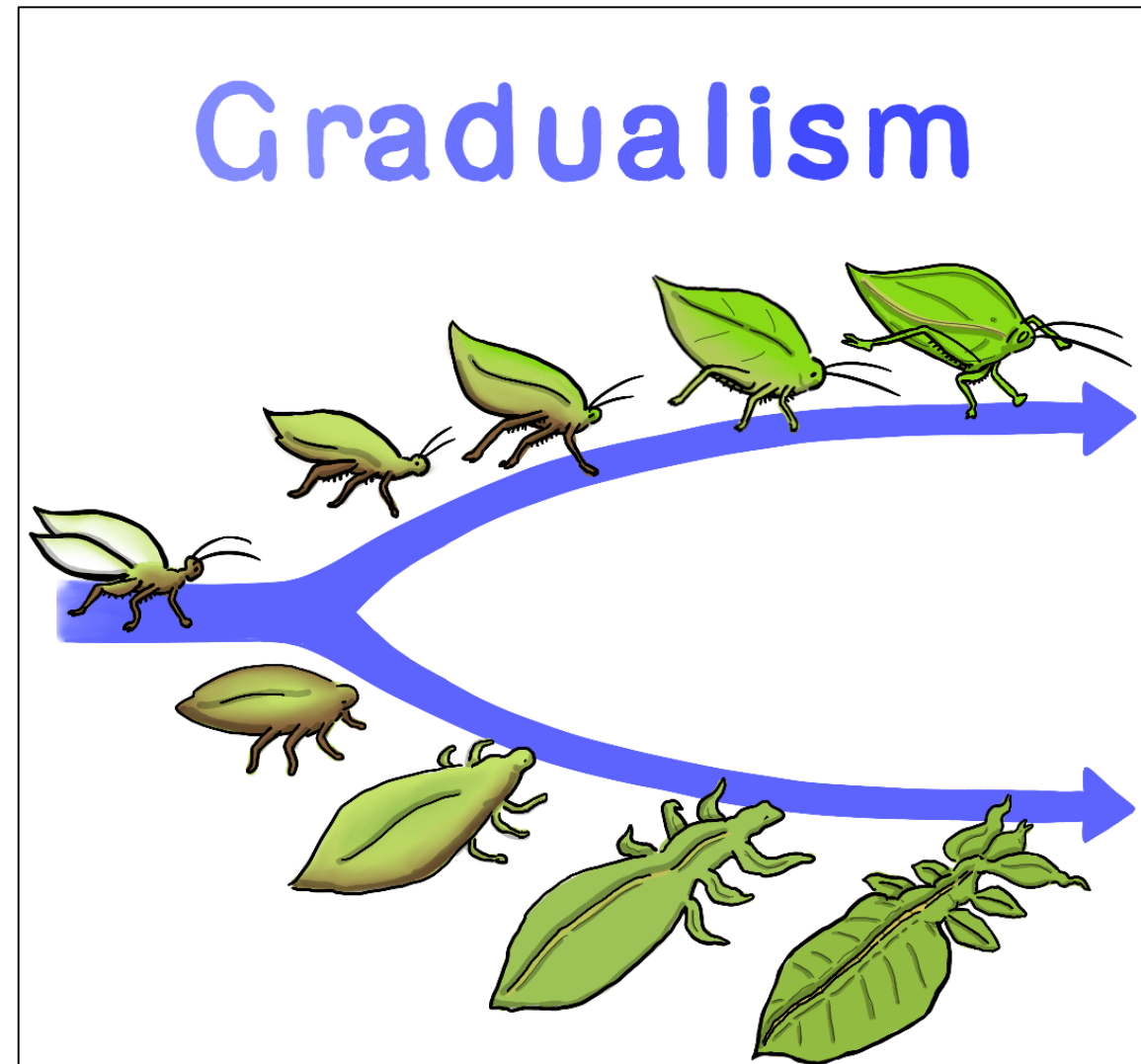
- All organisms had **one common ancestor** in the past and with the change in climate and habitat, they descended from that unknown ancestor.
- Over a period of time, the changes within organisms got collected resulting in their survival for the changing environment. This is also termed as **adaptation**.



2. Gradualism

Origin and evolution of new species depend on the accumulation of changes. Over a period of time upon accumulating different changes, results in adaptation, and a new species is evolved. Hence, **evolution is a slow and gradual process.**

The different forms show the species slowly changing over many years.
***There are many generations between each form.**



3. Multiplication



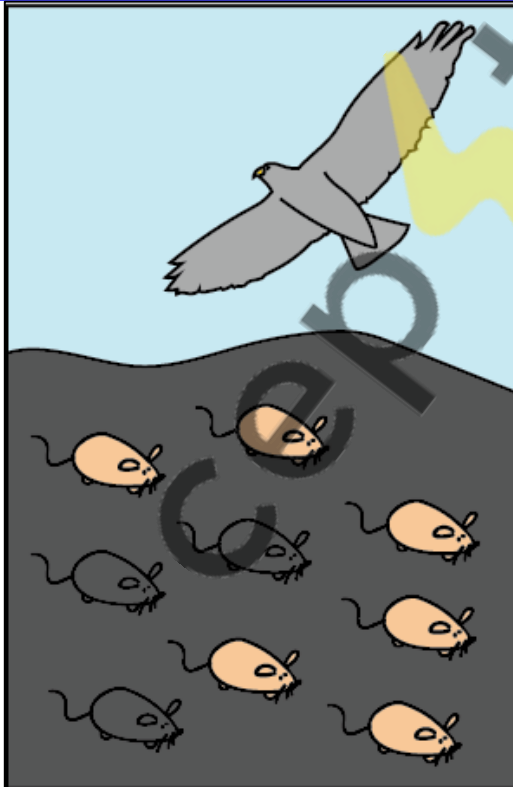
- Over a time period, changes with in species takes place.
- These changes can result in the species with similar in appearance or completely different in appearance and behavior.
- Hence, **species multiply** either having similar characteristics to their ancestor or different characteristics than them.

4. Natural Selection



- The evolutionary changes comes through the production of variation in each generation. The differential survival of individuals is based on different combinations of these variable characters.
- Individuals with characteristics which increase their probability of survival will have more opportunities to reproduce. **Their offsprings benefits from the inherited advantageous character which then spreads to their population.**

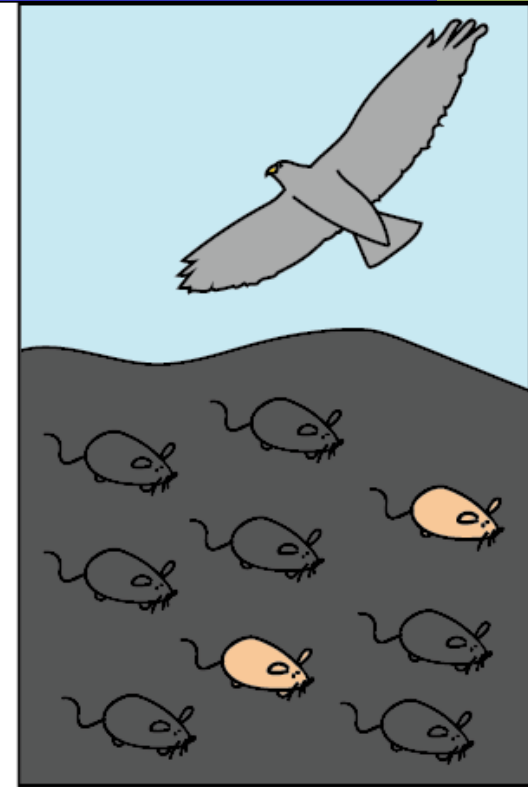
Natural Selection



Some mice are eaten by birds



Mice reproduce, giving next generation



A population of mice has moved into a new area where the rocks are very dark. Due to natural genetic variation, some mice are black, while others are tan.

Tan mice are more visible to predatory birds than black mice. Thus, tan mice are eaten at higher frequency than black mice. Only the surviving mice reach reproductive age and leave offspring.

Because black mice had a higher chance of leaving offspring than tan mice, the next generation contains a higher fraction of black mice than the previous generation.

How Natural Selection occurs?

Natural Selection occurs in 5 steps



1. Overproduction

Every species on this planet has potential to give birth to young ones and increase its number. This is known as **Overproduction**. There is no limit for any species for the number of offsprings.



2. Competition

More the number of species, higher will be the competition for their survival. It is because, the environment has finite resources that can support the limited number of species.

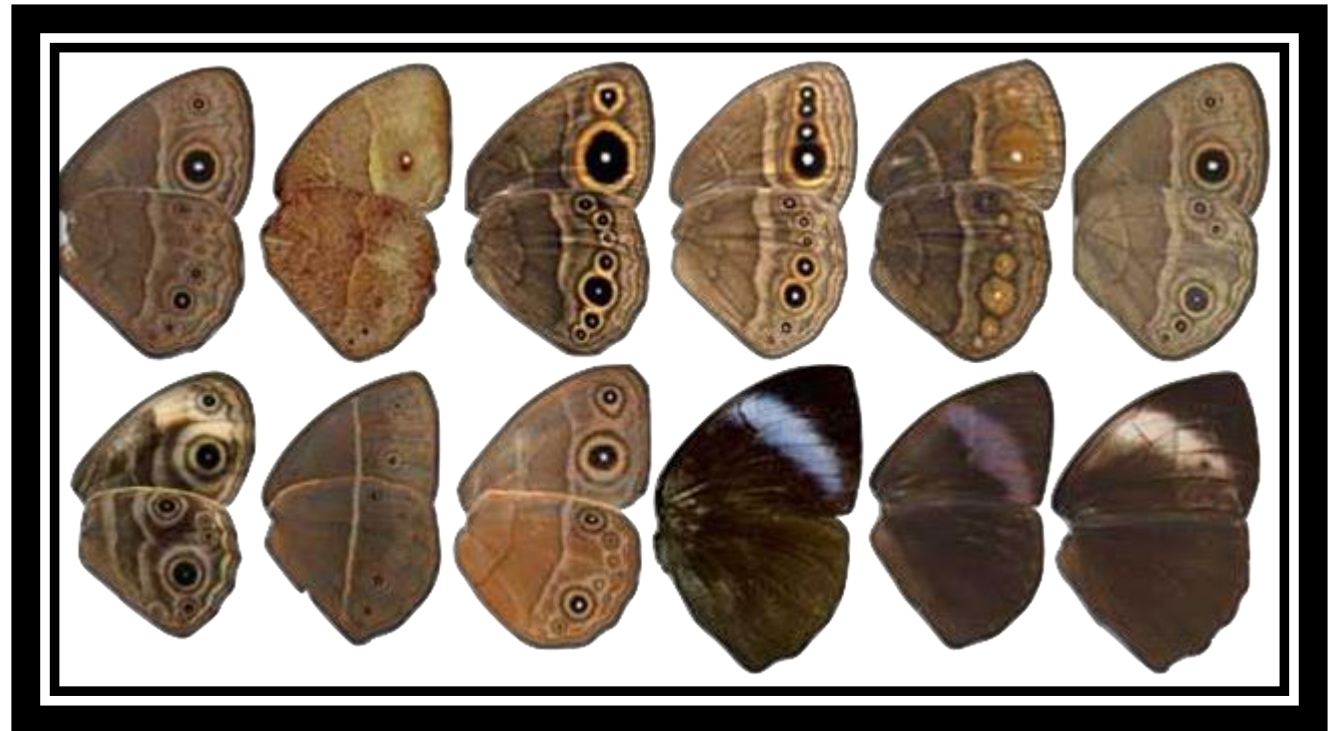
Hence, those offsprings who meet the environmental criteria can survive.



3. Variation

- Variation is the key ingredient for survival of the species.
- Those species which have the beneficial variations have higher chances of survival than those species without variation. Hence, as a result of variation, the species develop morphological and character differences.

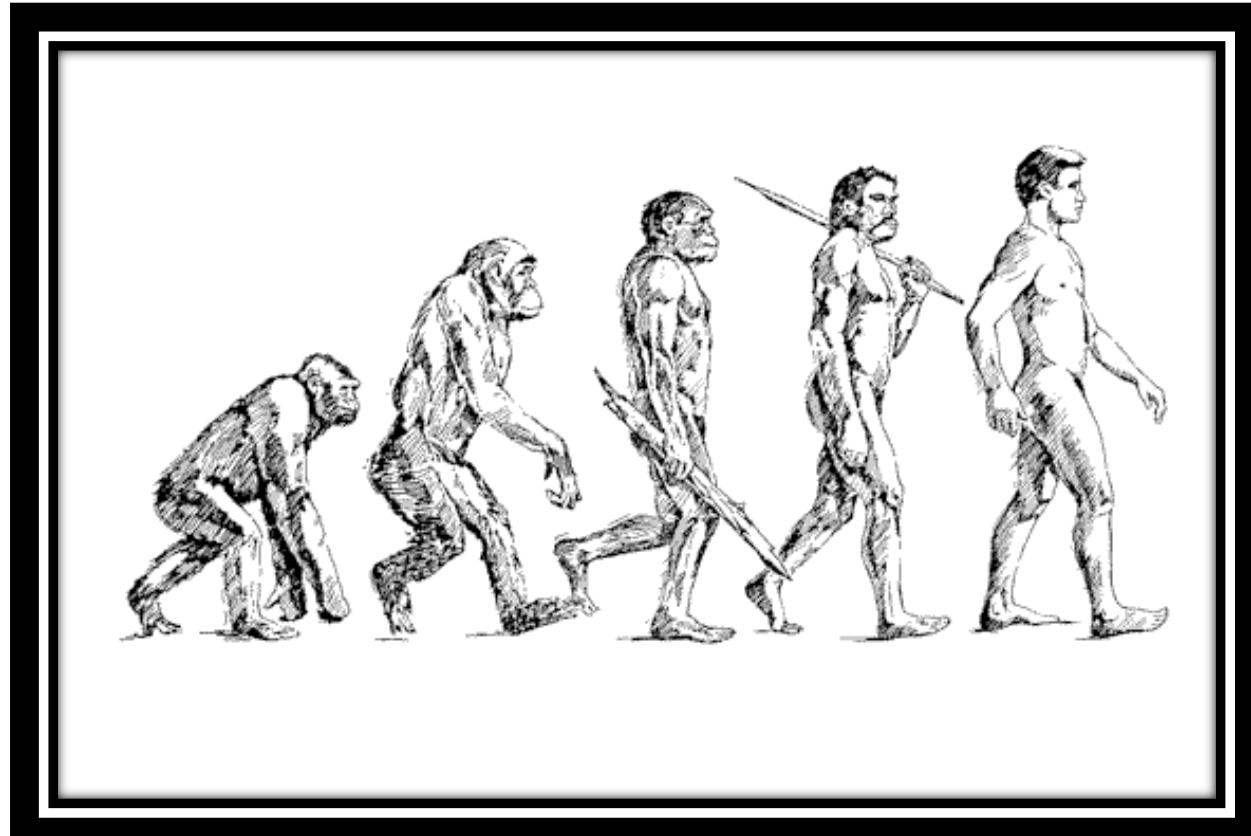
In predator-prey interaction, variation helps **butterflies having similar color wings as the dark background of tree bark** to hide from predators.



4. Survival of the fittest

Species which have developed the variation and can tackle the environmental conditions are fit for survival and reproduction. Hence, fitness depends upon the adaptation of species to a particular environment.

Therefore, any species with low fitness dies and that with higher fitness survives.

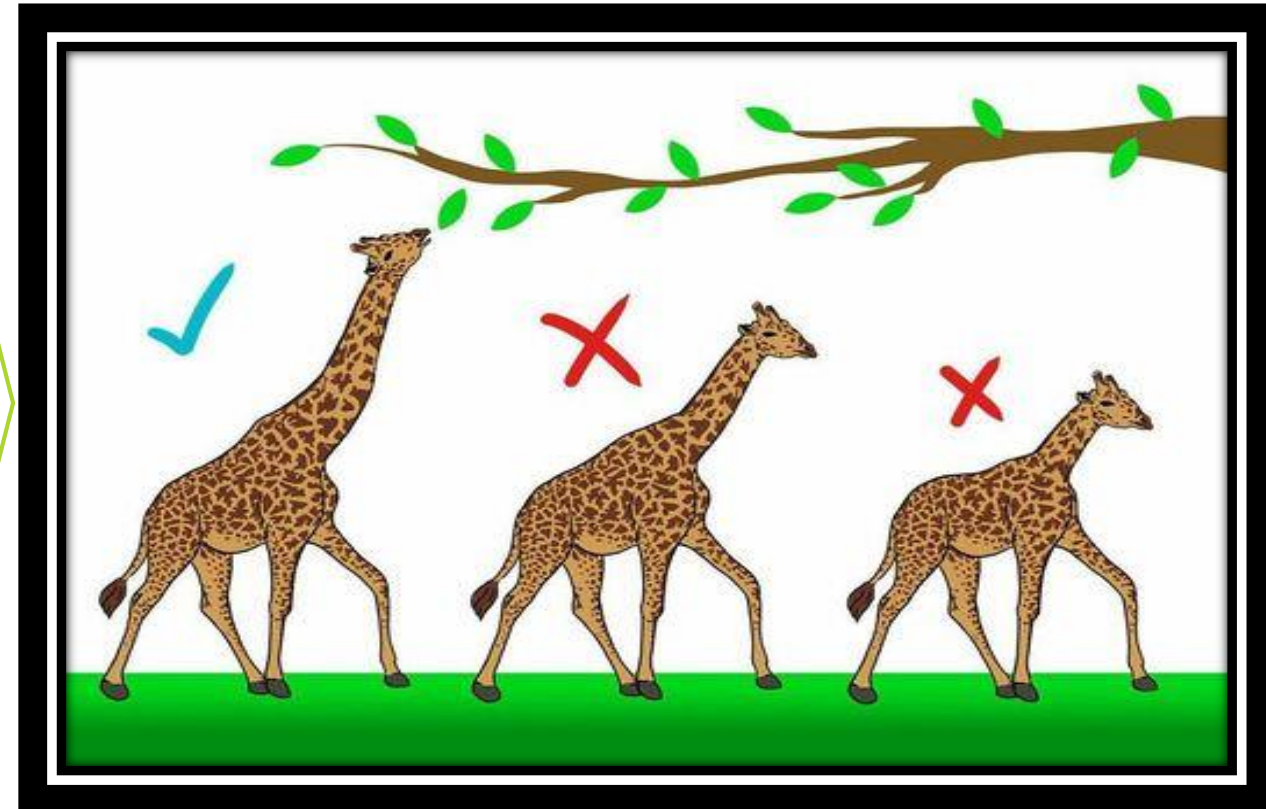


5. Natural selection

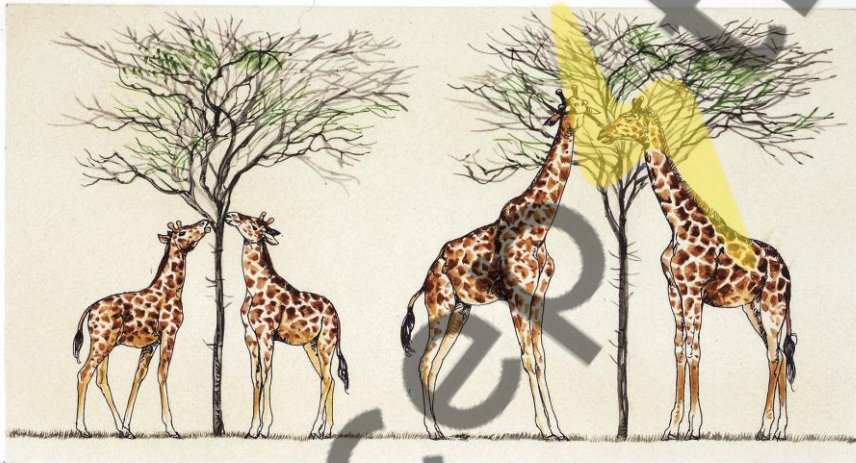
After a particular time period, the **changes or variation** acquired by the species results in the **Natural Selection**. This means that the nature selects the beneficial variation and increases the survival rate of the offspring.

According to **Darwin**, the long neck of Giraffe is the result of natural selection, acting through the animal's tree-feeding habit.

During dearth, the individuals who could reach even an inch or two above the others, survived.

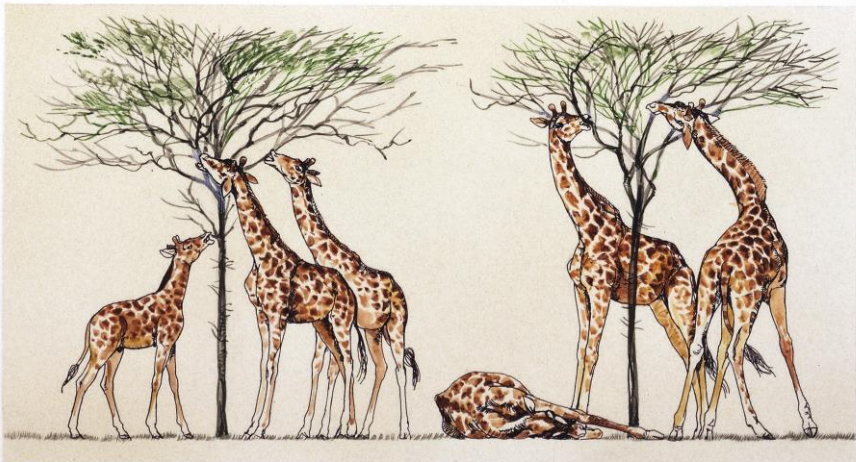


5. Natural selection



Theory of Jean-Baptiste Lamarck

and



Theory of Charles Darwin

Examples of Natural Selection

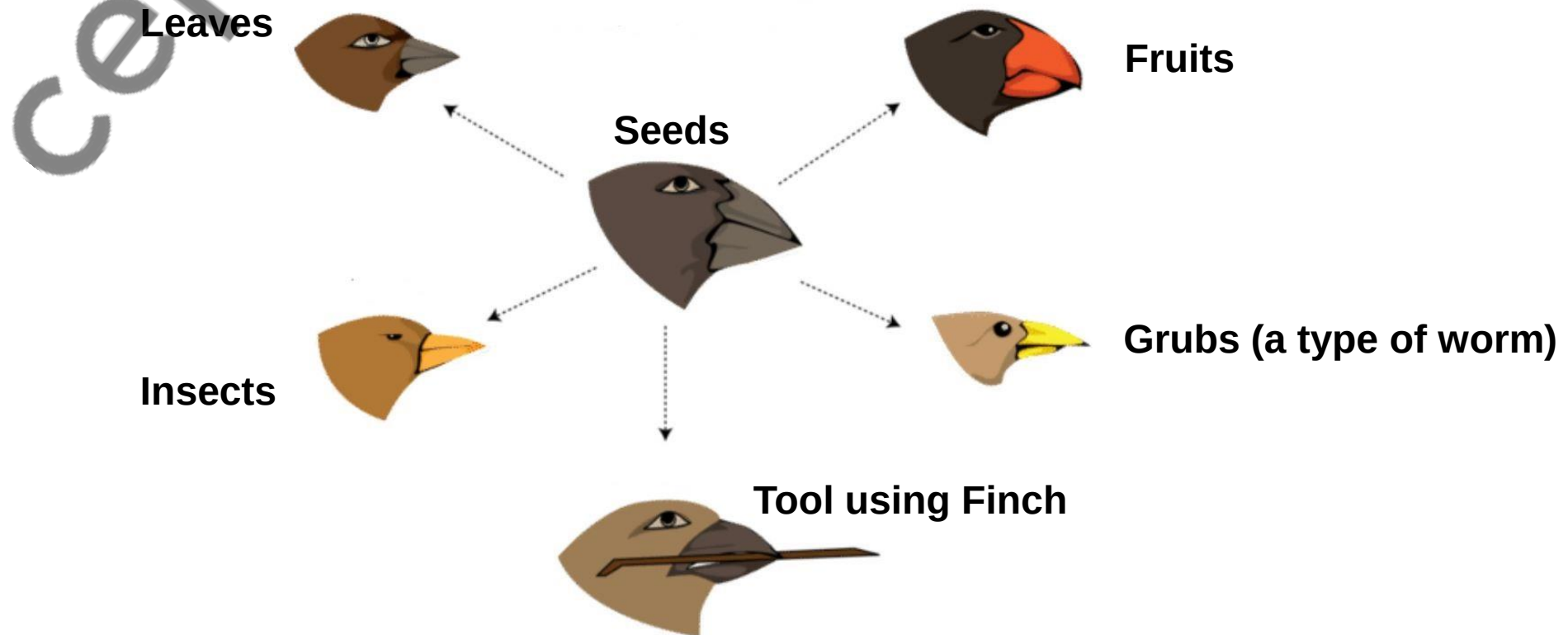


1. Drug Resistance in bacteria:

Antibiotic resistance is a consequence of **evolution** *via* natural selection. The **bacteria** which acquire mutation, allows them to survive and to reproduce. They pass this trait to their progeny, which will be a fully **drug resistant** generation.

Examples of Natural Selection

2. Shape of beak in finches: Darwin observed that the beak shape varies among Finch species. He postulated that the **beak** of an ancestral species had adapted over time to equip the **finches** for different food sources.



Example of Natural Selection

3. **The Peppered Moth:** **Evolution** of **peppered moth** is an **evolutionary** instance of directional color **change** in their population as a consequence of Air Pollution during the Industrial **Revolution**. The number of dark-color **moths** increased at that time, which is an example of **industrial melanism**.

Light gray peppered moth (*Biston betularia*) and a dark pigmented variant rest near each other on the trunk of a soot-covered oak tree.

Against this background, the light gray moth is more easily noticed than the darker variant.



Video



Important Links to see Videos



- ❑ **Darwin's theory of evolution:**
- ❑ <https://www.youtube.com/watch?v=urSNtaQKOAk>
- ❑ <https://www.youtube.com/watch?v=0SCjhI86grU>